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Barriers and Enablers for Implementing Nature-based Solutions in the Global North and Global South: A Comparative Analysis and Toolkit Proposal

María Isabela Villamizar Ariza

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María Isabela Villamizar Ariza

Supervisor: Malgorzata Blicharska

Subject Reviewer: Michael Jones

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MARÍA ISABELA VILLAMIZAR ARIZA

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Abstract:

Nature-based Solutions (NbS) have increasingly been recognized as a key strategy for addressing the triple planetary crisis of climate change, biodiversity loss, and pollution. However, despite the growing body of research on NbS, the literature remains strongly concentrated in the Global North (GN), while the differences between the Global North and the Global South (GS) regarding aspects affecting NbS implementation have not yet been extensively explored. In addition, practical guidance for implementers remains divided across repositories, and methodologies. This study aims to: (1) analyse how barriers and enablers for implementing NbS differ between the GN and the GS; and (2) explore if a practical toolkit for implementing NbS can be designed based on existing tools and methodologies.

To address these objectives, the research adopted a mixed-methods approach combining a literature review, semi-structured interviews, and the development of a toolkit prototype. The literature review analysed 45 articles using MAXQDA coding to identify barriers, enablers, and regional patterns related to NbS implementation. Additionally, five semi-structured interviews were conducted with NbS implementers and stakeholders from both the GN and the GS. The findings from the literature review, interviews, and additional grey literature were then used to design an initial user-oriented toolkit for NbS practitioners.

The results indicate that barriers to NbS implementation are highly context-dependent and differ between regions. In the GS, barriers were more frequently associated with governance weaknesses, policy misalignment, limited financial and technical capacity, rapid urbanization, and restricted access to technology and education. In contrast, the GN literature more commonly highlighted institutional inertia, regulatory complexity, and land availability constraints. At the same time, many enablers appeared more transferable across regions, particularly those related to collaboration, stakeholder engagement, trust-building, knowledge sharing, participatory governance, and continuous support processes. The research also revealed that many challenges surrounding NbS implementation are not solely technical, but strongly connected to governance systems, social dynamics, knowledge translation, and long-term stakeholder relationships.

Finally, the study demonstrates that it is possible to design a practical toolkit for NbS implementation by compiling existing tools, frameworks, and methodologies. However, the process also revealed that relevant NbS knowledge remains fragmented across repositories, disciplines, and languages. Therefore, the toolkit developed in this research should be understood as an initial and evolving prototype rather than a definitive resource. Overall, this research contributes to a better understanding of the regional dynamics shaping NbS implementation and highlights the importance of context-sensitive, collaborative, and adaptive approaches for scaling NbS across diverse contexts.

Keywords: Barriers, Enablers, Global North, Global South, Nature-based Solutions (NbS), Stakeholder Engagement, Toolkit Development

María Isabela Villamizar Ariza, Department of Earth Sciences, Uppsala University, Villavägen 16, SE- 752 36 Uppsala, Sweden

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Summary:

Humanity is currently facing major environmental challenges, including climate change, biodiversity loss, and pollution. In recent years, Nature-based Solutions (NbS) have gained increasing attention as a way of addressing these crises while also supporting human wellbeing. NbS are actions that work together with nature, such as restoring ecosystems, creating urban green spaces, improving water management, or supporting more sustainable agricultural practices. These solutions can provide environmental, social, and economic benefits at the same time.

However, implementing NbS is often complex. Different countries and regions face different social, political, financial, and environmental conditions, which influence how these solutions are developed and applied. This thesis focuses on understanding these differences by comparing the barriers and enabling factors affecting NbS implementation in the Global North (GN) and the Global South (GS). The study also explores whether existing tools and methodologies can be brought together into a practical toolkit to support people working with NbS.

To answer these questions, the research combined a literature review of 45 scientific articles with five interviews conducted with practitioners working with NbS projects in Colombia, Europe, and the United States. The interviews focused on projects related to ecosystem restoration, biodiversity conservation, water governance, and sustainable land management.

The findings show that many barriers to NbS are strongly connected to local realities. In the Global South, challenges such as limited funding, governance issues, and institutional weaknesses were more common. In contrast, many of the enabling factors identified across both regions were linked to collaboration, trust-building, stakeholder participation, knowledge sharing, and continuous learning processes. One of the strongest findings of the research is that successful NbS implementation depends not only on technical or environmental aspects, but also on people, relationships, and long-term cooperation.

Based on these findings, the thesis developed an initial prototype of a practical NbS toolkit that compiles tools, methodologies, frameworks, and knowledge platforms that may support NbS implementation. At the same time, the research highlights that many of the challenges surrounding NbS are structural and cannot be solved through technical tools alone. Instead, implementing NbS successfully requires approaches that are flexible, collaborative, and adapted to the specific realities of each territory.

Keywords: Barriers, Enablers, Global North, Global South, Nature-based Solutions (NbS), Stakeholder Engagement, Toolkit Development

María Isabela Villamizar Ariza, Department of Earth Sciences, Uppsala University, Villavägen 16, SE- 752 36 Uppsala, Sweden

1. Introduction

Humanity is currently facing what the United Nations Framework Convention on Climate Change (UNFCCC, 2022) describes as the "triple planetary crisis," composed of climate change, biodiversity loss, and pollution. These crises are deeply interconnected and collectively threaten the stability of Earth systems and the well-being of societies worldwide. In recent decades, increasing scientific evidence has demonstrated that human activities are pushing planetary systems beyond safe ecological boundaries, intensifying environmental degradation and increasing the vulnerability of both ecosystems and human communities (Stockholm Resilience Centre, 2025).

In response to these challenges, Nature-based Solutions (NbS) have emerged as an increasingly prominent strategy for addressing societal and environmental challenges while simultaneously supporting biodiversity and human well-being. The concept of NbS is rooted in the recognition that ecosystems provide essential services that can contribute to addressing issues such as climate change, food security, water management, disaster risk reduction, and biodiversity conservation (Cohen-Shacham et al., 2016). Over the last decade, NbS have gained significant attention within academic, political, and institutional agendas, leading to a rapid increase in related research, policies, and implementation initiatives worldwide (Cohen-Shacham et al., 2016; Mirsafa, Castaldo and Lemes De Oliveira, 2025).

Despite this growing attention, the research of NbS remain uneven across regions (Woroniecki et al., 2023). Existing literature highlights important differences between the Global North (GN) and the Global South (GS) regarding socioeconomic conditions, governance structures, institutional capacities, ecological contexts, and access to financial and technical resources (Kowalski, 2020) and these differences influence the context of NbS by shaping both the barriers and the enabling conditions affecting their implementation. While countries in the GN often face challenges related to institutional inertia, regulatory complexity, and integration within already consolidated urban and infrastructural systems, countries in the GS more frequently encounter barriers linked to weaker governance frameworks, limited financial capacity, rapid urbanization, and restricted access to technology and technical expertise (Nizamani and Torgersen, 2025).

At the same time, the Global South remains underrepresented within the NbS literature, despite often being more vulnerable to the impacts of climate change and environmental degradation. Chausson et al., (2020) reported that only 15% of NbS studies originate from the Global South, reflecting broader inequalities in climate research and environmental governance. This imbalance creates important knowledge gaps regarding how NbS are implemented across different contexts and how barriers and enablers may differ between regions. Furthermore, although numerous frameworks, methodologies, and operational tools related to NbS already exist, these resources remain highly fragmented across repositories, disciplines, and institutional platforms, making them difficult to access and operationalize for practitioners.

Recognizing these gaps, this research seeks to contribute to a better understanding of the regional dynamics influencing NbS implementation while also exploring practical ways to support implementers. Specifically, this study aims to answer the following research questions:

1. How do barriers and enablers for implementing Nature-based Solutions differ between the Global North and the Global South?

2. Can a practical toolkit for implementing Nature-based Solutions be designed from an analysis of existing tools and methodologies?

To address these questions, the study adopts a mixed-methods approach combining a literature review, semi-structured interviews with NbS implementers and stakeholders, and the development of a toolkit prototype. By comparing barriers and enablers across regions and compiling practical implementation resources, this research aims to contribute both to the academic understanding of

NbS implementation and to the practical support available for practitioners working across diverse contexts.

2. Background

2.1. Triple planetary crisis

The world is facing what is often referred to as a "triple planetary crisis", encompassing climate change, pollution, and biodiversity loss as the three main challenges humanity is currently confronting (UNFCCC, 2022). Addressing each of these crises is essential to securing a future on Earth. However, each has distinct drivers and consequences, which makes solving them a complex problem (UNFCCC, 2022).

Among these, biodiversity loss is increasingly recognized as a crisis of its own, and one that is deeply interconnected with the others. According to IPBES (2019), five direct drivers are responsible for the decline of natural terrestrial, freshwater, and marine ecosystems: land- and sea-use change; direct exploitation of organisms; climate change; pollution; and invasive alien species.

For some time now, planet Earth has been described as being "sick" or in crisis, prompting efforts to assess its overall health (Sakschewski et al., 2025). To better understand the state of the planet, the concept of planetary boundaries was proposed in 2009 as a way to highlight the processes triggered by human activities and how crossing certain thresholds can change the functioning of Earth systems beyond a safe operating space. These boundaries can be understood as limits which, if exceeded, may push humanity into a crisis for failing to respond adequately to the irreversible environmental changes we as a society are causing to the Earth system (Stockholm Resilience Centre, 2025).

Each year, the Potsdam Institute for Climate Impact Research applies this framework to assess the health of the Earth. The 2025 report (Sakschewski et al., 2025) is particularly concerning, as seven of the nine planetary boundaries - climate change, novel entities, stratospheric ozone depletion, atmospheric aerosol loading, ocean acidification, modification of biogeochemical flows, freshwater change, land system change, and biosphere integrity - have been crossed. This indicates that the Earth is under significant environmental stress, and that some of these changes could be irreversible. This assessment places the humanity in a high-risk zone, meaning that society must act urgently, as the window for returning to a safe operating space is rapidly closing.

In response to these pressing challenges, a range of initiatives has emerged. One of the most prominent is the reduction of CO_2 emissions, which contributes, at least partially, to mitigating all three crises (OECD, 2025). Another important approach to addressing the triple planetary crisis is the use of Nature-based Solutions (NbS; see definitions below) (Burgos Cuevas et al., 2025).

The following sections will further explore the concept and purposes of NbS, as well as their relevance to this thesis.

2.2. Nature-based Solutions (NbS)

Nature-based Solutions (NbS) have been defined by different organizations, and the concept has been subject to ongoing debate since it was first formally introduced by the International Union for the Conservation of Nature (IUCN) in 2016. According to the IUCN, NbS are defined as:

> *"…actions to protect, sustainably manage and restore natural or modified ecosystems, which address societal challenges (e.g. climate change, food and water security or natural*

> *disasters) effectively and adaptively, while simultaneously providing human well-being and biodiversity benefits" (Cohen-Shacham et al., 2016, p. xii).*

Another widely adopted definition is that of the European Commission (2025) which describes NbS as:

> *"Solutions that are inspired and supported by nature, which are cost-effective, simultaneously provide environmental, social and economic benefits and help build resilience. Such solutions bring more, and more diverse, nature and natural features and processes into cities, landscapes and seascapes, through locally adapted, resource-efficient and systemic interventions."*

Both definitions, along with others found in the literature, share common goals: through nature-based approaches, societal challenges can be addressed while simultaneously supporting biodiversity and generating environmental, social, and economic benefits.

In the report *Nature-based Solutions to Address Global Societal Challenges*, the IUCN not only outlines the societal challenges that NbS can address but also categorizes the ecosystem-based approaches within their definition into restoration, issue-specific, infrastructure-related, management, and protection approaches (Cohen-Shacham et al., 2016).

In this sense, NbS should not be interpreted as a one-size-fits-all solution, but rather as a flexible umbrella concept that encompasses a variety of ecosystem-based approaches adapted to different ecosystems, societal challenges, and territorial contexts. As such, NbS can support transformations across urban, rural, industrial, and energy systems while simultaneously enhancing resilience, supporting biodiversity, and contributing to human well-being (Cohen-Shacham et al., 2016; WWF International, 2020; The Swedish Environmental Protection Agency, 2022; Regmi and Gheewala, 2025).

In recent years, both public and private institutions have increasingly considered NbS as a key strategy for addressing climate change and biodiversity loss. The term began to develop through the recognition of the services that ecosystems provide in addressing societal challenges. Although this relationship between nature and people has long been acknowledged within Indigenous and traditional knowledge systems, it was only in the 1970s that the broader scientific community recognized the need for a more systematic approach to understand and value these connections. This growing awareness was later reinforced by the Millennium Ecosystem Assessment (2005), which provided scientific evidence supporting the conservation, restoration, and sustainable management of ecosystems while recognizing the increasing pressures placed on ecosystem services. Building on this perspective, the term *Nature-based Solutions* emerged in the late 2000s, reflecting a shift from viewing people solely as beneficiaries of nature to recognizing their active role in protecting, managing, and restoring ecosystems as a way to address major societal challenges (Cohen-Shacham et al., 2016).

To further operationalize the concept, the IUCN also proposes a set of guiding principles that complement the NbS definition and help clarify its practical application. These eight principles emphasize that NbS: (1) align with nature conservation norms; (2) can be implemented either independently or in combination with other solutions, including technological and engineering approaches; (3) are context-specific, integrating local, traditional, and scientific knowledge; (4) deliver societal benefits in a fair and equitable manner, promoting transparency and participation; (5) maintain biological and cultural diversity, as well as the capacity of ecosystems to adapt over time; (6) are applied at a landscape scale; (7) acknowledge and address trade-offs between short-term economic gains and the long-term provision of ecosystem services; and (8) are embedded within broader policy frameworks and actions aimed at addressing societal challenges (Cohen-Shacham et al., 2016). To complement these principles, the IUCN also developed the *IUCN Global Standard for Nature-based Solutions™ (NbS-GS)*, a framework designed to assess NbS in a structured and systematic manner. The standard aims to ensure that NbS are implemented effectively and sustainably while complying with the previously mentioned principles (IUCN, 2026).

Over the past years, there has been a significant increase in the number of studies and publications focusing on NbS. The concept has evolved from only two published articles in 2012 to more than 2,500 in 2024, with

the majority of contributions coming from researchers in Europe, North America, and Asia (Mirsafa, Castaldo and Lemes De Oliveira, 2025). The trend also highlights an imbalance in the geographical distribution of research, evidencing a lower number of studies focused on the Global South, regions that are often more vulnerable to climate change. For instance, research on NbS in Latin America and Africa remains limited, and existing approaches related to ecosystem services are often not sufficiently developed (Marchetti et al., 2026). This imbalance is further reflected in important evidence gaps identified in the literature, including a lack of peer-reviewed NbS studies from low- and lower-middle-income countries in the Global South, as well as limited research on specific intervention types, ecosystems, and climate impacts (Chausson et al., 2020).

The recognition of these gaps highlights the need to better understand the challenges posed by the division between the Global North and Global South. Therefore, the following section explores this distinction and examines how it may influence the implementation of NbS.

2.3. Global North (GN) - Global South (GS) divide

The distinction between the Global North (GN) and the Global South (GS) is commonly used to describe persistent global inequalities related to economic development, political power, technological advancement, infrastructure, and access to resources (Kowalski, 2020). Although these terms are often used interchangeably with “developed” and “developing” countries, they also reflect broader historical, geopolitical, and structural dynamics shaping relations between regions of the world. The concepts emerged as an evolution of earlier classifications such as the “First World” and “Third World,” shifting the focus from purely economic development toward a more complex understanding of global inequalities and interdependencies (Kowalski, 2020; CIRIS, 2026).

This division is also deeply rooted in the historical exploitation of labour and natural resources in Latin America, Africa, and Asia by European colonial powers, processes that contributed to the creation of enduring socioeconomic and political inequalities. As a result, the Global North is generally associated with higher levels of income, technological development, and institutional stability, while the Global South is more frequently linked to structural challenges such as poverty, inequality, limited technological access, and weaker institutional capacity (Kowalski, 2020). However, this distinction is neither geographically absolute nor homogeneous, as significant variations exist within both regions. These inequalities are reflected across multiple socioeconomic dimensions, including wealth, health, education, infrastructure, and access to technology, posing important challenges for development processes, international relations, and climate-related policies and practices (Blicharska et al., 2017; Kowalski, 2020; CIRIS, 2026).

According to the World Bank classification (Kowalski, 2020), the Global North includes countries such as the United States, Canada, Europe, developed parts of Asia, as well as Australia and New Zealand. In contrast, Global South (GS) encompasses regions such as Africa, Latin America, and developing parts of Asia, including the Middle East. This division is also reflected in geographical and ecological characteristics: countries in the Global North are more commonly associated with temperate, subarctic, and arctic climates, while those in the Global South tend to be characterized by tropical and subtropical ecosystems (Kamal Uddin, 2017).

These structural differences become particularly relevant in the context of climate change and environmental governance, where the North–South divide adds an additional layer of complexity to international cooperation and policymaking. As highlighted by Blicharska et al., (2017), climate research and the global climate agenda have historically been dominated by Northern countries, which continue to shape much of the scientific knowledge and policy frameworks surrounding climate action. This has also influenced the prioritization of climate mitigation strategies within global climate governance, while adaptation, often more urgent for countries in the Global South due to their higher vulnerability to climate impacts, has received comparatively less attention until recently (Persson, 2019).

The Global North, due to its higher levels of industrialization and economic development, has historically been responsible for the majority of greenhouse gas emissions, contributing significantly to the current climate crisis. At the same time, these countries possess greater financial, technological, and institutional capacities to address and adapt to its impacts. In contrast, countries in the Global South, partly due to their geographical and ecological characteristics, are often more vulnerable to the effects of climate change, including extreme weather events, sea-level rise, droughts, floods, and food insecurity (Kamal Uddin, 2017). Despite having contributed less to the causes of the crisis, they face higher impacts and must respond to these challenges with comparatively fewer resources than their counterparts in the Global North (Blicharska et al., 2017; Vasques, Nakaoshi and Fortunato, 2022; Arruda Filho, Torres and Jacobi, 2024).

As mentioned earlier, countries in the Global South are largely located in tropical and subtropical ecosystems and face challenges such as growing inequalities, rapid urbanization, and heightened vulnerability to the impacts of the climate crisis. At the same time, the Global South is a highly heterogeneous group of low- and middle-income countries, with diverse social, economic, and environmental realities that differ significantly from those of the Global North (Castaldo et al., 2025). For example, Latin American cities are frequently characterized by rapid urbanization, socioeconomic inequalities, fragmented green spaces, and weak governance structures, which influence socio-ecological dynamics differently from many Global North contexts (Marchetti et al., 2026).

This diversity highlights the need to address existing gaps in research and to adopt a more nuanced understanding of the Global North–South divide. It is therefore important not only to acknowledge this division, but also to examine both the similarities and differences between the two regions, and to understand how these differences shape the challenges and opportunities associated with addressing the climate crisis.

There is a high proportion of NbS-related publications focused on the Global North. In these regions, particularly in Europe, NbS have become relatively well established and are often integrated into urban planning and development processes, contributing to the creation of green and open spaces, biodiversity corridors, recreational areas, and flood mitigation strategies. This widespread implementation is supported by the availability of technical expertise, sufficient financial resources, and high-quality, accessible data (Medina Rivera and Faggian, 2024; Castaldo et al., 2025).

Although attention to the Global South has been increasing, it remains a relatively recent area of study, and there is still a lack of comprehensive overviews of NbS research in this region. This imbalance is reflected in the findings of Chausson et al., (2020), who reported that only 15% of NbS studies originate from the Global South. This gap is especially significant given that many of the challenges present in the Global South could be effectively addressed through the multiple benefits that NbS can provide. However, in contrast to the Global North, many developing countries face structural barriers to the adoption and implementation of NbS, including limited financial capacity, restricted access to data, and gaps in technical expertise (Medina Rivera and Faggian, 2024; Castaldo et al., 2025).

Despite the potential of NbS to address the triple planetary crisis in both the Global North and the Global South (Burgos Cuevas et al., 2025), knowledge and implementation of the concept remain uneven in both practical and economic terms. For example, evidence suggests a divergence in adoption barriers between the two regions. In the Global North, challenges such as high costs, regulatory complexity, and data limitations often create institutional inertia that hinder the implementation of NbS (Nizamani and Torgersen, 2025). In contrast, barriers in the Global South are more frequently associated with limited awareness, misaligned policy priorities, and weaker governance frameworks, which can result in NbS receiving lower political and institutional priority (Nizamani and Torgersen, 2025). Recognizing these contextual differences is essential for developing strategies that respond to the specific realities of each region. This is consistent with one of the IUCN guiding principles, which states that “NbS are determined by site-specific natural and cultural

contexts that include traditional, local and scientific knowledge" (Cohen-Shacham et al., 2016, p. xxi).

These differences highlight the importance of identifying the key barriers and enablers of NbS in diverse contexts in order to develop effective strategies for scaling these initiatives (Ershad Sarabi et al., 2019). Although several studies have explored these factors, much of the existing literature is grounded in a Global North perspective, with a notable gap in research focused on the Global South (McPhearson et al., 2025) and there are, to best of my knowledge, no studies that would explicitly compare the barriers and enablers affecting NbS implementation across the Global North and the Global South in a systematic way.

3. Methods

This study adopts a mixed-methods approach combining a literature review and semi-structured interviews with NbS implementers and other relevant stakeholders. Through these methods, the research aims to identify key barriers, enablers, and regional dynamics influencing the adoption and implementation of NbS. As part of the literature review, the study also includes a comparative analysis of barriers and enablers between the Global North and the Global South in order to address the first research question.

In addition, to respond to the second research question, this study explores how a toolkit for NbS implementers can be developed using existing tools and methodologies. The toolkit design combines insights from the literature review, the semi-structured interviews, and a review of grey literature containing practical resources, frameworks, and methodologies relevant to NbS implementation. The following sections describe the methods used for the literature review, the semi-structured interviews, and the design of the toolkit. Fig. *1* provides a graphical overview of how both research questions are addressed through this methodological approach.

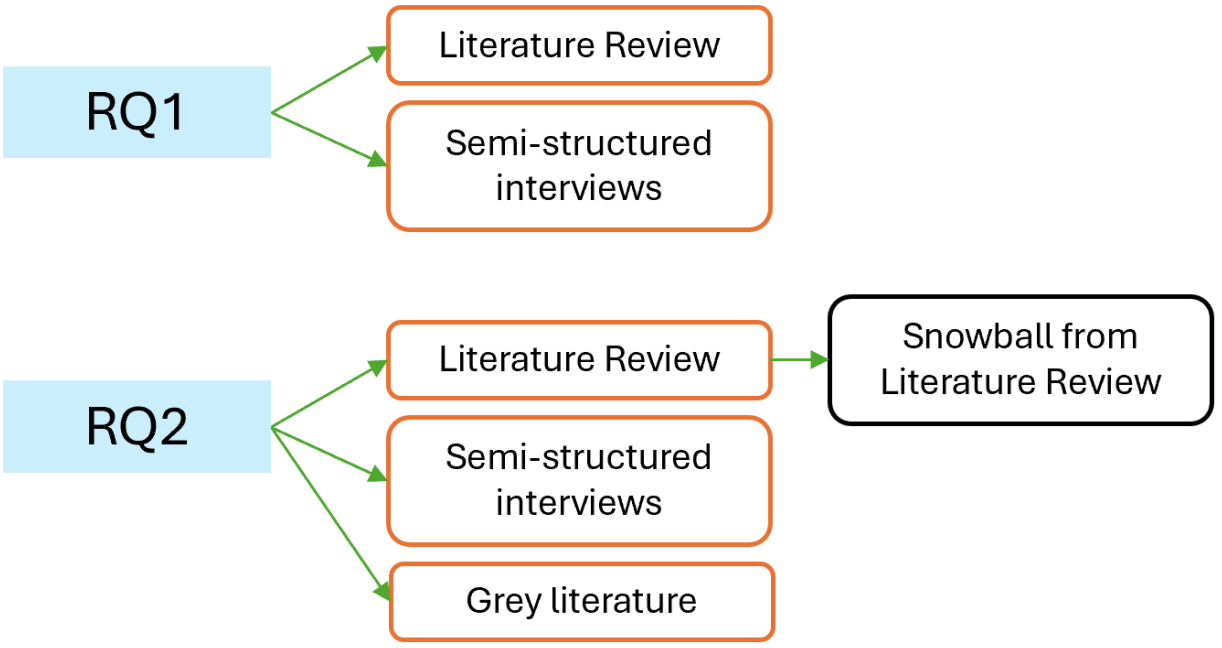


Fig. 1. Summary of the Methods

3.1. Literature review

To conduct the literature review, an initial search was performed in the Web of Science database using the following query: ("nature-based solution*" OR "nature based solution*" OR NbS) AND ("barrier*" OR "enabl*" OR "driver*" OR "challenge*" OR "facilitat*") AND ("global north*" OR "global south*" OR "developed countr*" OR "developing countr*" OR "low-income countr*" OR "high-income countr*"). Web of Science was selected due to its wide coverage of peer-reviewed, high-quality academic literature, ensuring a robust and reliable sample of studies. This search resulted in a total of 179 articles.

A first screening was conducted based on titles and abstracts, leading to the exclusion of 100 articles according to the following criteria: (E1) lack of focus on NbS; (E2) absence of barriers or enablers analysis; (E3) highly technical focus on NbS[1], e.g., "Hybrid smart bio-reactor and modular wetland system for blackwater treatment in tidal coastal settlements"; (E4) narrow or single-problem focus; and (E5) off-topic content. These criteria were defined to ensure that the selected studies were directly relevant to the research objectives and provided sufficient analytical depth.

Subsequently, a full-text analysis was carried out on the remaining articles. At this stage, an additional 40 articles were excluded: two due to lack of access to full text, and 38 because they did not present clear barriers or enablers and/or maintained an overly technical focus. Through a snowballing approach, 6 additional relevant articles were identified and included, allowing for the inclusion of key studies that may not have appeared in the initial search. In total, 45 articles were selected for analysis. This process can be also seen in Fig. *2*.

The selected articles were analysed using the qualitative data analysis software MAXQDA. A coding process was applied to identify and organize relevant information. While the primary aim was to extract barriers and enablers to implement NbS, additional categories were included to capture broader insights relevant to the research, particularly regarding conceptual understanding, regional comparisons and the design of the toolkit. The coding categories were Innovation toolkit; NbS purposes; NbS concept; Global North–South comparison; Barriers (Global North, Global South, General); and Enablers (Global North, Global South, General), in Table *1* there is an explanation of each category. This approach enabled a structured and comparative analysis of the literature across different contexts.

Table 1. Description of coding categories.

Coding Category	Explanation
Enabler – General Enabler – GN Enabler – GS Barrier – General Barrier – GN Barrier – GS	For the enablers and barriers, three independent categories were established: enablers/barriers identified in the Global North (GN), enablers/barriers identified in the Global South (GS), and general enablers/barriers, i.e. enablers and barriers that did not explicitly specify whether they were associated with the GN or the GS.
Innovation toolkit	This category was created exclusively to address the second research question. The aim was also to identify potential tools, methodologies, and frameworks that could support the implementation and scaling of NbS.
NbS purposes NbS concept	These categories were created to collect information related to how NbS are conceptualized in the literature, as well as the different

[1]This exclusion category includes NbS studies with a strong engineering or technical focus, particularly those that primarily address technical barriers and/or enablers related to their direct implementation.

Coding Category	Explanation
	purposes and objectives associated with their implementation. The aim was to identify recurring understandings, definitions, and intended outcomes of NbS that could later support the analysis and interpretation of the results.
Comparison GS GN	This category was created to identify comparisons made in the literature review between the Global South and the Global North, both to support the background section and to inform the subsequent analysis of the results.

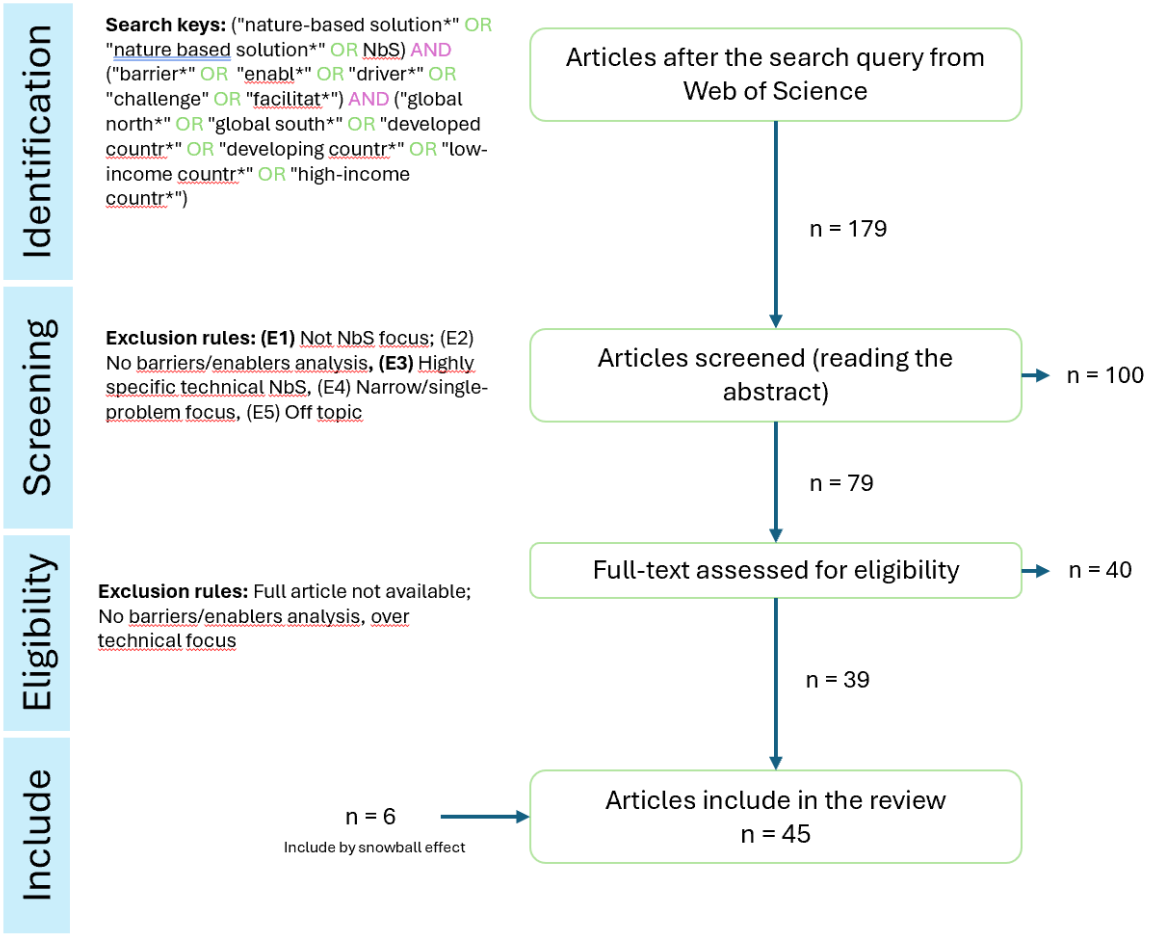


Fig. 2. Description of the selection of the articles for the literature review

Following the coding of barriers and enablers, a deeper analysis was conducted to identify patterns and correlations across the selected references. This involved reviewing each coded segment and examining how different types of barriers and enablers were described and categorized within the literature. Across the reviewed articles, barriers and enablers were initially classified under a wide range of categories, including political, social, financial, institutional and policy-related, ecological, geographical, and environmental factors, as well as issues related to technology access and affordability, awareness, knowledge and skills, governance structures, regulations, actors and networks, power relations, stakeholder engagement, and inclusivity, among others.

Several of the reviewed articles already proposed their own categorizations for barriers and enablers, which were initially considered as a starting point for the analysis. At this stage, approximately 10 to 12 preliminary categories emerged. However, after a deeper comparative analysis, similarities and overlaps

between categories were identified, and broader thematic groupings were progressively developed in order to reduce fragmentation and create a more manageable analytical structure. Consequently, six main categories were established through an iterative and interpretative process that considered: (1) the categorizations proposed within the reviewed literature, (2) recurring patterns across the coded segments, and (3) the researcher's qualitative interpretation of thematic similarities between barriers and enablers.

Once the six main categories were established, a similar process was followed for the development of the subcategories. Existing classifications identified in the literature were again used as references, while overlapping or closely related subcategories were progressively merged to reduce redundancy and facilitate the comparison between the Global North and the Global South. A similar process was later applied to the enablers. However, at this stage, the six main categories had already been established through the barrier analysis, and the enablers helped reinforce and refine these categories. For example, the term "Engagement" was incorporated into the category "Social, Engagement & Perception" due to the strong prominence of collaboration and stakeholder engagement throughout the enabler analysis.

This categorization process was inherently complex and involved a degree of subjective interpretation, particularly because several coded segments simultaneously reflected more than one category or subcategory. In such cases, analytical judgement was required to determine the most representative classification or, when necessary, to divide the coded segment across multiple categories.

For the comparison between the Global North and the Global South, the number of coded segments identified in MAXQDA was considered in order to evaluate how frequently each category and subcategory appeared across regions, and to analyse their relative predominance within the GN, GS, and General categories.

3.2. Interviews

The study planned to conduct semi-structured interviews with practitioners involved in the implementation of NbS, including two from the Global South and three from the Global North. This selection aimed to capture diverse perspectives across different regional contexts.

The primary purpose of the interviews was to validate the findings from the literature review and to gather additional insights to inform the design of the proposed toolkit. As the main focus of the thesis lies in the literature review, a relatively small sample size was selected, allowing for in-depth exploration of each case while remaining aligned with the scope of the study.

Semi-structured interviews were chosen as the preferred method because they provide a balance between consistency and flexibility. While an interview guide ensures that key topics, such as barriers, enablers, and perceptions of NbS success, are systematically addressed, the open-ended nature of the format allows participants to share context-specific experiences and insights that may not emerge through more structured approaches.

To identify suitable participants, organizations and networks working with NbS in practice were contacted via email to compile a database of potential candidates; additionally the report "Applying the IUCN Global Standard for Nature-based Solutions" from Cohen-Shacham, Cabecinha and Andrade, (2025) was used to identify other possible stakeholders to interview. From this pool, participants were selected based on their availability, experience and involvement in NbS implementation.

The interview data was analysed using a thematic analysis approach using the same coding categories used for the literature review, aligned with the primary purpose of the interviews, which is to validate the findings from the literature review and to generate additional insights for the design of the proposed toolkit. Interview recordings were transcribed using Microsoft Word transcribing feature, and the transcripts will be reviewed to identify recurring patterns and themes across participants,

particularly in relation to barriers, enablers, and contextual factors influencing NbS implementation.

A total of five interviews were conducted via Zoom, with durations ranging from 23 to 60 minutes. The initial interviews followed the guide provided in Appendix 1. However, after the first interview, the guide was refined to enable a more natural and simplified conversation that could elicit the same information without introducing additional context that might influence participants' responses. The revised version of the guide, used for the remaining interviews, is included in Appendix 1 under the title *Interview Guide 2.0*.

In particular, Question No. 7, which asked about a suggestion of tools, frameworks or methodologies for implementing NbS, proved challenging during the first interview, as the interviewee requested clarification. In providing this clarification, some unintended cues were introduced, which may have influenced the response. For the subsequent interviews, greater care was taken when explaining this question in order to minimize bias and avoid leading the participants. Despite these adjustments, Question No. 7 remained challenging, as it was interpreted differently across interviewees and generated a wide range of responses depending on the organization.

3.3. Design of the practical toolkit

The design of the toolkit was informed by multiple sources of evidence and methodological approaches. First, within the literature review, a specific category labelled "innovation toolkit" was established to identify relevant tools, methodologies, and frameworks. These elements were selected based on their potential to address the barriers and leverage the enablers identified in the analysis. In addition, a snowballing approach was applied to the reviewed articles to trace further sources and platforms that provide useful tools for NbS implementation.

The literature review led to the identification of several tools and concepts, which were later complemented with grey literature to provide more accessible and detailed explanations for users. Although these additional sources were used to enhance understanding, the selection and inclusion of the tools were primarily based on the results of the literature review.

These findings were complemented by the results of the semi-structured interviews, in which practitioners were explicitly asked to suggest tools that have supported their NbS implementation processes, as well as tools or resources they perceived as missing.

The toolkit is therefore developed through an integrative process, combining theoretical insights and practical knowledge. Its aim is to provide stakeholders with practical tools to help overcome barriers and/or achieve identified enablers. The resulting toolkit is an initial prototype and a work in progress.

3.4. Methodological limitations

One of the main limitations of this research is related to the literature search process and the resulting sample distribution. Despite the intention to compare the Global North (GN) and the Global South (GS), the search equation produced a considerably larger number of GS-focused articles than GN-focused studies. This may partially be explained by the fact that researchers from the GN do not always explicitly frame their studies using terms such as "Global North," "developed countries," or similar classifications, but instead refer to specific countries or regions such as Europe or the United States. Consequently, some potentially relevant GN studies may not have been captured by the search strategy.

Another limitation relates to the selection criteria applied during the literature review. In order to identify broader barriers and enablers applicable across different types of NbS, highly technical studies were excluded from the analysis. As a result, the research prioritized articles discussing more general implementation dynamics rather than highly specialized technical aspects of specific NbS interventions. While this approach supported the comparative and cross-cutting nature of the study, it may also have excluded important insights related to particular NbS typologies. A more extensive or specialized study could therefore include a broader range of highly technical NbS cases and compare barriers and enablers across different types of NbS interventions.

In this same sense, the final sample showed a strong predominance of articles focused on urban NbS and urban management challenges. This may indicate that some of the identified barriers and enablers are more strongly associated with urban contexts than with other NbS applications. However, the five semi-structured interviews conducted for this research focused on different NbS contexts, including conservation, restoration, agriculture, and water governance, none of them within urban settings. Interestingly, these interviews still validated many of the barriers and enablers identified throughout the literature review, suggesting that several of the findings may extend beyond urban NbS contexts.

Another important methodological limitation relates to the inherently subjective nature of the qualitative coding and categorization process. Both the coding process itself and the development of the categories and subcategories required interpretative decisions by the researcher. In many cases, barriers and enablers were transversal and overlapped across multiple categories simultaneously, making classification particularly complex. Consequently, decisions regarding where to locate certain barriers or enablers ultimately relied on the analytical judgement of the researcher. Although efforts were made to reduce fragmentation and maintain coherence throughout the analysis, a different researcher may have categorized some elements differently.

Additionally, the comparative analysis between the GN and the GS relied partly on the quantification of coded segments identified through MAXQDA. However, this comparison was limited by the unequal proportion of articles representing each region, which may have influenced the predominance of certain categories or subcategories within the results.

Finally, due to the time constraints and scope of this thesis, the proposed toolkit should be understood as an initial prototype rather than a comprehensive repository of all existing NbS tools, methodologies, and frameworks. The toolkit explores only a small portion of the extensive number of resources currently available across different repositories, disciplines, and languages. Therefore, the further development, expansion, and evaluation of practical NbS toolkits represents an important area for future research.

4. Results

4.1. Enabler and Barriers and comparison GN/GS

A total of 45 articles were included in the analysis. The geographical distribution of the case studies shows a strong representation of countries from the Global South. The most frequently represented countries include Ghana (4 articles), China (3), Thailand (2), and Viet Nam (2), while other countries such as India, Ethiopia, Malawi, Colombia, Brazil, Mexico, and South Africa are each represented by a smaller number of studies. In addition, several articles adopt a broader regional focus, including studies categorized as “Global South” (5), “Latin America” (1), “Europe” (2), “Caribbean Small Island Developing States” (1), and “Asia-Pacific Islands” (1).

In terms of regional focus, 26 articles primarily address the Global South, compared to 4 focused on

the Global North. Additionally, 8 articles adopt a general perspective without explicitly differentiating between regions, while 7 studies provide a comparative analysis or address both regions simultaneously.

Regarding thematic focus, most of the studies (25 articles) concentrate on urban settings, often addressing multiple societal challenges, with a predominant emphasis on climate change mitigation and adaptation. Other thematic areas identified in the literature include water security (6 articles), food security (6 articles), disaster risk reduction (4 articles), environmental degradation and biodiversity loss (3 articles), and general NbS applications without a specific thematic focus (2 articles). The comparison of this data can be seen in Fig. 3

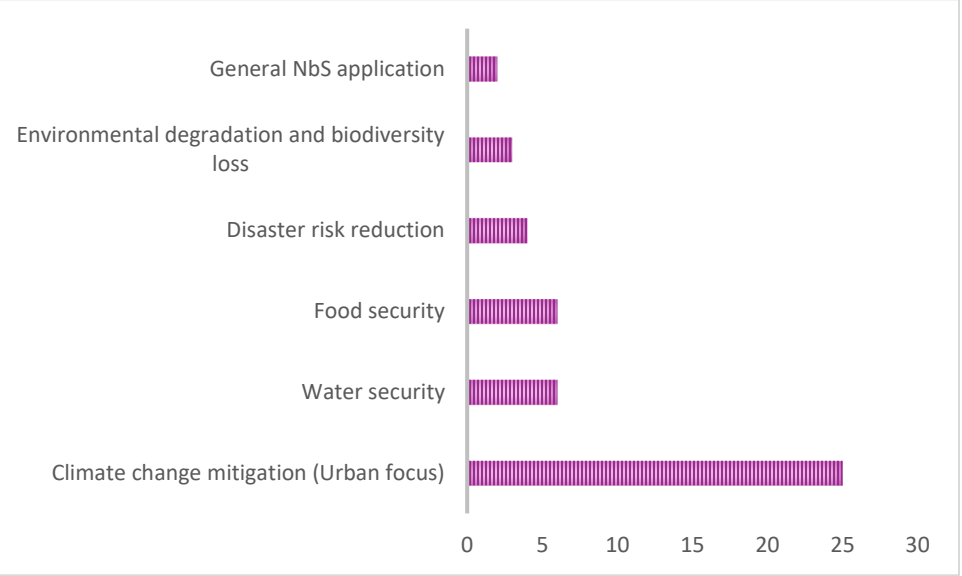


Fig. 3. Literature review article classification based on IUCN global societal challenges (Cohen-Shacham et al., 2016)

The coding analysis results initially in more than 1000 coded segments among the 10 coding categories and reveals important patterns in the distribution of themes across the selected literature. As shown in Fig. 4, a significantly higher number of coded segments were associated with barriers in the Global South (283) compared to the Global North (49) and general contexts (96). A similar trend is observed for enablers, with 280 coded segments for the Global South, compared to 87 for the Global North and 180 in general contexts.

This distribution is largely influenced by the composition of the sample, as the majority of the selected articles focus on the Global South. This, in turn, reflects a broader trend in the literature, where it is comparatively more difficult to find studies that explicitly analyse NbS barriers and enablers within a Global North context.

Despite the diversity of terminology and approaches used across the reviewed articles, the analysis revealed several recurring themes among the identified barriers and enablers. Given this diversity and overlap, a process of categorization was undertaken to group similar barriers and enablers into broader and more manageable themes. As a result, six main categories of barriers and enablers were proposed:

1. Governance & Institutional
2. Financial
3. Planning & Spatial
4. Technical & Capacity
5. Social, Engagement & Perception
6. Knowledge & Information

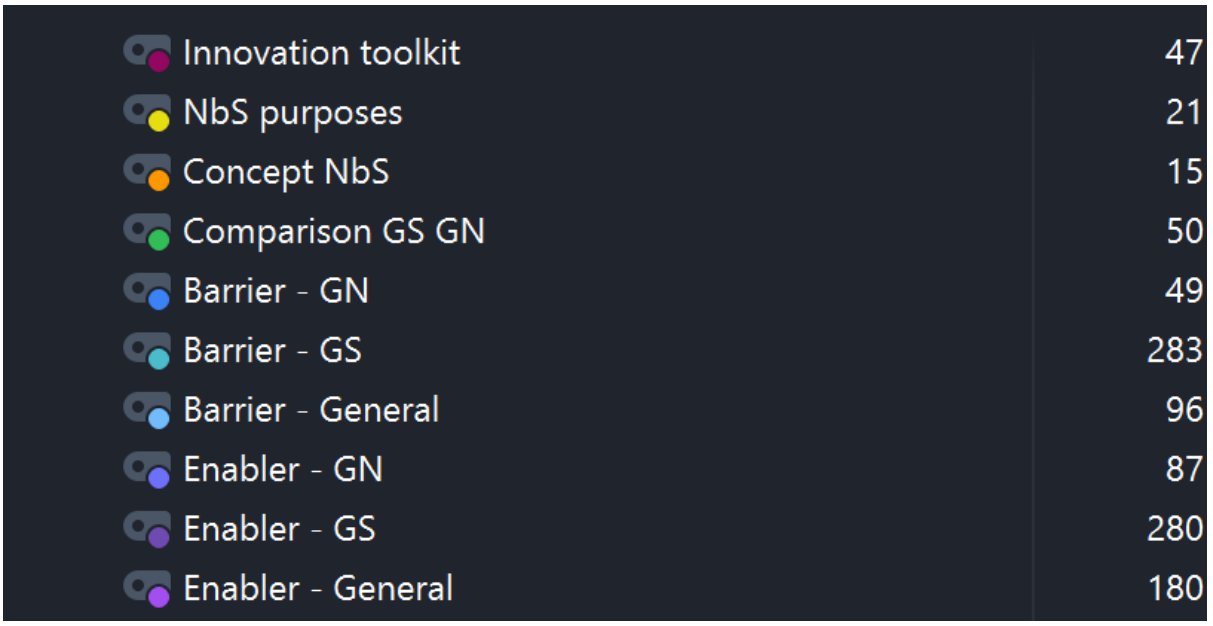


Fig. 4. Coding categories in MAXQDA

This classification aims to provide a more structured framework for analysing barriers and enablers across different contexts.

After classifying the barriers and enablers into the six main categories, sub-categories were developed to provide a more nuanced understanding of each category and to facilitate comparison between the results from the Global South and the Global North. In addition, sub-categories were identified within each of these groups to allow for a more nuanced analysis.

Given the smaller number of Global North articles within the sample of 45 studies, the comparison between the Global North and the Global South was conducted by considering both the proportion of articles representing each region and the distribution of coded segments identified through MAXQDA, which were subsequently analysed using an Excel spreadsheet. Some coded segments were divided, as several initially contained more than one barrier and/or enabler. After this division, a very similar number of coded segments was identified for both enablers and barriers, which is particularly interesting considering that the search query was not designed to obtain an equal number of articles focused on these two categories. This result facilitated a more balanced comparison between barriers and enablers.

The quantification process also revealed that most coded segments were concentrated in the Global South (GS), largely because most of the selected articles focused on this region. However, the analysis also revealed an initial distinction between barriers and enablers. In the case of barriers, the difference between the Global North (GN) and the Global South (GS) was more pronounced, whereas for enablers the distribution of coded segments was more balanced across the three categories (GN, GS, and General), with smaller differences between them, as shown in Table 2.

It is important to note that some of the articles identified in the literature review were classified as "General," meaning that they discussed barriers and enablers without specifying a particular regional context. These articles were therefore analysed separately, as it was not possible to determine whether the identified barriers or enablers were specific to a particular region. The following table presents a quantitative overview of the literature review findings and supports the comparison between regions.

Table 2 presents both the number of coded segments identified for each category and their corresponding percentages to facilitate a clearer comparative analysis. Fig. 5 also offers a graphic representation of these results.

Table 2. Number of coded segments resulting from the literature review

	Barriers				Enablers			
	General	**GS**	**GN**	**Total**	**General**	**GS**	**GN**	**Total**
Governance & Institutional	31	80	12	**123**	22	46	13	**81**
Financial	28	59	10	**97**	33	18	13	**64**
Planning & Spatial	8	48	5	**61**	7	24	10	**41**
Technical & Capacity	14	43	4	**61**	33	45	17	**95**
Knowledge & Information	29	41	7	**77**	29	50	10	**89**
Social, Engagement & Perception	12	53	10	**75**	35	58	19	**112**
Total	122	324	48	**494**	159	241	82	**482**
	Barriers				**Enablers**			
	General	**GS**	**GN**	**Total**	**General**	**GS**	**GN**	**Total**
Governance & Institutional	6%	16%	2%	**25%**	5%	10%	3%	**17%**
Financial	6%	12%	2%	**20%**	7%	4%	3%	**13%**
Planning & Spatial	2%	10%	1%	**12%**	1%	5%	2%	**9%**
Technical & Capacity	3%	9%	1%	**12%**	7%	9%	4%	**20%**
Knowledge & Information	6%	8%	1%	**16%**	6%	10%	2%	**18%**
Social, Engagement & Perception	2%	11%	2%	**15%**	7%	12%	4%	**23%**
Total	**25%**	**66%**	**10%**	**100%**	**33%**	**50%**	**17%**	**100%**

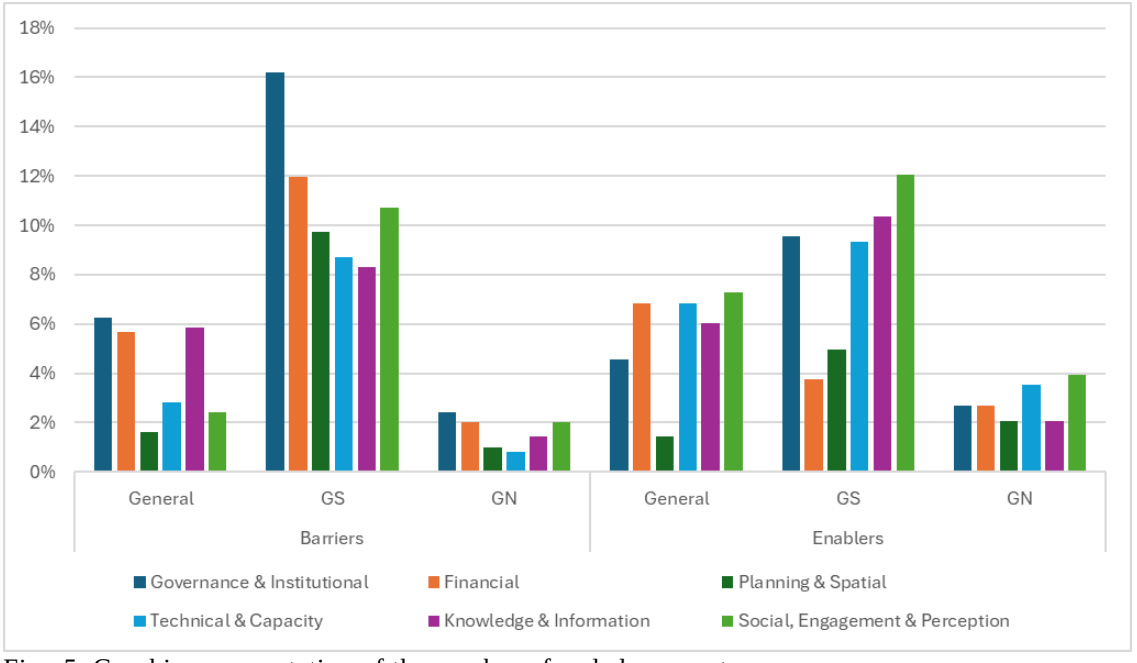


Fig. 5. Graphic representation of the number of coded segments

This analysis makes the differences in the distribution of barriers across the three contexts (Global South, Global North, and general) more evident, particularly when compared to the distribution of enablers. This distinction will be explored further in the respective barriers and enablers sections and later discussed in greater depth in the discussion section.

4.1.1. Barriers

Most of the identified barriers were concentrated within the **Governance and Institutional** category, followed by **Financial**, and **Knowledge & Information**. Table 3 presents the sub-categories within each main category, the articles in which each sub-category is mentioned, and a general comparison indicating whether each barrier appears to be more predominant in one of the two regions or in the general articles. It is important to note that this indication of predominance does not imply that a sub-category is only exclusive to a specific region, but rather that it was identified more frequently or more prominently within the literature associated with that region.

Some sub-categories may be transversal across multiple main categories; however, for analytical purposes they were assigned to the category in which they appear to have the greatest influence or prominence. Likewise, some sub-categories may overlap, but they are analysed from the perspective of the category in which they are situated. Fig. 6 visualises the coded segment distribution for the barriers.

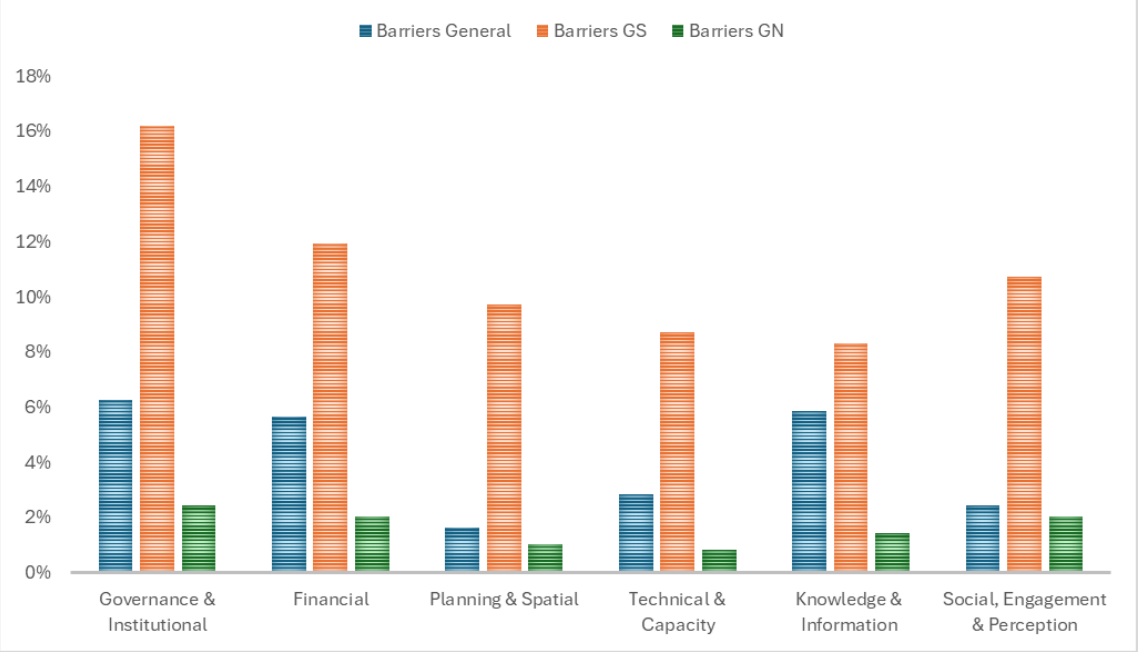


Fig. 6. Coded segment distribution for the barriers section

Table 3. Categories and sub-categories of the barriers

Barriers	Articles mentioning this*	Predominance
Governance & Institutional		
1. Lack of clear policies and regulatory frameworks	11, 21, 23, 26, 28, 36, 37, 41, 64, 82, 84	**GS, GN and General**
2. Weak enforcement of existing policies	14, 19, 28, 37, 39, 42	**GS**
3. Fragmented and siloed institutional structures	19, 21, 23, 26, 28, 30, 42, 64, 81, 82, 84	**GS, GN and General**
4. Poor coordination across stakeholders and governance levels	2, 5, 13, 14, 19, 20, 21, 24, 26, 28, 37, 38, 39, 42, 44, 80, 84	**GS**
5. Misalignment between national and local policies and decisions	5, 13, 19, 28, 82	**GS**
6. Restrictive regulations and conventional planning approaches	11, 13, 14, 21, 26, 30, 36, 39, 63, 64, 84	**GS, GN and General**
7. Lack of long-term political commitment	5, 6, 19, 23, 26, 28, 30, 36, 38, 39, 43, 44, 64, 81, 84	**GS, GN and General**

Barriers	Articles mentioning this*	Predominance
Financial		
1. Lack of funding or investment	5, 14, 16, 19, 20, 21, 26, 28, 36, 37, 38, 41, 42, 47, 63, 64, 77, 78	**GS> GN and General**
2. High upfront costs of NbS	11, 14, 30, 34, 36, 37, 39, 43, 44, 47, 59, 84	**GS, GN and General**
3. Limited access to and mechanisms for mobilizing investment	21, 26, 37, 42, 44, 81, 84	**GS, GN and General**
4. Lack of financial incentives (subsidies, tax benefits)	11, 19, 21, 44, 81	**GS, GN and General**
5. Difficulty monetizing NbS benefits	11, 19, 21, 22, 42, 78, 81	**GS, GN and General**
6. Dependence on external funding (donors, public)	5, 26, 39, 77, 82	**GS**
7. Mismatch between short-term investment horizons and long-term economic returns	36, 37, 47, 82, 84	**General**
8. Prioritizing the allocation of financial resources for conventional projects or other challenges	5, 84	**GS and General**
9. Lack of viable business models for NbS	21, 37, 42, 63, 64, 81	**GS**
Planning & Spatial		
1. Limited land availability and plot size constraints	5, 13, 14, 24, 26, 41, 43, 47, 64, 81, 84	**GS, GN, General**
2. Land tenure insecurity	24, 26, 39	**GS**
3. Conflicts with existing land uses and management (housing, infrastructure)	5, 13, 19, 26, 39, 64, 81, 84	**GS, GN and General**
4. Difficulty integrating NbS into legal and planning systems	5, 13, 14, 21, 28, 64, 84	**GS**
5. Environmental and biophysical constraints on NbS implementation	13, 20, 21, 24, 26, 41, 43, 64, 77	**GS**
6. High population density and land-use pressure	13, 20, 39, 42, 84	**GS**
Technical & Capacity		
1. Limited technical skills, training and capacity building for NbS	4, 5, 11, 14, 19, 21, 24, 34, 36, 37, 38, 39, 44, 63, 64, 77, 81, 84	**GS**
2. Lack of standardized guidelines and metrics	1, 19, 20, 37, 41, 64, 84	**GS and General**
3. Limited monitoring, maintenance and evaluation systems	5, 11, 13, 19, 24, 36, 37, 38, 40, 41, 77, 81, 84	**GS, GN and General**
4. Limited access to appropriate technologies and infrastructure for NbS	23, 28, 37, 41, 43, 64	**GS**
Social, Engagement & Perception		
1. Preference for grey infrastructure, conventional solutions, and other alternatives	10, 23, 24, 44, 77, 81	**GS**
2. Low perceived relevance and understanding of NbS among stakeholders	4, 5, 11, 13, 14, 19, 21, 23, 24, 26, 36, 38, 42, 43, 81, 84	**GS, GN and General**
3. Lack of participation or engagement/resistance from community or stakeholders	2, 5, 19, 24, 26, 36, 38, 41, 42, 43, 44, 63, 64, 81, 84	**GS and General**
4. Conflicts between stakeholders / inequitable distribution of benefits	5, 19, 21, 24, 26, 36, 38, 41, 42, 43, 44, 63, 64, 81, 84	**GS>GN and General**
5. Path dependency and resistance to change in planning and decision-making practices	7, 11, 82	**GN and General**
6. Competing socio-economic priorities over environmental goals	24, 42	**GS**
Knowledge & Information		

Barriers	Articles mentioning this*	Predominance
1. Lack of data on NbS implementation and performance	2, 11, 13, 14, 19, 20, 21, 23, 26, 28, 34, 36, 42, 64, 82	**GS and General**
2. Limited evidence on NbS effectiveness and long-term benefits	1, 7, 11, 19, 23, 24, 26, 80, 81	**GS, GN and General**
3. Lack and underutilization of local and context-specific knowledge	11, 19, 37, 42, 84	**GS and General**
4. Limited knowledge sharing, transfer, and management across sectors and institutions	19, 24, 38, 84	**GS, GN and General**
5. Research-practice gap	1, 26, 44, 81, 82	**GS, GN and General**
6. Limited awareness and understanding of NbS	9, 11, 14, 16, 19, 21, 23, 26, 28, 34, 36, 37, 38, 41, 42, 47, 82	**GS and General**

*The article corresponding to each number can be seen in Appendix 2

4.1.1.1. Governance & Institutional

Within this category, it was evident that both the Global North and the Global South face a **lack of clear policies and regulatory frameworks** that support NbS. For example, in urban contexts, many regulations continue to be shaped by a grey infrastructure perspective, while environmental regulations often do not explicitly incorporate initiatives framed under the NbS concept (Ershad Sarabi et al., 2019; Medina Rivera and Faggian, 2024; Bauer and Titz, 2025; Debie, 2025). This limits the integration of NbS practices more broadly and can hinder the allocation of resources and institutional support (Debie, 2025).

Related to this gap, **restrictive regulations and conventional planning approaches** can further obstruct the adoption of NbS. Bureaucratic barriers such as zoning laws, land-use constraints (Enu et al., 2025; Zarei and Shahab, 2025), and rigid regulatory procedures can make implementation difficult. In the Global South, for instance, regulations related to organic certifications can be highly restrictive and often favour more conventional methods, creating challenges for entering markets and generating economic benefits (Zhu et al., 2024).

Also, highly present in both regions**, lack of long-term political commitment** and **fragmented and siloed institutional structures** emerge as important barriers preventing the implementation and scaling of NbS. In many cases, different departments and agencies operate according to their own mandates, procedures, and sector-specific perspectives, with limited coordination across sectors. This fragmentation can hinder collaboration and reduce opportunities to integrate innovative approaches into NbS planning and management processes (Ershad Sarabi et al., 2019; Sarabi et al., 2020; Bauer and Titz, 2025). In addition, the short-term cycles of national and local administrations are often misaligned with the long-term nature of NbS. This can contribute to weak long-term political commitment, as frequent shifts occur not only in the actors responsible for these processes but also in governmental priorities over time (Sarabi et al., 2020; Castaldo et al., 2025; McPhearson et al., 2025; Zarei and Shahab, 2025). This instability, combined with a limited sense of urgency among policymakers to implement such initiatives, can further constrain the development and continuity of NbS (Sarabi et al., 2020; Medina Rivera and Faggian, 2024).

Inside this category there were barriers that were more predominant for and present in the Global South. One of the challenges this region faces is the **weak enforcement of existing policies** due to political interference, the lack of understanding of NbS benefits, the previously mentioned siloed mentality, weak regulatory structures, lack of transparency in governmental decision-making, and other priorities focused on economic and political concerns (Adu Boateng et al., 2023; Enu et al., 2025; McPhearson et al., 2025; Regmi and Gheewala, 2025; Zarei and Shahab, 2025). These aspects

also cause, and are caused at the same time by, the **poor coordination across stakeholders and governance levels** and the **misalignment between national and local policies and decisions.** Conflicting interests, differing priorities, and challenges in building consensus among multiple actors can delay implementation and hinder effective collaboration, particularly in larger-scale initiatives. This barrier is further reinforced by misalignment between local, national, and global governance levels, as well as fragmented coordination among institutions and stakeholders, which can weaken policy implementation and constrain the adoption of NbS (Medina Rivera and Faggian, 2024; Castaldo et al., 2025; McPhearson et al., 2025; Mirsafa, Castaldo and Lemes De Oliveira, 2025).

4.1.1.2. Financial

Within the **Financial** category, several barriers appear to be present regardless of regional context. One of the most prominent is the **high upfront cost of NbS**, as both in the Global North and the Global South significant initial investments are often required for implementation, accompanied by substantial maintenance needs, particularly in the early stages (Thuy et al., 2019; Nizamani and Torgersen, 2025; Regmi and Gheewala, 2025; Zyoud and Zyoud, 2025). This is closely linked to a **mismatch between short-term investment horizons and long-term economic returns**, which emerges as a major challenge for NbS, as stakeholders often prioritize investments with quicker and more predictable returns (Ershad Sarabi et al., 2019; Regmi and Gheewala, 2025; Zarei and Shahab, 2025; Zyoud and Zyoud, 2025).

Related to this, **limited access to investment and mechanisms for mobilizing finance**, together with a **lack of financial incentives**, such as subsidies or tax benefits for NbS implementers, further constrain implementation in both regions (Sarabi et al., 2020; Marchetti et al., 2026). These challenges are often exacerbated by the **difficulty of monetizing NbS benefits**, as many of their social, ecological, and long-term economic values are not easily captured through conventional financial metrics and the cost-effectiveness and co-benefits of NbS are not recognized fully (Sarabi et al., 2020; McPhearson et al., 2025; Marchetti et al., 2026). Closely linked to this is the **uncertainty about economic returns**, which can limit the perceived marketability of NbS beyond their ecological or social value. Gaps in cost-effectiveness data, uncertainty regarding long-term performance under climate change, and difficulties in pricing benefits through mechanisms such as carbon markets or Payments for Ecosystem Services can complicate financial assessments and increase perceptions of risk (Regmi and Gheewala, 2025). In many cases, returns on investment in NbS are perceived as longer-term and higher risk than those of conventional alternatives, which can discourage private investment, particularly where viable business models capable of generating returns have not yet been established (Frantzeskaki et al., 2019; Bauer and Titz, 2025).

Other barriers closely connected to those mentioned above, but identified mostly in the Global South literature, also emerged. First, **the prioritization of financial resources toward conventional projects or other pressing challenges** was frequently highlighted. In many contexts within the Global South, where multiple urgent social and development needs coexist, NbS may be overlooked or assigned lower priority compared to more immediate concerns(Huynh Thi Ngoc et al., 2024; Rauf et al., 2024; Zarei and Shahab, 2025). Second, **limited funding and investment for NbS** appeared as a particularly prominent barrier in the Global South. While this is a common challenge more broadly, in these contexts institutional resources are often directed toward addressing pressing social needs, while less conventional or innovative approaches such as NbS may not be seen as priority investments. This is further compounded by perceptions of uncertainty around NbS outcomes, which can make private investors hesitant and lead them to favour more traditional alternatives (Sarabi et al., 2020; Huynh Thi Ngoc et al., 2024; Bauer and Titz, 2025; Marchetti et al., 2026). A third barrier relates to **dependence on external funding**, such as support from donors or the public sector. Even when funding is secured, the continuity of these financial flows is often uncertain, and available resources may be insufficient for long-term project development, generating additional pressure on implementation and maintenance(Ershad Sarabi et al., 2019; Bauer and Titz, 2025; Ojeda-Revah and Muñoz-Meléndez, 2026). Finally, many of these barriers appear interconnected with the **lack of viable business models for NbS**, particularly models capable of articulating a clear value proposition

and sustaining implementation over time. This challenge is especially relevant given the multi-stakeholder and multidisciplinary nature of NbS, which often requires business models that can accommodate complexity while supporting long-term value creation (Du Plessis et al., 2022; Lee and Song, 2024).

4.1.1.3. Planning & Spatial

This category considers barriers related to both the planning dimension of NbS and the spatial conditions required for their implementation. Overall, many of the sub-categories within this theme appear more prominent in the Global South, largely due to contextual factors such as high population density, access constraints, informal settlements, land tenure conditions, and broader urban development dynamics (Bauer and Titz, 2025; Enu et al., 2025). In particular, the urban planning context in many Global South countries is shaped by colonial legacies and rapid urbanization processes, which have contributed to infrastructure deficits and settlement patterns that differ significantly from those of the Global North (Abera et al., 2025). These conditions create broader challenges for infrastructure planning and are particularly relevant for NbS implementation.

Many of the barriers identified within this category relate to land. First, **high population density and land-use pressure** emerge as important challenges, as the rapid expansion of urban areas increases demand for housing and places pressure on land that could otherwise be allocated to NbS or other ecological functions (Huynh Thi Ngoc et al., 2024; Abera et al., 2025; Mirsafa, Castaldo and Lemes De Oliveira, 2025). This is further complicated by the presence of informal settlements, which are often not fully integrated into formal planning processes and can intensify land management pressures (Abera et al., 2025; Enu et al., 2025; McPhearson et al., 2025). In many peri-urban areas, heterogeneous and often informal land-use patterns can also complicate infrastructure development, service provision, and the incorporation of NbS into planning processes (Abera et al., 2025).

These dynamics contribute to **difficulties integrating NbS into legal and planning systems**, due to concerns over the spatial requirements of NbS, conflicts with existing land-use priorities, and limited recognition of NbS as a legitimate planning solution (Hale et al., 2023; Rauf et al., 2024; Mirsafa, Castaldo and Lemes De Oliveira, 2025). Closely related to this is **land tenure insecurity**, which was particularly present in the Global South literature. Unclear ownership arrangements and fragmented land governance can hinder the adoption of NbS, especially when such solutions require long-term planning or coordinated management across territories (Abera et al., 2025; Bauer and Titz, 2025; Enu et al., 2025).

In addition, spatial planning for NbS in the Global South can be further complicated by **environmental and biophysical constraints** that can pose particular challenges for NbS implementation. Factors such as highly contaminated areas, the specific ecological conditions of tropical rainforests, monsoon dynamics, mosquitoes, pests and diseases, high levels of water and waste pollution, and the proliferation of invasive species can create context-specific challenges that are less commonly reported in the Global North (Huynh Thi Ngoc et al., 2024; Lee and Song, 2024; Abera et al., 2025; Mirsafa, Castaldo and Lemes De Oliveira, 2025).

Finally, two barriers commonly identified in both the Global North and the Global South are **limited land availability and plot size constraints**, as well as **conflicts with existing land uses and management priorities** such as housing and infrastructure. One of the most frequently cited constraints in the literature was the limited availability of suitable land for NbS implementation. Factors such as space limitations, soil conditions, terrain slope, and the need for retrofitting existing urban areas can all affect the viability of NbS interventions (Ershad Sarabi et al., 2019; Sarabi et al., 2020; Abera et al., 2025).

Related to this, conflicts often arise over competing land uses, as NbS may be perceived as competing with other development priorities. This tension can be reinforced by concerns over possible gentrification associated with certain NbS interventions (Abera et al., 2025), which may further

complicate land management decisions.

4.1.1.4. Technical & Capacity

Similar to **Planning & Spatial**, several barriers within this category appear more prominently in the Global South and are closely linked to the technical capacity required to support NbS implementation. **Limited technical skills, training, and capacity building** emerged as a recurring challenge across several Global South articles, particularly in relation to insufficient technical expertise, limited education in the engineering and scientific dimensions of NbS, lack of practical know-how, and inadequate site-specific technical knowledge (Thuy et al., 2019; Adu Boateng et al., 2023; Lee and Song, 2024; Nizamani and Torgersen, 2025). One factor contributing to this challenge is that much of the available knowledge and technical guidance is produced in English, which can create an additional barrier for non-English-speaking contexts (Abera et al., 2025).

Closely related to this, **limited access to appropriate technologies and infrastructure for NbS** also emerged as a challenge in both regions, though more strongly in the Global South. Affordability constraints around technologies such as GIS systems, cost-effective urban technologies, or even technical requirements associated with interventions such as rooftop gardens can hinder implementation (Artmann and Sartison, 2018; Zhu et al., 2024; Regmi and Gheewala, 2025). These challenges are often amplified by the particular environmental and socio-technical conditions found in parts of the Global South (Telwala, 2023; Huynh Thi Ngoc et al., 2024).

A barrier commonly identified across both regions was the **lack of standardized guidelines, design standards, and metrics** for NbS. The absence of consistent approaches for planning, implementation, maintenance, and monitoring has made it difficult to systematize NbS practices, compare results across cases, or assess performance relative to conventional alternatives (Fu, 2023; Huynh Thi Ngoc et al., 2024; Regmi and Gheewala, 2025). This challenge is compounded by the lack of context-specific guidelines adapted to local institutional, socio-cultural, and ecological conditions, which are critical for responding to place-based challenges (Sarabi et al., 2020).

This lack of standardization is also linked to **limited monitoring, maintenance, and evaluation systems**, as well as unreliable or insufficient data, which can affect long-term stewardship and generate hidden costs and uncertainties in predicting outcomes (Regmi and Gheewala, 2025; Zarei and Shahab, 2025). Such uncertainty can complicate the economic valuation of NbS, their integration with existing planning systems, and ultimately their development as a credible business case, as information on performance remains limited, an important constraint for scaling these solutions (McPhearson et al., 2025; Zyoud and Zyoud, 2025).

4.1.1.5. Social, Engagement & Perception

This category is present in both the Global North and Global South literature review, although some challenges appear more prominent in one region than the other due to contextual differences. Broadly, this category covers two main dimensions: **awareness and perceptions of NbS**, and **stakeholder participation and engagement** throughout the different stages of implementation.

One of the main barriers identified, also briefly touched upon in the Technical & Capacity category, is the **low perceived relevance and limited understanding of NbS among stakeholders**. This challenge appears across both regions and relates to how different actors perceive NbS compared to more traditional alternatives, as well as concerns regarding their effectiveness and potential consequences (Sarabi et al., 2020; Zarei and Shahab, 2025). In the Global South, this barrier can be reinforced by perceptions that NbS may conflict with aesthetic norms, attract unwanted plants or animals in urban areas, or negatively affect property values (Lee and Song, 2024; Rauf et al., 2024; Nizamani and Torgersen, 2025). In some contexts, NbS have even been associated with reproducing legacies of domination or environmental imperialism, generating distrust toward approaches perceived as externally imposed and raising concerns about exclusion, segregation, or threats to land

rights (Abera et al., 2025; McPhearson et al., 2025; Mirsafa, Castaldo and Lemes De Oliveira, 2025).

Related to this, a **preference for grey infrastructure and conventional solutions** was identified as another important barrier. While this was mentioned in one Global North study (Sarabi et al., 2020), it seemed more prevalent in the Global South, where a tendency toward hard engineering approaches, short-term solutions, and cultural preferences for "modern" infrastructure often shape decision-making (Villarroel-Lamb et al., 2024; Abera et al., 2025; Boafo et al., 2025). In the Global North, this challenge appeared more as **path dependency and resistance to change** within planning and decision-making systems, where systemic inertia and growth-oriented priorities can limit the uptake of NbS despite the existence of resources and technical capacity (Ershad Sarabi et al., 2019; Collier et al., 2023).

Closely connected to these barriers is the **lack of participation or resistance from communities and stakeholders**, which appeared mainly in Global South and general studies. A recurring challenge is limited involvement of local communities in NbS projects, influenced by factors such as perceptions that participatory processes are time- and resource-intensive (McPhearson et al., 2025), low interest or motivation regarding environmental issues (Liu and Martens, 2023; Abera et al., 2025), and top-down governance approaches that constrain engagement (Bauer and Titz, 2025). This can generate resistance or lack of acceptance among communities (Rauf et al., 2024), reproduce inequalities, and create a disconnect between citizens' needs and the NbS proposed (Marchetti et al., 2026).

Related to participation challenges, **conflicts between stakeholders and inequitable distribution of benefits** also emerged as barriers, particularly in the Global South. Stakeholders may hold different views regarding roles, priorities, and responsibilities, and disagreements across sectors or disciplines can hinder collaboration and lead to slow implementation (Liu and Martens, 2023; Rauf et al., 2024). In addition, concerns about green gentrification, mentioned previously, reflect fears that access to urban nature may reinforce inequalities related to class and income, potentially leading to unequal distribution of NbS benefits and exacerbating social injustice if not carefully addressed(Abera et al., 2025; Marchetti et al., 2026). In the Global North, the lack of stakeholder engagement and participation were also mentioned, though more from a broader governance perspective, emphasizing the challenges of general stakeholder coordination and support rather than active community involvement as a central concern (Sarabi et al., 2020; Du Plessis et al., 2022).

Finally, another related barrier, present only in the Global South literature, is **competing socio-economic priorities over environmental goals**. In contexts marked by poverty, inequality, and limited access to basic services such as housing, water, and health, immediate cover of basic necessities often takes precedence over ecological restoration or long-term resilience objectives. As a result, NbS planning may compete with more pressing development priorities and can be deprioritized in favour of investments that generate quicker economic returns. This challenge also highlights the importance of considering socio-cultural contexts, as public acceptance and local development needs can strongly influence whether NbS are recognized as relevant and viable solutions (Abera et al., 2025; Marchetti et al., 2026).

4.1.1.6. Knowledge & Information

Within the category **Knowledge & Information**, there is broad consensus in the literature that a major barrier is the lack of accessible information about NbS, with a persistent **research–practice gap**. Knowledge on NbS remains largely concentrated within academic and expert spheres, with limited diffusion among the broader public, which can affect awareness and acceptance (Ershad Sarabi et al., 2019; Sarabi et al., 2020). Several studies point to a disconnect between research and implementation, between *knowing* and *doing*, where scientific knowledge is not sufficiently translated into practical application (Fu, 2023; Villarroel-Lamb et al., 2024; Bauer and Titz, 2025).

Closely related to this is **limited evidence on NbS effectiveness and long-term benefits**, a barrier

identified across both regions. As discussed in the Financial category, difficulties in monetizing NbS benefits are partly linked to fragmented evidence and insufficient standardized information regarding their long-term performance. This can hinder return-on-investment assessments and the integration of NbS into policy and planning frameworks (Collier et al., 2023; Fu, 2023; Nizamani and Torgersen, 2025). In addition, some benefits remain under-researched, such as the relationship between NbS and human well-being (Abera et al., 2025; McPhearson et al., 2025).

Another commonly identified barrier is **limited knowledge sharing, transfer, and management across sectors and institutions**. A lack of centralized knowledge repositories, tools, and mechanisms for sharing lessons from previous NbS experiences can make it difficult for planners and policymakers to evaluate performance and learn from practice. Instead, knowledge often remains siloed across disciplines, organizations, or stakeholder groups, limiting broader transfer and uptake (Abera et al., 2025; McPhearson et al., 2025; Zarei and Shahab, 2025).

Connected to this is the **lack of data on NbS implementation and performance**, a challenge particularly visible in Global South and general studies. Beyond limited evidence on effectiveness, there are gaps in technical, economic, and design-related data that can constrain informed decision-making (Bousquet et al., 2023; Mirsafa, Castaldo and Lemes De Oliveira, 2025). The absence of empirical evidence and indicators can also hinder optimization and scalability of NbS interventions (McPhearson et al., 2025; Zyoud and Zyoud, 2025). In remote or underserved Global South contexts, logistical difficulties in data collection and limited resources for long-term monitoring can further restrict the availability of reliable information for planning and evaluation (Mirsafa, Castaldo and Lemes De Oliveira, 2025). This challenge is reinforced by the high cost of data generation and monitoring, as well as limited knowledge of alternative approaches, which can leave countries reliant on conventional solutions (Medina Rivera and Faggian, 2024; Mirsafa, Castaldo and Lemes De Oliveira, 2025).

This lack of knowledge and data also translates into **limited awareness and understanding of NbS**, particularly in the Global South. While stakeholders may recognize the urgency of environmental issues such as climate change, awareness of NbS as a potential response, how these solutions work and what benefits they can provide, often remains moderate, especially at the community level (Thuy et al., 2019; Lee and Song, 2024; Medina Rivera and Faggian, 2024; Velayo et al., 2024; Castaldo et al., 2025).

Finally, another barrier identified mainly in the Global South is the **lack and underutilization of local and context-specific knowledge**. In some cases, local data is unavailable, forcing implementers to rely on external datasets that may not be fully relevant to the context in which they are applied, making one-size-fits-all approaches ineffective in practice (Regmi and Gheewala, 2025). This challenge is reinforced by broader geographical biases in NbS knowledge production, as much of the evidence and research continues to be concentrated in Europe and the United States (McPhearson et al., 2025). As a result, context-appropriate knowledge for Global South settings remains limited, particularly regarding how locally grounded NbS can be adapted, replicated, and scaled effectively (McPhearson et al., 2025). In the same sense, the literature also highlights limited integration of indigenous and ancestral knowledge in the development and implementation of NbS, despite its potential relevance for context-sensitive approaches (Zarei and Shahab, 2025; Marchetti et al., 2026).

4.1.2. Enablers

A notable difference between the barriers and enablers identified in the literature review is that, for enablers, the contrast between the Global North and the Global South is not as pronounced as it is for barriers as it can be seen in and Table 2 and Fig. 5Fig. 7. In addition, many of the coded segments identified as enablers were from the General articles, suggesting a greater degree of overlap across regional contexts.

Differing from the barriers, most enablers were concentrated within the **Social, Engagement & Perception,** followed by **Technical & Capacity** and **Knowledge & Information**. Notably, many of the enablers identified are associated with some form of collaboration, engagement, or participatory processes among stakeholders, a pattern that will be further explored in the following sections.

Table 4 presents the sub-categories within each main category, the articles in which each sub-category is mentioned, and a general comparison indicating whether each enabler appears to be more predominant in one of the two regions or in the "General" articles. It is important to note that this indication of predominance does not imply that a sub-category is only exclusive to a specific region, but rather that it was identified more frequently or more prominently within the literature associated with that region.

It is important to note that some sub-categories may be transversal across multiple main categories; however, for analytical purposes they were assigned to the category in which they appear to have the greatest influence or prominence. Likewise, some sub-categories may overlap, but they are analysed from the perspective of the category in which they are situated. Fig. 7 demonstrates the coded segment distribution for the enabler section.

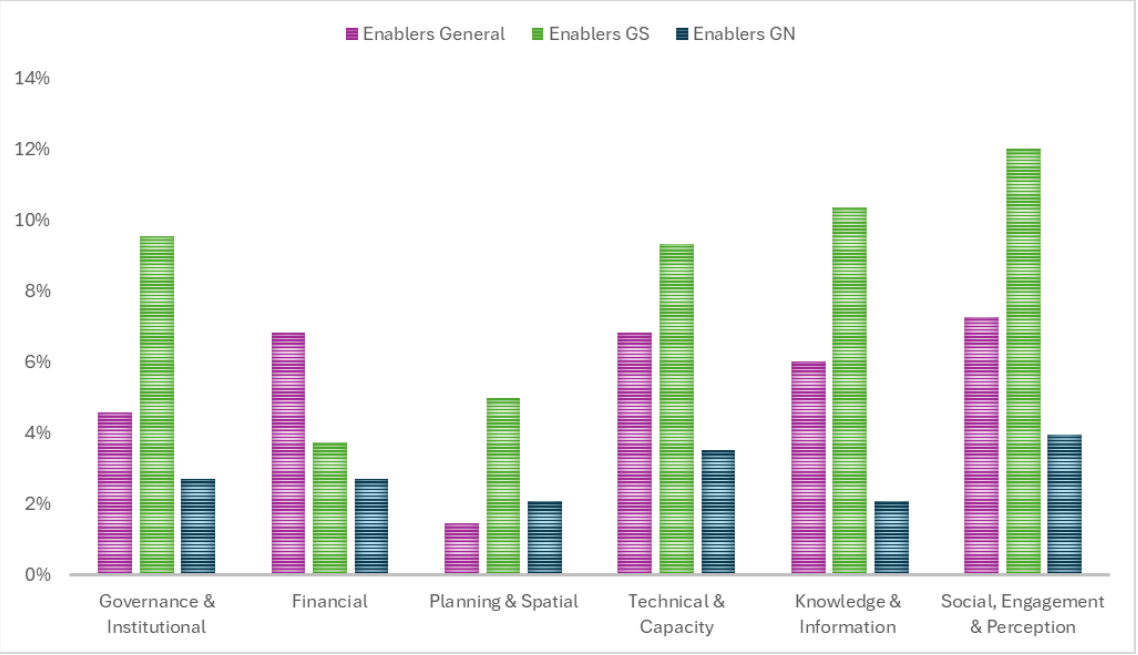


Fig. 7. Coded segment distribution for the enablers section

Table 4. Categories and sub-categories of the enablers

Enablers	Articles mentioning this*	Predominance
Governance & Institutional		
1. Supportive policy frameworks and regulations	7, 11, 12, 14, 16, 20, 21, 23, 24, 26, 28, 30, 36, 37, 39, 40, 42, 43, 59, 64, 79, 80, 82, 84	**GS, GN and General**
2. Collaboration across disciplines and governance levels	1,7, 13, 21, 24, 28, 37, 42, 80	**GS, GN and General**
3. Supportive governance structures to NbS processes	9, 11, 12, 13, 20, 28, 42	**GS**
4. Strong and sustained political will and support	3, 4, 11, 13, 20, 21, 30, 40 42, 64, 77, 79, 80	**GS, GN and General**
5. Inclusive and participatory governance processes	5, 10, 19, 30, 42	**GS and general**

Enablers	Articles mentioning this*	Predominance
Financial		
1. Clear and appropriate business models	3, 7, 11, 21, 37, 64, 76, 79, 80	**GS, GN and General**
2. Financing mechanisms, economic instruments and incentives	7, 10, 14, 19, 20, 24, 28, 30, 36, 37, 38, 42, 43, 63, 82, 84	**GS, GN and General**
3. Financial feasibility and economic sustainability of NbS	7, 13, 19, 20, 30 37, 76, 80, 84	**GS, GN and General**
4. Economic valuation of NbS benefits	11, 13, 76, 80	**GS, GN and General**
5. Public-private partnerships (PPP)	7, 18, 20, 23, 26, 39, 76, 80, 82, 84	**GS, GN and General**
Planning & Spatial		
1. Integration of NbS into planning processes	7, 19, 23, 26, 64, 77, 82	**GS, GN and General**
2. Adopting an integrated, context-specific and data-driven approach in planning and implementation n	3, 7, 10, 11, 13, 19, 20, 24, 42, 47	**GS**
3. Combining NBS with traditional approaches	14, 20, 24, 82	**GS**
4. Land availability, acquisition and management strategies	6, 13, 43, 63, 64	**GN>GS and General**
5. Supportive planning rules and practices to NbS	7, 19, 20, 24, 26, 28, 44, 79, 84	**GS, GN and General**
Technical & Capacity		
1. Technical knowledge and expertise in implementing NBS projects	7, 16, 19, 24, 42, 84	**GS, GN and General**
2. Guidelines and standards for NbS	1, 7, 11, 13, 19, 20, 24, 37	**GS, GN and General**
3. Effective monitoring, maintenance and evaluation systems and protocols	1, 2, 4, 13, 19, 20, 24, 37	**GS, GN and General**
4. Learning-by-doing approach	1, 3, 7, 11, 13, 14, 19, 30, 80, 82	**GS, GN and General**
5. Delivery of multiple benefits	13, 24, 26, 39, 41, 43, 45, 84	**GS**
6. The integration of technologies and tools in NbS	19, 21, 37, 64	**General**
7. Education and training programs	3, 4, 11, 14, 16, 20, 24, 42, 44, 63, 64, 82	**GS**
Social, Engagement & Perception		
1. Involvement/engagement of the community in the NbS process	3, 4, 5, 6, 7, 10, 13, 16, 19, 21, 24, 37, 38, 40, 42, 43, 44, 47, 63, 64, 81, 82, 84	**GS, GN and General**
2. Stakeholders' engagement and collaboration in NbS planning and development	1, 6, 7, 10, 13, 16, 19, 21, 24, 37, 39, 42, 79, 80, 82, 84	**GS, GN and General**
3. Partnerships among stakeholders	4, 7, 24, 36, 79, 80, 82	**GS, GN and General**
4. Participatory methods and approaches	7, 10, 13, 19, 21, 40, 42, 44, 81	**GS, GN and General**
5. Enhanced public awareness and capacity building for NbS	6, 7, 9, 10, 11, 13, 42 ,63	**GS**
6. Creation and nurture of NbS networks	6, 19, 30, 63, 79	**GS, GN and General**
Knowledge & Information		
1. Integration of local knowledge and conditions	7, 10, 19, 24, 26, 28, 30 37, 42, 81	**GS, GN and General**

Enablers	Articles mentioning this*	Predominance
2. Knowledge sharing mechanisms and technologies	3, 4, 11, 13, 19, 20, 24, 28, 37, 40, 42, 44, 79, 80, 82, 84	**GS, GN and General**
3. Identifying and communicating benefits, co-benefits of NbS	3, 7, 10, 11, 16, 19, 20, 26, 28, 42, 45, 64, 79, 80, 84	**GS, GN and General**
4. Knowledge production and co-production	1, 19, 20, 21, 36, 42, 44, 80	**GS and General**
5. Effective knowledge translation and dissemination	4, 16, 37, 40, 42, 44, 80	**GS**
6. Availability of reliable data on NbS	7, 13, 20, 21, 30, 42, 44, 84	**GS and General**

*The article corresponding to each number can be seen in Appendix 2

4.1.2.1. Governance & Institutional

As a response to one of the main barriers identified, the **lack of clear policies and regulatory frameworks**, one of the key enablers emerging from the literature is the presence of **supportive policy frameworks and regulations**. This enabler was identified in both regions and underscores how the implementation of NbS is closely linked to the existence of enabling institutional conditions that allow these solutions to develop and scale effectively (Collier et al., 2023, 2023). In this regard, some governments have begun to recognize the need for guidelines that encourage municipalities to adopt and mainstream NbS (Hale et al., 2023; Huynh Thi Ngoc et al., 2024; Lee and Song, 2024). Supportive regulatory frameworks can contribute to creating enabling environments by increasing awareness of NbS and expanding opportunities for funding (Lee and Song, 2024). In addition, aligning these policies with international frameworks, such as the SDGs and the IUCN's NbS standards, can further strengthen their effectiveness (Simelton et al., 2021). The integration of NbS into sectoral policies, including health, land, and water management, was also identified as an important factor in reinforcing institutional support (Marchetti et al., 2026).

Closely related to this, **strong and sustained political will and support** emerged as another key enabler, particularly given the long-term nature of NbS (Simelton et al., 2021). For instance, in Thailand, the implementation of vetiver grass, as an NbS for landslide risk reduction, was strongly supported by the monarchy, which contributed to both the successful implementation of the NbS and its acceptance among local communities (Leknoi et al., 2026). Especially in the Global South, successful NbS cases are often associated with good governance and long-term institutional commitment (Agaton and Guila, 2024; Mirsafa, Castaldo and Lemes De Oliveira, 2025; Marchetti et al., 2026). This highlights the importance of **supportive governance structures** that enable systematic and coordinated implementation processes, grounded in democracy, equity, and scientific evidence (Marchetti et al., 2026). In some cases, the presence of a leading organization responsible for coordinating stakeholders, defining shared responsibilities, and establishing clear implementation procedures has been identified as beneficial (Velayo et al., 2024; Castaldo et al., 2025; Mirsafa, Castaldo and Lemes De Oliveira, 2025).

Finally, as previously mentioned, one of the most prominent enablers across the literature is the importance of collaboration, engagement, and participatory processes, supported by appropriate policy frameworks. From a governance perspective, **collaboration across disciplines and governance levels** can facilitate NbS implementation, as these solutions are inherently cross-sectoral, multi-scale, and inclusive. This requires moving beyond siloed institutional approaches and fostering interaction among diverse actors to build a shared vision (Collier et al., 2023; Castaldo et al., 2025). In this context, effective institutional capacities are needed to support collaboration across administrative, organizational, and jurisdictional boundaries, while also helping to bridge the research–practice gap (Abera et al., 2025; Mirsafa, Castaldo and Lemes De Oliveira, 2025; Marchetti

et al., 2026).

In a similar vein, the development of **inclusive and participatory governance processes,** identified particularly in the Global South, but also present in the General studies, is critical for the long-term success of NbS. Such approaches can help address issues of social exclusion and unequal distribution of benefits, while promoting more context-sensitive and equitable solutions. These processes should be developed through transdisciplinary, interdepartmental, and cross-sectoral collaboration to fully leverage the potential of NbS within different regions (McPhearson et al., 2025; Marchetti et al., 2026).

Further discussion on the role of engagement and participation will be explored in more depth within the **Social, Engagement & Perception** category.

4.1.2.2. Financial

The **Financial** category represents one of the less prominent categories in terms of the number of enablers identified; however, it is notable for its consistency across regions, as most of the enablers were strongly represented in both the Global North and the Global South literature. This suggests that financial support is a fundamental requirement for the implementation and scaling of NbS, regardless of context.

One of the most frequently mentioned enablers is the presence of **financing mechanisms, economic instruments, and incentives**. NbS require both initial investment and sustained long-term funding to be effectively developed and maintained (Adu Boateng et al., 2023; McPhearson et al., 2025). In this sense, diversified funding sources are particularly important to reduce dependency on single streams of finance (19). Financial incentives can play a key role in unlocking the potential of NbS (Boafo et al., 2025), especially when these initiatives are integrated into national and local budgets (Huynh Thi Ngoc et al., 2024) and complemented by funding from other mechanisms, such as Corporate Social Responsibility (CSR) and Environmental, Social, and Governance (ESG) schemes (Simelton et al., 2021). Furthermore, aligning NbS policies with financial strategies is essential for their long-term viability (Liu and Martens, 2023; Regmi and Gheewala, 2025). Instruments such as subsidies, tax incentives, hybrid financial models, and private investment were identified as important mechanisms to support NbS implementation (Collier et al., 2023).

A commonly highlighted mechanism within this category is **public–private partnerships (PPPs)**, which can also be understood as bridging governance and financial dimensions. PPPs were identified in both regions as an effective approach to financing NbS, as they involve collaboration between public institutions and private actors, allowing for shared risks and benefits while leveraging complementary resources and expertise (Wickenberg, McCormick and Olsson, 2021; Huynh Thi Ngoc et al., 2024).

Another key enabler across both regions is the development of **clear and appropriate business models** that can support and sustain NbS over time. Such models are essential for attracting investment and integrating NbS into mainstream markets (Frantzeskaki et al., 2019; Zhu et al., 2024). Effective business models can support clearer processes, utilizing existing resources more efficiently, and creating incentives that enable NbS to thrive (Lee and Song, 2024; Regmi and Gheewala, 2025). Central to this is the need to ensure the **financial feasibility and economic sustainability** of NbS beyond the initial investment phase, allowing them to remain viable throughout their life cycle (Mirsafa, Castaldo and Lemes De Oliveira, 2025). Tools such as life-cycle assessments and cost–benefit analyses were identified as important in supporting decision-making processes and strengthening the economic case for NbS (Huynh Thi Ngoc et al., 2024; McPhearson et al., 2025; Regmi and Gheewala, 2025).

Finally, the **economic valuation of NbS benefits** emerges as a crucial enabler. Recognizing and assigning economic value to the multiple benefits generated by NbS, across environmental, social,

and economic dimensions, can help justify investments and demonstrate long-term returns. These benefits may include the creation of sustainable livelihoods, promotion of tourism, contributions to food production, climate change adaptation, and other co-benefits that enhance overall societal well-being (Frantzeskaki et al., 2019; Mirsafa, Castaldo and Lemes De Oliveira, 2025; Nizamani and Torgersen, 2025).

4.1.2.3. Planning & Spatial

The **Planning & Spatial** category included the fewest enablers in the literature review and, similarly to the barriers, primarily focuses on land availability and the role of planning processes in NbS implementation. Across both regions, a key enabler is the **integration of NbS into planning processes**, particularly within urban contexts, which were the dominant focus of the reviewed articles. This enabler highlights the importance of embedding NbS within urban planning to enhance resilience and to position urban nature as a central component of policy and development agendas (Frantzeskaki et al., 2019; Medina Rivera and Faggian, 2024; Bauer and Titz, 2025). For example, integrating urban agriculture (i.e. communal gardens) within existing food systems (Zhu et al., 2024) and recognizing green infrastructure as a viable strategy for climate change adaptation (77) were identified as relevant approaches.

In line with this, **supportive planning rules and practices** can facilitate the incorporation of NbS into national and local policies. A recurring theme in the literature is the need for innovative, holistic, and inclusive planning approaches, where nature is considered a central element in urban development. Experimenting with planning concepts that integrate ecological functions into the built environment can contribute to the development of more resilient cities (Van Der Jagt et al., 2017; Abera et al., 2025; McPhearson et al., 2025). This requires the design of planning frameworks that actively embed nature within urban systems (Zarei and Shahab, 2025).

A particular advantage identified in some Global South contexts is that, in areas where infrastructure systems are still developing, there may be greater flexibility in planning processes. This can create opportunities to integrate NbS more easily, without the constraints of retrofitting pre-existing systems (Abera et al., 2025).

Another important enabler, identified across both regions but more prominently in the Global North, is the development of **land availability, acquisition, and management strategies**. Securing land for NbS implementation is a key driver, which can be supported through land-use analysis, the identification of protected areas, and innovative approaches to utilizing available urban space like using spaces along the streets as green corridors, green walls or vacant lands for urban agriculture (Du Plessis et al., 2022; Tyllianakis, Martin-Ortega and Banwart, 2022; Zhu et al., 2024; Mirsafa, Castaldo and Lemes De Oliveira, 2025).

Two additional enablers were particularly prominent in the Global South. First, the **adoption of integrated, context-specific, and data-driven approaches** to planning and implementation was identified as fundamental for effective NbS integration. While the concept of NbS has gained global traction, transferring approaches from the Global North without adapting them to local conditions can lead to ineffective or even maladaptive outcomes. Therefore, place-based solutions that consider ecological diversity, cultural contexts, and socio-economic conditions are essential (McPhearson et al., 2025; Mirsafa, Castaldo and Lemes De Oliveira, 2025). For instance, incorporating endemic plant species, engaging community institutions, and understanding local social norms can significantly enhance the success of NbS (McPhearson et al., 2025).

Finally, **combining NbS with traditional approaches** was identified as a valuable strategy mainly in the Global South. Rather than viewing green and grey solutions as mutually exclusive, hybrid approaches can improve acceptance, reduce uncertainty, and enhance the overall effectiveness of interventions, while considering environmental, social and economic factors. This integration can support climate adaptation efforts while enabling the scaling up of NbS by demonstrating their

practical value within existing systems (Ershad Sarabi et al., 2019; Adu Boateng et al., 2023; Huynh Thi Ngoc et al., 2024; Abera et al., 2025).

4.1.2.4. Technical & Capacity

In the analysis of barriers, this category was identified as one of the least prominent and was largely found in the Global South literature. In contrast, within the enablers this category emerges as one of the most significant, with a more balanced distribution between the Global North and Global South and a strong representation in the General articles.

Two of the most prominent and complementary enablers are the availability of **guidelines and standards for NbS** and the presence of **effective monitoring, maintenance, and evaluation systems and protocols**. These enablers are relatively evenly distributed across regions and emphasize the importance of standardizing NbS processes throughout their entire lifecycle, from planning and implementation to long-term stewardship. For instance, post-implementation performance assessments using standardized metrics have been identified as key to enhancing the credibility and acceptance of NbS (Fu, 2023). Similarly, the use of tools such as Geographic Information Systems (GIS) to support decision-making during the planning phase (Bousquet et al., 2023), as well as environmental impact assessments to evaluate outcomes after implementation (Collier et al., 2023), contribute to more robust and evidence-based approaches. Continuous monitoring and maintenance, supported by clear protocols, enable NbS projects to remain effective over time and adapt to changing conditions (McPhearson et al., 2025; Mirsafa, Castaldo and Lemes De Oliveira, 2025). Moreover, the availability of standardized performance data can facilitate access to funding by reducing uncertainty for investors and decision-makers (Abera et al., 2025). Initiatives like collaborative monitoring through tools like the Internet of Things (IoT) can give valuable information to decision makers about aspects as the efficiency of the project (Ershad Sarabi et al., 2019). It was recognized how important it is for NbS to have a systems approach and guiding frameworks that help with its implementation (Fu, 2023; McPhearson et al., 2025).

In the same line, one enabler that supports those mentioned previously and that was mainly mentioned in the General articles is the **integration of technologies and tools in NbS** as it can help with the implementation of NbS by gathering important data, making the delivery of NbS more efficient and effective and allowing to replicate NbS in different contexts (Lee and Song, 2024; Zhu et al., 2024; McPhearson et al., 2025).

Another important enabler identified across both regions is the **learning-by-doing approach**. The literature shows that pilot and demonstration projects can significantly increase acceptance and support for NbS, as they allow stakeholders to observe tangible results and benefits in controlled or real-world settings (Ershad Sarabi et al., 2019; Collier et al., 2023; Mirsafa, Castaldo and Lemes De Oliveira, 2025; Leknoi et al., 2026).

In addition to standards and monitoring systems, the presence of **technical knowledge and expertise** is critical for identifying opportunities for NbS, improving their effectiveness, and enabling their scaling (Hale et al., 2023; Abera et al., 2025; McPhearson et al., 2025; Zarei and Shahab, 2025). In this regard, one enabler particularly prominent in the Global South is the development of **education and training programs** tailored to different stakeholders involved in NbS processes, including local communities. These programs support capacity building, facilitate knowledge dissemination, and contribute to the long-term sustainability of NbS initiatives (Huynh Thi Ngoc et al., 2024; Wolff et al., 2025; Leknoi et al., 2026). Furthermore, integrating NbS concepts into education programs, especially among younger generations, has been suggested as a strategy to strengthen future capacity and awareness (Villarroel-Lamb et al., 2024; Abera et al., 2025; Marchetti et al., 2026).

Finally, from a technical perspective, the literature highlights that NbS are more likely to gain acceptance when they can be shown to deliver **multiple co-benefits**, a finding that was particularly prominent in the Global South. The perceived value of NbS increases when, in addition to their

primary function, they contribute to social well-being, community interaction, eco-tourism, recreational opportunities, and aesthetic improvements, among other benefits (Debie, 2025; Enu et al., 2025).

4.1.2.5. Social, Engagement & Perception

While in the barriers section this category primarily focused on the perception of NbS among different actors, preferences for alternative solutions, and challenges related to stakeholder engagement and conflict, the enablers present a contrasting picture. Most enablers in this category are concentrated around engagement, collaboration, and partnerships. This is the category with the highest number of enablers, and although several sub-categories overlap conceptually, they have been organized around two main dimensions: engagement and collaboration. Notably, most of these enablers were found across both the Global North and the Global South literature.

Stakeholder engagement and collaboration in NbS planning and implementation are widely recognized as fundamental drivers of success. Active involvement of researchers, policymakers, practitioners, and communities is identified as a key factor supporting NbS implementation (Tyllianakis, Martin-Ortega and Banwart, 2022; Fu, 2023). When this engagement is embedded systematically within decision-making processes, it fosters a sense of ownership and long-term commitment, which is critical for the sustainability of NbS (McPhearson et al., 2025; Regmi and Gheewala, 2025). This engagement extends into cross-sectoral collaboration**,** where co-creation with diverse actors functions as a form of collaborative governance. Given the multidisciplinary nature of NbS, coordination across sectors is essential to generate synergies between climate, biodiversity, and development agendas (Collier et al., 2023; Lee and Song, 2024; Abera et al., 2025; Bauer and Titz, 2025; Marchetti et al., 2026).

A particularly prominent enabler, across both regions, is the **Involvement/engagement of the community**. The literature strongly emphasizes that community engagement is central to the successful implementation and scaling of NbS. Practices such as integrating community-driven and bottom-up initiatives with national and municipal governance structures are highlighted as key to achieving transformative outcomes (Boafo et al., 2025; Leknoi et al., 2026). Involving communities throughout the design, implementation, and maintenance phases helps ensure that NbS respond to local needs, enhances their perceived relevance, and builds long-term support (Tyllianakis, Martin-Ortega and Banwart, 2022; Rauf et al., 2024). Effective engagement also requires acknowledging community preferences, concerns, and knowledge systems, while simultaneously contributing to local capacity building (Mirsafa, Castaldo and Lemes De Oliveira, 2025). In the Global South, there is additional emphasis on the involvement of Indigenous communities and the integration of their knowledge, which can significantly strengthen NbS implementation (Villarroel-Lamb et al., 2024; Regmi and Gheewala, 2025). This aspect is further explored in the Knowledge & Information category.

To facilitate engagement and collaboration, several mechanisms are identified as key enablers across both regions. First, the establishment of **partnerships among stakeholders** plays a crucial role. Partnerships create structured spaces for collaboration among governments, NGOs, communities, academia, and the private and financial sectors, helping to navigate power dynamics and define roles and responsibilities (Collier et al., 2023; Wolff et al., 2025). Such partnerships can support co-stewardship arrangements, improve coordination, and reduce trade-offs in NbS implementation (Collier et al., 2023; Abera et al., 2025).

Second, the use of **participatory methods and approaches** is highlighted as an important driver of co-creation, co-production, and co-design processes. Mechanisms such as workshops, multi-actor platforms, and living labs enable stakeholders to collaboratively develop NbS, align goals, and ensure that solutions are connected with the public (Bousquet et al., 2023; Collier et al., 2023; McPhearson et al., 2025; Mirsafa, Castaldo and Lemes De Oliveira, 2025). These approaches foster inclusive spaces for dialogue, enabling more informed, transparent, and systematic decision-making processes

(McPhearson et al., 2025).

Third, the **creation and nurture of NbS networks** is identified as a key mechanism to support engagement, collaboration, and knowledge exchange. These networks can enhance awareness, provide technical and political support, build capacity, and facilitate open dialogue among stakeholders, ultimately contributing to more resilient and well-informed NbS initiatives (Van Der Jagt et al., 2017; Simelton et al., 2021; McPhearson et al., 2025).

Finally, particularly in the Global South, the importance of **enhancing public awareness and capacity building** is emphasized. Awareness campaigns that communicate the benefits of NbS and address misconceptions can play a critical role in increasing acceptance and support for these solutions (Boafo et al., 2025; Nizamani and Torgersen, 2025). In this sense it was also identified that higher levels of positive perception and awareness of NbS can be associated with an increased awareness of climate change which can result in better support for this kind of initiatives (Velayo et al., 2024).

4.1.2.6. Knowledge & Information

This category is the third most prominent in terms of enablers and shows a stronger concentration in the Global South, while still maintaining a considerable representation in the General literature.

One of the main enablers identified is the **identification and communication of NbS benefits and co-benefits**, which is present across both regions. As discussed in the **Financial** category, the valuation of NbS benefits is essential; however, this enabler goes further by emphasizing the need to actively communicate and leverage these benefits. This includes raising public awareness, formulating clear value propositions, and recognizing both monetary and non-monetary values to attract funding and stakeholder support (Collier et al., 2023; Hale et al., 2023; Huynh Thi Ngoc et al., 2024; Marchetti et al., 2026). Presenting NbS as accessible, cost-effective in the long term, and multifunctional, delivering environmental, social, and economic benefits, can significantly enhance their acceptance, sustainability, and overall success (Huynh Thi Ngoc et al., 2024; Boafo et al., 2025).

Other key enablers relate to knowledge processes, which in this category are approached from the perspective of production, sharing, translation, and integration. In terms of **knowledge sharing mechanisms and technologies**, the literature, particularly from the Global South and General studies, highlights the importance of exchanging successful practices through platforms such as repositories and digital tools. These mechanisms facilitate feedback, idea exchange, issue mapping, and the dissemination of best practices, case studies, and lessons learned (Ershad Sarabi et al., 2019; Abera et al., 2025; Zarei and Shahab, 2025). Effective knowledge sharing not only supports NbS adoption but also strengthens stakeholder engagement, enhances awareness, and contributes to the resilience of NbS initiatives (Agaton and Guila, 2024; Marchetti et al., 2026).

Closely related is the enabler of **knowledge production and co-production**, which is also prominent in the Global South and General literature. There is a recognized need for more research on NbS globally, as increased knowledge production can support the development of evidence-based standards and policies, ultimately facilitating implementation and scaling (Zyoud and Zyoud, 2025). This is particularly relevant for the Global South, where a significant proportion of the world's biodiversity is located, yet where more context-specific social and ecological research is needed to address local realities (Huynh Thi Ngoc et al., 2024; McPhearson et al., 2025).

The **integration of local and context-specific knowledge** is another important enabler, especially in the Global South. Incorporating local practices and strategies can make NbS more inclusive, contextually appropriate, and resilient (Collier et al., 2023; Boafo et al., 2025). In particular, Indigenous and ancestral knowledge systems can provide valuable insights and, in some cases, offer more effective solutions than directly transferring approaches from the Global North (McPhearson et al., 2025). Recognizing and valuing these knowledge systems, often transmitted verbally through

generations, can inspire innovative and culturally grounded NbS approaches (McPhearson et al., 2025). For example, in South Africa, historical local practices used to stabilize mobile sand have demonstrated how existing ecological knowledge and remnant nature can support multiple social and environmental benefits (McPhearson et al., 2025).

In line with this, **effective knowledge translation and dissemination** is highlighted as a crucial enabler, particularly in the Global South. Translating scientific knowledge into inclusive formats for the community and establishing a shared language among stakeholders can help bridge the gap between research and practice. This, in turn, enables broader participation by local actors and supports the successful implementation of NbS (Hale et al., 2023; Marchetti et al., 2026).

Finally, the **availability of reliable data on NbS** is identified as an important enabler, especially in the Global South. Access to high-quality data supports informed decision-making, as well as the planning, implementation, and long-term stewardship of NbS projects (Mirsafa, Castaldo and Lemes De Oliveira, 2025). The development of accessible databases and data-sharing platforms is therefore essential to support the wider adoption and scaling of NbS (Huynh Thi Ngoc et al., 2024).

4.2. Insights from the interviews

Table 5 presents a summary of the five interviews, including the country, the focus of the NbS project, and a brief profile of each interviewee. This is followed by a more detailed description of the interviewed organizations and the main findings from each interview.

The interviews were assigned a code, as shown in Table 5, indicating GS (Global South) or GN (Global North) to identify the region to which they correspond. The findings were divided into five sections based on the main interview questions relevant to this study: Context, identified barriers, identified enablers, what is mostly needed in the regional context, and suggested tools, frameworks, or methodologies. The findings from the section “what is mostly needed in the regional context” could also be interpreted as barriers or enablers depending on the response; however, this category was maintained separately to facilitate a clearer analysis.

Table 5. Profile of the interviews

Interview code	Country	Project	Profile interviewee
GS1	Colombia	Implementation of nature-based solutions for climate change adaptation in cocoa systems in the Chucurí River basin	Project Manager - Participatory Climate Monitoring
GS2	Colombia	Water governance and wildlife supply chains	Environmental Engineer and Researcher of the Water, Biodiversity and Climate Network
GN1	Netherlands	Advancing Nature-based Solutions in River Systems	Nature advisor
GN2	Spain	Using extensive livestock for vegetation control under high voltage power lines	Social Innovation Specialist Sustainability Office
GN3	United States of America	Conservation and restoration: Supporting bird-friendly practices on working lands	Project Coordinator

4.2.1. GS1

4.2.1.1. Context

The first implementer interviewed (GS1) is a Colombian NGO that has been working for more than 40 years to transform society’s relationship with nature towards more sustainable practices. The

organization is committed to the conservation of Colombia's natural wealth through an ecosystem services approach.

Although the organization has been implementing practices aligned with Nature-based Solutions (NbS) for many years, it has only recently begun to explicitly frame some of its initiatives under this concept. This shift was influenced by its participation in a funding call promoting NbS in Colombia, supported by the German government.

The three main NbS-related projects currently being developed by the organization are:

- Conservation of natural areas through passive restoration
- Management and development of agroforestry systems
- Diversification of forestry into productive systems, including the integration of beekeeping.

According to the project coordinator the organization *"had been working on those three initiatives in other processes, but not under the umbrella framework of the concept of nature-based solutions".*

4.2.1.2. Barriers identified

The main barrier identified in this interview is the widespread misconception that productivity and conservation are mutually exclusive *(Social, Engagement & Perception; Knowledge & Information).* According to the interviewee, in rural areas of Colombia, forested land has historically been perceived as unproductive. In fact, past public policies encouraged the clearing of forests to convert them into agricultural land. As a result, many farmers still hold this perception, which continues to hinder the adoption of NbS approaches that seek to integrate conservation with productive activities.

"A farm with patches of stubble and wooded areas is considered an abandoned farm. It's not seen as an ideal producer; people view it as lazy and unwilling to utilize the entire area to develop a productive system"

The main challenge, therefore, lies in shifting producers' mindsets. The organization has found that this is a gradual process that requires building trust and providing tangible evidence over time. As producers begin to see that practices benefiting ecosystems can also generate economic and social benefits for them, they become more open to adopting these approaches.

These barriers can be classified within the following sub-categories:

- Low perceived relevance and understanding of NbS among stakeholders (Social, Engagement & Perception)
- Limited awareness and understanding of NbS (Knowledge & Information)

4.2.1.3. Enablers identified

In terms of enablers, several were identified in this interview. First, the importance of building trust with producers was strongly emphasized, as it is fundamental to enabling change *(Social, Engagement & Perception*). Closely linked to this is the need to provide stakeholders with relevant information and evidence *(Knowledge & Information).* In this case, the organization supports farmers by demonstrating the benefits of Nature-based Solutions, particularly in relation to forest conservation, biodiversity, and alternative productive activities such as beekeeping. Making these benefits visible is key to overcoming historical barriers, such as the perception that conservation and productivity are incompatible, and to fostering greater acceptance of more sustainable practices.

Another key enabler identified is the provision of continuous, context-sensitive support to rural communities (*Knowledge & Information; Social, Engagement & Perception*). The interviewee emphasized the importance of translating technical concepts, strategies, and NbS-related knowledge

into accessible and practical language that farmers can understand and internalize. One-off workshops were considered insufficient; instead, sustained engagement and follow-up are necessary to ensure that knowledge is effectively absorbed and applied in practice. This ongoing support enables farmers to gradually integrate NbS principles into their activities, reinforcing both learning and long-term adoption.

"Of course, for us it is very important to have the possibility of support; when you have support in the processes, people begin to become familiar with the concept, which is the first need that is required."

"To achieve this approach, the concept must be presented pragmatically so that people can internalize it within the development of their daily farm activities."

An additional enabler identified in this interview is the use of technology to support knowledge generation and decision-making (*Technical & Capacity; Knowledge & Information*). The interviewee highlighted that the integration of digital tools in rural contexts is both feasible and valuable, particularly in strengthening farmers' access to relevant, real-time information. For example, in a separate project, the organization has implemented a participatory climate monitoring system in which farmers use handheld sensors to collect climate data and upload it to an application. This data is then processed using artificial intelligence to provide insights on how climatic conditions may affect their crops. Such approaches not only enhance technical capacity but also promote experiential learning, enabling farmers to engage directly with climate-related knowledge and apply it in their practices.

"To have instant, truthful information that they themselves are collecting and feeding"

These enablers can be classified within the following sub-categories:

- Learning-by-doing approach (Technical & Capacity)
- The integration of technologies and tools in NbS (Technical & Capacity)
- Community engagement (Social, Engagement & Perception)
- Stakeholders' engagement and collaboration in NbS planning and development (Social, Engagement & Perception)
- Participatory and collaboration methods and approaches (Social, Engagement & Perception)
- Identifying and communicating benefits, co-benefits of NbS (Knowledge & Information)
- Effective knowledge translation and dissemination (Knowledge & Information)

4.2.1.4. What is mostly needed in the regional context

Regarding what is most needed in the regional context, the interview highlighted the importance of starting at the farm level. While the scale of the climate crisis demands large-scale action, the interviewee emphasized that effective solutions must begin with small, localized interventions. Building awareness among stakeholders, particularly producers, about the relevance and benefits of NbS in their daily practices was identified as a critical step.

"Let's help people and producers understand the benefits, and that's done through individual support. This is achieved through processes and projects that can provide resources so people can understand the advantages2.

4.2.1.5. Suggested tools, frameworks, or methodologies

In terms of tools and methodologies, the interviewee noted that the organization does not currently rely on specific formal tools in the NbS process. However, a clear need was identified for tools that support the commercialization of products derived from agroecological and NbS initiatives. Examples include marketing honey, developing value-added products such as handicrafts from cocoa by-products, and identifying appropriate market channels. Strengthening capacities in product

development, value chain integration, and market access was highlighted as essential to enhance the economic viability of NbS and incentivize their adoption among producers.

4.2.2. GS2

4.2.2.1. Context

The second implementing organization interviewed is a Colombian network composed primarily of NGOs, as well as public institutions, academic organizations, and private sector actors. The network focuses on strengthening water governance and is currently implementing a Nature-based Solutions (NbS) project in the Lebrija River basin. In addition, it works on developing wildlife corridors to promote economic and social development through biodiversity conservation in the region.

The organization's activities are supported through international funding, including contributions from the Dutch government, and it is currently participating in a funding call aimed at promoting NbS implementation in Colombia with support from the German government.

4.2.2.2. Barriers identified

One of the main barriers mentioned in the interview is the disconnect between formal education and local biodiversity. School curricula tend to prioritize globally recognized species over native flora and fauna, limiting students' awareness of and connection to their immediate natural environment *(Technical & Capacity).* This, coupled with a profound lack of knowledge about the country's flora and fauna and the absence of precise studies, makes it difficult to define how to work to protect the species that inhabit Colombian territory, one of the most biodiverse places in the world *(Knowledge & Information).*

"The lack of information and the fragmentation of information is the major bottleneck for everything."

Another barrier identified in this interview relates to the misalignment between global NbS concepts and local practices (*Knowledge & Information*). The interviewee noted that many practices in the Global South that align with NbS principles have long been implemented but are not formally recognized under this terminology, which often originates in the Global North *(Social, Engagement & Perception).* In some cases, these externally developed conceptualizations fail to account for local socio-ecological contexts. For example, in the project region, an NbS cluster[2] did not classify wildlife corridors as NbS, despite their clear contribution to biodiversity conservation and the generation of social and economic benefits. This disconnect can limit recognition, support, and funding opportunities, ultimately reinforcing the gap between local practice and formal NbS frameworks.

"A number of concepts that, when you start working with grassroots communities, they obviously say, "But what are you talking about?" We talk about agroecology, "and that's a nature-based solution? Oh, yes, of course", and we talk about wetland conservation in the Middle Magdalena region. "Is that a solution? Oh, of course". So, instead of uniting, these terms actually create distance."

These barriers can be classified within the following sub-categories:

[2] Cluster refers to a regionally organized stakeholder network focused on a specific sector or theme, as defined by Colombia's national competitiveness policy (Cámara de Comercio de Bogotá, 2026)

- Limited technical skills, training and capacity building for NbS (Technical & Capacity)
- Low perceived relevance and understanding of NbS among stakeholders (Social, Engagement & Perception)
- Limited awareness and understanding of NbS (Knowledge & Information)
- Lack and underutilization of local and context-specific knowledge (Knowledge & Information)
- Research-practice gap (Knowledge & Information)

4.2.2.3. Enablers identified

One of the most prominent enablers in the interview, and one that was mentioned several times, is the direct work with the farmers communities *(Social, Engagement & Perception)* because they are the ones who have the knowledge and they are the ones who share day by day with the flora and fauna of the territory.

"I believe that our greatest allies are the communities, and that's why I think one of the biggest shortcomings of the nature-based solutions cluster in Santander, Huila, and Córdoba is that it's basically made up of academia… And non-governmental organizations can help support and accompany social processes in the territory, because that is really where the changes happen."

Another enabler identified in this interview is the use of technologies such as Artificial Intelligence and drones to support data collection and analysis (*Technical & Capacity*). In this case, these technologies are implemented through participatory approaches, where local farmer communities are actively involved in gathering information (*Social, Engagement & Perception*). This combination not only enhances the quality and availability of data but also strengthens community engagement and ownership in NbS implementation processes.

"We work with drone technologies and artificial intelligence, involving farming communities so that they themselves generate information about the conservation of the white-tailed deer."

These enablers can be classified within the following sub-categories:

- The integration of technologies and tools in NbS (Technical & Capacity)
- Learning-by-doing approach (Technical & Capacity)
- Involvement/engagement of the community in the NbS process (Social, Engagement & Perception)
- Stakeholders' engagement and collaboration in NbS planning and development (Social, Engagement & Perception)
- Participatory and collaboration methods and approaches (Social, Engagement & Perception)
- Knowledge production and co-production (Knowledge & Information)

4.2.2.4. What is mostly needed in the regional context

Regarding what is most needed in the regional context, the interview emphasized the importance of strengthening research efforts, particularly applied and co-produced research with local communities (*Knowledge & Information; Social, Engagement & Perception*). Expanding knowledge of local flora, fauna, and social resources was identified as essential for informing territorial planning that is economically, culturally, socially, and environmentally sustainable. In this regard, the interviewee also highlighted the importance of integrating this knowledge-building process into educational systems, particularly by initiating research and awareness efforts within schools.

"That girls and adolescents begin to research more about what they have in their territory, within their territory, and not remain confined in classrooms, in front of a board that teaches them, first,

about things that do not exist in their territory, and second, about production chains that are not the ones they have around them."

4.2.2.5. Suggested tools, frameworks, or methodologies

One of the tools highlighted in the interview is governance building as a foundational approach across different contexts. The interviewee emphasized that by strengthening governance structures, more inclusive and effective solutions can be developed. In their case, governance is built through a three-layer methodology, originally developed in the Netherlands and later adapted to the Latin American context. This three-layer approach consists of:

- **Knowledge layer:** Focuses on building a solid baseline of knowledge around the project and ensuring that this information is sufficient, relevant, and accessible to all stakeholders. Strategies such as participatory monitoring are used to support data generation and shared understanding.
- **Institutional layer:** Involves harmonizing the policies, instruments, and tools surrounding the project. This includes understanding which regulations influence decision-making, who the decision-makers are, and what public policies govern the process.
- **Relational layer:** Addresses how stakeholders collaborate within the same territory. This layer is influenced by both the knowledge and institutional dimensions, shaping how collective action is carried out.

4.2.3. GN1

4.2.3.1. Context

The third implementing organization interviewed is a public entity from the Netherlands responsible for the management of national roads, major waterways, and large water systems. The organization applies Nature-based Solutions (NbS) primarily within river systems, with a focus on enhancing climate resilience by implementing a 'city-in-nature' approach by implementing floodplains and blue-green corridors.

Its projects typically require collaboration with multiple stakeholders, including provinces, water boards, municipalities, and nature organizations. In some cases, international organizations such as the IUCN are also involved in the implementation process.

4.2.3.2. Barriers identified

One of the main barriers identified in this interview is limited land availability, particularly in the Netherlands, where space constraints are a significant challenge (*Planning & Spatial*). This issue is further compounded by the complexity of implementing NbS in areas located below sea level, where physical and environmental conditions impose additional constraints. Climate change intensifies these challenges, as increasing variability and fluctuations in environmental conditions make planning and implementation more uncertain and complex.

"So it's very challenging over there to manage the water levels. And yeah, basically because we see bigger fluctuations in rivers due to climate change, extra safety measures are, yeah, are sought after"

Another barrier identified is the mismatch between the long-term nature of NbS and the preference for short-term results, particularly in relation to economic resources (*Financial*).

"Yeah, it's always short term versus long term, because I know a lot of research is showing that nature-based solutions can actually be beneficial if you look in the long term. But on a short term, it costs money to transform an area."

Finally, the interview highlighted terminology and language barriers as a challenge (*Knowledge & Information*). Many of the key concepts related to NbS are developed and communicated in English and are often not translated or adapted to local contexts. As a result, stakeholders may have different interpretations of terms such as Nature-based Solutions, nature-positive projects, nature-inclusive projects, and ecosystem-based approaches. This lack of shared understanding can lead to confusion and disagreements regarding what qualifies as NbS, ultimately hindering alignment and implementation efforts.

"And a lot of people are saying, well, we shouldn't separate them so much. Nature-based solutions is also nature inclusive. And then there are a lot of people who are very strict and saying, no, it's very separate. Nature-based solutions is solutions based. So that means it's often a big project, whereas nature inclusive measures can be very tiny, can also just be putting an insect hotel in your garden. So, I feel like that might also be a little bit of a barrier."

These barriers can be classified within the following sub-categories:

- Mismatch between short-term investment horizons and long-term economic returns (Financial)
- Limited land availability and plot size constraints (Planning & Spatial).
- Environmental and biophysical constraints on NbS implementation (Planning & Spatial).
- Low perceived relevance and understanding of NbS among stakeholders (Social, Engagement & Perception)
- Limited awareness and understanding of NbS (Knowledge & Information)

4.2.3.3. Enablers identified

One key enabler identified in this interview is the visibility of climate change, which has contributed to a shift in public perception and awareness (*Social, Engagement & Perception*). In the Netherlands, this is particularly evident in the paradox of experiencing both excess water and water scarcity, which has made climate-related challenges more tangible for society. As a result, there has been a growing recognition of the need to rethink the relationship with water and to adopt more adaptive approaches, including NbS.

Complementing this, the presence of strong and well-established water governance institutions, such as water boards, plays a fundamental role in enabling NbS implementation (*Governance & Institutional*).
"We cannot always be draining water. We also need to start retaining. So slowly, more and more water is being retained in the Netherlands. And also, this is where the water boards come in."

Another enabler mentioned is the recognition at the state level of the value of NbS (*Governance & Institutional*). The interview highlighted how public authorities increasingly acknowledge that the conservation and restoration of ecosystems, particularly river systems, generate significant environmental and societal benefits. This growing institutional recognition has positioned NbS as a viable and necessary approach.

"We have a problem with biodiversity loss. We have a problem with water shortage but also flooding events. Climate change is unpredictable. All of this comes together, and nature-based solutions are promising results in all aspects."

These enablers can be classified within the following sub-categories:

- Supportive policy frameworks and regulations (Governance & Institutional)
- Supportive governance structures to NbS processes (Governance & Institutional)

- Enhanced public awareness and capacity building for NBS (*Social, Engagement & Perception*)
- Identifying and communicating benefits, co-benefits of NbS (Knowledge & Information)

4.2.3.4. What is mostly needed in the regional context

Regarding what is most needed in the regional context, the interview emphasized the importance of time as a critical enabling condition. Given the long-term nature of NbS, sufficient time is required for projects to be implemented, monitored, and to generate visible results. Allowing this time is essential to demonstrate the effectiveness of NbS, build confidence among stakeholders, and support their continued adoption and scaling.

"So we need time for monitoring of projects that we have done. We need more research showing this was a really good move, this is the way to go. and maybe even couple it with finances, show how much money you save in the end. Yeah. Because there are projects running, but yeah, it could be increased, it could be scaled up."

4.2.3.5. Suggested tools, frameworks, or methodologies

As a suggested resource, the interviewee highlighted the NL2120 webpage, which compiles case studies and examples of Nature-based Solutions (NbS) in the Netherlands.

"I wouldn't say it's like a good toolbox, but this is an organization, it's more of a strategic organization looking into the future of the Netherlands. And that organization has made the case also for scaling up nature-based solutions. So they are compiling all kinds of projects and they are showing an overview of it. It's not yet a toolbox, but it is a start"

4.2.4. GN2

4.2.4.1. Context

The fourth implementing organization interviewed is a company responsible for the transmission and operation of the electricity system in Spain. One of its core functions is infrastructure maintenance. As part of its legal obligations, the company must manage safety corridors, areas beneath power lines that must meet specific vegetation conditions to reduce fire risk.
In 2021, the company initiated a Nature-based Solutions (NbS) project that integrates extensive livestock grazing as a method for vegetation control. This approach complements traditional mechanical maintenance practices, contributing to both fire prevention and the promotion of more sustainable land management practices.

This project is implemented in collaboration with a company that provides integrated agricultural and environmental services, acting as an intermediary between the power grid operator and livestock farmers. Additional stakeholders include the municipality and the regional government, reflecting a multi-actor governance structure.

Through this arrangement, extensive livestock grazing contributes to maintaining vegetation within safety corridors at optimal levels, supporting both infrastructure safety and biodiversity enhancement.

4.2.4.2. Barriers identified

One of the main barriers identified in this interview is the need to align the NbS with strict regulatory and operational requirements (*Governance & Institutional*) and the reluctancy among actors to accept

these kinds of solutions *(Social, Engagement & Perception).* As the project is embedded within a highly regulated infrastructure system, there were initial concerns regarding whether the proposed solution would comply with existing standards, safety conditions, and maintenance protocols. This uncertainty led to reluctance among the personnel traditionally responsible for maintaining these areas. As a result, a key challenge was demonstrating that the NbS approach met regulatory requirements and could ensure the same level of safety and reliability as conventional practices.

"So, the main challenge was convincing our colleagues, that is, the company's facility maintenance team, to accept the project."

Another major barrier identified is the limited availability of extensive livestock systems required for the implementation of this NbS (*Planning & Spatial)*. The interviewee noted that intensive livestock farming predominates, resulting in a smaller number of farmers practicing extensive grazing. This creates a mismatch between the locations of power line corridors and the availability of suitable farmers, as not all areas have nearby actors capable of participating in the project.

Additionally, the technical requirements of the project, such as specific grazing conditions and operational constraints, are not accepted by all farmers, further limiting participation (*Technical & Capacity*). These factors together constrain the scalability and feasibility of the solution.

"Farmers always raise the issue of mobility with us. This mobility refers to movement within their own municipality or movement between autonomous communities, because in our case, there is a lot of legislation, particularly between autonomous communities."

These barriers can be classified within the following sub-categories:

- Restrictive regulations and conventional planning approaches (Governance & Institutional)
- Path dependency and resistance to change in planning and decision-making practices (Social, Engagement & Perception)
- Limited land availability and plot size constraints (Planning & Spatial)

4.2.4.3. Enablers identified

One of the key enablers identified in this interview is the presence of strong individual commitment to the project (*Social, Engagement & Perception)* The interviewee highlighted how the conviction of a specific team member in the value and potential of the project played a crucial role in driving its implementation and success.

Another important enabler has been the role of the intermediary organization (*Social, Engagement & Perception; Technical & Capacity*). This organization supported the process by engaging and convincing farmers to participate, as well as by developing practical tools, such as implementation guidelines and mechanisms to calculate payments for farmers. These contributions helped reduce uncertainty, structure the process, and enable more effective collaboration among stakeholders.

"The other factor is this intermediary company. We've also been lucky, maybe, I don't know. The profile of the company we work with is also important because it provides a guide."

Additionally, in the interview it was mentioned that a key driver is the active engagement and empowerment of livestock farmers and maintenance personnel (*Social, Engagement & Perception*). While the relationship with farmers is often facilitated through the intermediary organization, the project also includes participatory mechanisms such as workshops and thematic sessions to address concerns, answer questions, and incorporate stakeholder perspectives.

The interview emphasized the importance of giving farmers and the workers that do the maintenance in the power lines a leading role in the project. Fostering a sense of ownership, belonging, and shared

responsibility among both groups was identified as essential for achieving a balanced and collaborative process. This inclusive approach strengthens stakeholder commitment and supports the long-term sustainability of the initiative.

"We have a good exchange with them, a relaxed chat, and then every year we hold a meeting, a conference focused on a specific theme. For example, last year we talked about ecosystem services, so we had a day-and-a-half conference where we invited livestock farmers to participate."

Another key enabler identified in this interview is the use of a pilot project approach combined with systematic monitoring and evaluation (*Technical & Capacity*). The initiative was initially implemented as a pilot to test its feasibility, during which data were collected to assess its impact using technologies such as drones and monitoring systems. This process also enabled the evaluation of biodiversity benefits associated with the intervention.

The evidence generated through the pilot phase contributed to building credibility and reducing uncertainty, facilitating the replication of the project in other regions of Spain. Building on these results, a feasibility study is currently planned to support the potential scaling of the initiative at the national level.

"We have replicated this... and they are now convinced that, well, it's a project that fits well with maintenance, with the upkeep of the installations."

Finally, the interview highlighted the importance of recognizing non-monetary and co-benefits as key drivers of NbS implementation (*Financial; Knowledge& Information*). Although the initiative is not driven by profit generation, it was noted that the costs of the NbS approach are comparable to those of traditional maintenance methods. The value of the project therefore lies in its intangible benefits, which are strongly recognized by those involved.

These benefits are linked to a close connection with the land and the local community, as well as to the broader contribution of the project to territorial development. In addition, the initiative has enhanced the perceived value of extensive livestock farming by demonstrating its role beyond food production, particularly in delivering ecosystem services. This broader value proposition strengthens the relevance and acceptance of the NbS approach.

"But of course, it's an intangible, that is, what's interesting for a value that we see and that our colleagues in line maintenance see, which is also why they liked it, that is, they like the project, they want to integrate it, is the closeness you have to the territory with the people of that municipality"

These enablers can be classified within the following sub-categories:

- Education and training programs (Technical & Capacity)
- Delivery of multiple benefits (Technical & Capacity)
- Learning-by-doing approach (Technical & Capacity)
- The integration of technologies and tools in NbS (Technical & Capacity)
- Effective monitoring, maintenance and evaluation systems and protocols (Technical & Capacity)
- Guidelines and standards for NbS (Technical & Capacity)
- Technical Knowledge and Expertise in implementing NBS Projects (Technical & Capacity)
- Combining NBS with traditional approaches (Planning & Spatial)
- Stakeholders' engagement and collaboration in NbS planning and development (Social, Engagement & Perception)
- Partnerships among Stakeholders (Social, Engagement & Perception)
- Participatory methods and approaches (Social, Engagement & Perception)
- Identifying and communicating benefits, co-benefits of NbS (Knowledge & Information)

4.2.4.4. What is mostly needed in the regional context

Regarding what is most needed in the regional context, the interview emphasized the importance of maintaining and revitalizing extensive livestock farming systems. The interviewee noted a growing trend of rural abandonment, driven in part by generational shifts and a lack of interest among younger populations in working in the countryside. This poses a significant risk to the continuity of NbS initiatives that depend on these land-use practices.

As a result, creating incentives and opportunities for younger generations to engage in rural livelihoods was identified as a critical need. Supporting this transition is essential not only for sustaining extensive livestock systems but also for ensuring the long-term viability of NbS that rely on these socio-ecological practices.

4.2.4.5. Suggested tools, frameworks, or methodologies

The interview highlighted technology as a key supporting tool for NbS implementation (*Technical & Capacity*). In particular, the use of drones for vegetation monitoring and control, as well as digital systems for tracking and managing livestock, was identified as valuable for improving efficiency and oversight. These tools support more precise management of the intervention and enhance the capacity to monitor outcomes over time.

4.2.5. GN3

4.2.5.1. Context

The fifth implementing organization interviewed is an education and research institution based in the United States that has been leading a Nature-based Solutions (NbS) project since 2021, with the goal of translating research into conservation action. The project focuses on supporting biodiversity conservation practices on working lands, collaborating directly with farmers in Virginia, where a significant proportion of the landscape is dedicated to agriculture. In this context, engaging landowners is essential for advancing conservation efforts.

4.2.5.2. Barriers identified

One of the main barriers identified in this interview is limited human and organizational capacity to keep up with existing demand (*Technical & Capacity; Governance & Institutional*). The interviewee noted that, although there is strong interest and willingness among farmers to collaborate, the organization needs personnel and resources to effectively scale its activities and cover more ground. Given the large number of farmers and rural landowners in Virginia, the extent of the territory to be covered poses a significant challenge, limiting the organization's ability to expand its reach and impact.

I think the number one challenge is keeping up with demand for what we do. You know, we don't call farmers. We don't have to go knock on doors. There's a demand for support, for technical assistance, for knowledge, for collaboration. And so, you know, at the end of the day, Capacity equals conservation. So, the more people we can have on staff, the more we can do.

Another barrier identified is the instability in funding and capacity from some of their partners, influenced by changes in the broader political and administrative context (*Financial; Governance & Institutional*). The interviewee noted that shifts in the current, more volatile national administration affect funding availability and reliability, which in turn impacts the capacity of partner organizations and affects the project.

"So, money is moving, it's getting cut, it's unreliable. And so, when all of a sudden, we don't have

our partners there, some of our programming is very partner dependent, and so that is challenging."

These barriers can be classified within the following sub-categories:

- Lack of long-term political commitment (Governance & Institutional)
- Misalignment between national and local policies and decisions (Governance & Institutional)
- Lack of funding or investment (Financial)
- Limited access to and mechanisms for mobilizing investment (Financial)
- Dependence on external funding (donors, public) (Financial)

4.2.5.3. Enablers identified

The project is managed through a partnership of five organizations from different disciplines, and the interviewee highlighted the value of this multidisciplinary collaboration, particularly in terms of combining diverse expertise and networks (*Social, Engagement & Perception; Governance & Institutional*). Additionally, funding is sourced from multiple channels, which was identified as an important enabling factor, as it reduces dependency on a single funding source and enhances the project's financial stability (*Financial*).

"And the idea there is that we all bring different networks of people, different expertise, different funding. And also if something happens to funding to one of our organizations, the initiative doesn't just hit a wall and stop. You know, there's still all these other groups pushing it forward"

Another enabler identified in this interview is the strong commitment and intrinsic motivation of the individuals involved in the project (*Social, Engagement & Perception*). The interviewee emphasized that the passion for conservation and the shared belief in the project's goals play a key role in driving its progress. This high level of engagement and dedication among team members and partners of the project acts as an important enabling factor, supporting collaboration and resilience.

"The people are everything, right? At the end of the day, if the people's hearts aren't in it, if the passion isn't there, if the drive, the work ethic isn't there, then nothing's going to happen. So yeah, that's, that's, that's the big advantage right now."

Additionally, the interview highlighted that strong relationships with farmers can act as both an enabler and a challenge (*Social, Engagement & Perception; Financial*). On one hand, farmers are generally willing to collaborate and express a genuine interest in doing what is best for their land, which facilitates engagement. On the other hand, uncertainty regarding potential profit loss can discourage the adoption of conservation practices. To address this, the organization works to demonstrate practices that generate both environmental and economic benefits, helping to reduce perceived risks. Building trust and clearly communicating shared goals are essential in this process, reinforcing the idea that conservation and productivity can be aligned rather than conflicting objectives.

"So the farmers are fantastic. It's really on us to get creative in how we build those relationships and to make sure they understand we're there to work with them alongside them and to listen and to make sure we're trying to answer questions that they have, not just our questions, but where do they need support? So there's a lot of that. Yeah. So the farmers are a fantastic ally in all this."

Another enabler identified in this interview is the development of strong, trust-based relationships with farmers, which has facilitated access and collaboration (*Social, Engagement & Perception*). The interviewee highlighted that these relationships have enabled the organization to engage farmers in conservation practices in a way that is perceived as positive rather than threatening. In particular, the return of wildlife has been well received by farmers, who value the visible ecological improvements resulting from their efforts.

These positive outcomes not only reinforce farmer engagement but also help build broader support for NbS initiatives. As a result, the organization's work has generated a ripple effect, opening the door for additional stakeholders and encouraging other organizations to replicate or expand similar initiatives.

"And so what we've realized is that because we have this really unique way of getting onto farms and having farmers open their doors to us, which is, you know, the biggest thing for them to let us on their property and to start having these conversations is that we're able to then open that door to a lot of other partners. Those can be state and federal agencies. They might be other land trusts. They might be invasive species groups. They might be cooperative extension. There's all these other great groups who have all these other skill sets and free money and free help that want to help farmers. But somebody's got to make that initial contact and create a relationship and open that door. So we found that we're pretty good at doing that."

Finally, the interview highlighted the project's ability to integrate research, conservation design, and implementation within a single process (*Knowledge & Information*). By conducting research in-house and directly translating findings into conservation practices and on-the-ground implementation, the organization maintains control over the entire process. This integration enables a more efficient transition from knowledge generation to action, reducing delays and ensuring that research outcomes are rapidly applied in practice.

"It's really fun to have all this happening in here, which means we can move pretty fast, which I think, most people would agree that we need to be a little faster with conservation because things are changing quickly. And so we don't want to sit on this research for 15 years and think about it. We want to put it into action, which is great."

These enablers can be classified within the following sub-categories:

- Supportive governance structures to NbS processes (Governance & Institutional)
- Collaboration across disciplines and governance levels (Governance & Institutional)
- Financial feasibility and economic sustainability of NbS (Financial)
- Public-private partnerships (PPP) (Financial)
- Economic valuation of NbS benefits (Financial)
- Technical Knowledge and Expertise in implementing NBS Projects (Technical & Capacity)
- Learning-by-doing approach (Technical & Capacity)
- Delivery of multiple benefits (Technical & Capacity)
- Stakeholders' engagement and collaboration in NbS planning and development (Social, Engagement & Perception)
- Partnerships among Stakeholders (Social, Engagement & Perception)
- Participatory methods and approaches (Social, Engagement & Perception)
- Creation and nurture of NbS networks (Social, Engagement & Perception)
- Identifying and communicating benefits, co-benefits of NbS (Knowledge & Information)
- Effective knowledge translation and dissemination (Knowledge & Information)
- Availability of reliable data on NbS (Knowledge & Information)

4.2.5.4. What is mostly needed in the regional context

Regarding what is most needed in the regional context, the interview emphasized the importance of stable and sustained funding, particularly to support staff capacity (*Financial; Technical & Capacity*). Financial resources were identified as essential for expanding the team and meeting the growing demand for engagement with farmers and landowners.

The interviewee noted that many NbS initiatives rely heavily on grants, and the process of continuously applying for funding is time-consuming, resource-intensive, and often uncertain. This creates challenges for maintaining continuity, as funding cycles frequently prioritize novelty over the

continuation of effective ongoing work. As a result, more consistent and long-term funding mechanisms were identified as critical to sustaining and scaling NbS initiatives.

"You know, in the grant world, everybody always wants to see something new. They always want to fund something new. And that's a huge problem because sometimes what you're already doing just needs continued support and you're already doing great things. And the pressure to create something new every time is exhausting because then you keep adding more to your plate and then you do a worse job at all these things, right? Because you have too many things to do. Funding that is consistent and that pays groups to continue just doing the good work they do."

4.2.5.5. Suggested tools, frameworks, or methodologies

When asked about suggested tools or frameworks, the interviewee did not identify any specific formal methodology. Instead, they emphasized the need for regional or state-level coordination mechanisms to support NbS implementation (*Governance & Institutional*). In particular, the interview highlighted the importance of having dedicated coordinators responsible for overseeing projects, aligning different groups, securing funding, and communicating opportunities across geographically distributed stakeholders.

The organization already works through structures referred to as "regional conservation partnerships" which operate at smaller geographic scales (e.g., specific landscapes such as mountain ranges) and involve collaboration among local actors. However, the interviewee noted that these partnerships would benefit from stronger coordination and support, particularly in terms of resource mobilization and strategic alignment. Strengthening these coordination roles was identified as key to ensuring continuity of these kinds of initiatives.

"But what would be great is to have a bigger coordinator that helps support all of those and finds bigger money to funnel it down to all those small groups, you know, because that kind of support would just be invaluable to have somebody at a higher level really advocating for all of us, you know. So we've dreamed about that. I just don't know what organization would do that. We don't have that figured out yet, but in theory it could. It could work that way".

4.2.6. Differences between the GN and the GS identified in the interviews

The interviews also sought to identify whether the interviewees recognized potential differences between the Global South (GS) and the Global North (GN) regarding the implementation of NbS. This section synthesizes some of the main differences identified throughout the interviews, which emerged primarily during interview GN1.

The interview GN1 highlighted several differences in NbS implementation between the Global North and the Global South, many of which are shaped by the specific characteristics of the territory. In the case of the Netherlands, NbS are predominantly oriented toward water management. In addition, spatial conditions play a significant role. The Netherlands is a highly densely populated country, where limited land availability constrains the design and implementation of NbS.

"It's very much always water related. So, nature-based solutions outside of the water management are rare."

Another difference identified, which can also function as an enabler, is the presence of environmental legislation, particularly in European contexts. The interviewee noted that regulatory frameworks often require institutions to incorporate environmental considerations into their projects, and in some cases, large-scale initiatives that appear to be strongly environmentally oriented are, in fact, driven by compliance with existing laws and regulations. This highlights the role of policy frameworks as both a driver and an enabling condition for NbS implementation.

"We have, especially in Europe, we have a lot of legislation. Yeah. So, nature-based solutions also overlaps a little bit with the Water Directive framework. just an end to nature restoration law. So, we're also by law required to do more. And I'm not sure if the Global South has as much legislation on nature and water regulation."

Interview GN2 highlighted similarities in the challenges faced within livestock farming systems, which the interviewee identified during a recent international grazing event, although no specific comments were made directly regarding NbS. In turn, interview GN3 suggested that some of the differences between regions may be related to the relationship each country has with farming, as well as the relationships established between farmers and other stakeholders involved in these processes. Finally, interview GS2 emphasized that one of the main differences between the regions lies in biodiversity, highlighting that the Global South, particularly countries such as Colombia, possesses some of the highest levels of biodiversity in the world, whereas many countries in the Global North have historically exploited and significantly depleted much of their biodiversity.

4.3. Toolkit

Based on the literature review process, several tools, methodologies and frameworks were identified for inclusion in the proposed toolkit. One of the most relevant findings is the Connecting Nature Resource Centre , developed as part of the Connecting Nature project. This platform compiles a wide range of resources aligned with the project's framework, and consequently, several of the tools included in this toolkit are drawn from this source and by following a snowballing effect from this source.

In addition, the literature review identified other resource centres and concepts that were incorporated into the toolkit, broadening the range of available materials and approaches. To complement these resources, insights from the interviews were considered.

Rather than proposing entirely new tools, this toolkit represents a curated and structured integration of existing resources, adapted to support NbS practitioners in navigating the complexity of implementation across different regional contexts. In particular, it aims to bridge gaps between theory and practice, and to provide a practical guide to achieve the enablers or address the barriers identified in the literature.

The toolkit is provided in Table 6, each tool is accompanied by a code in parentheses indicating the initial source where it was mentioned. In some cases, this code refers either to the article(s) included in the literature review (with corresponding references provided in Appendix 2) or to the interview in which the tool was identified.

In most of the cases, tools were identified through a snowballing process that originated from the **Connecting Nature Resource Centre**, which was initially identified in article 7 of the literature review. This platform led to additional resource centres that were also incorporated into the toolkit. In such cases, the initials **CNRC** are used to denote the initial source.

The decision to incorporate resources from the Connecting Nature Resource Centre and similar repositories was based on the fact that they already provide practical tools and methodologies specifically designed for NbS implementation. This is particularly beneficial for practitioners, as these tools can be applied more directly within implementation processes, potentially facilitating greater accessibility, usability, and acceptance among NbS implementers.

Finally, certain tools are not associated with a source code, indicating that they were identified through grey literature. These tools were incorporated to complement the toolkit and to address a wider range of enablers identified throughout the research. In the barriers column in brackets the

initials of the category can be found as some barriers respond to enablers from different categories.

Table 6. Practical toolkit

Enablers	Tools	Barriers
Governance & Institutional (G&I)		
1. Supportive policy frameworks and regulations	Urban Governance Atlas (CNRC)	Lack of clear policies and regulatory frameworks (G&I)
2. Collaboration across disciplines and governance levels	Guidelines for Co-Creation and Co-Governance of Nature-based Solutions -Insights from EU-funded Projects (CNRC) Stakeholder mapping (42) Co-production guidebook (CNRC)	Poor coordination across stakeholders and governance levels (G&I)
3. Supportive governance structures to NbS processes	Adaptive Governance for Successful Urban Nature-based Solutions (CNRC) Aligning nature-based solutions with strategic priorities: a practical guide (CNRC) Co-production guidebook (CNRC) Governance Guidebook (CNRC)	Misalignment between national and local policies and decisions (G&I) Fragmented and siloed institutional structures (G&I)
4. Strong and sustained political will and support	Aligning nature-based solutions with strategic priorities: a practical guide (CNRC)	Restrictive regulations and conventional planning approaches Lack of long-term political commitment (G&I) Weak enforcement of existing policies (G&I)
5. Inclusive and participatory governance processes	Guidelines for Co-Creation and Co-Governance of Nature-based Solutions -Insights from EU-funded Projects (CNRC) Co-production guidebook (CNRC) Governance Guidebook (CNRC)	Poor coordination across stakeholders and governance levels (G&I) Lack of long-term political commitment (G&I) Fragmented and siloed institutional structures (G&I)
Financial (F)		
1. Clear and appropriate business models	Capturing the Values and Making the Business Case for Nature-Based Solutions - a step by step guide (CNRC) Nature-Based Solutions Business Model Canvas (CNRC)	Lack of viable business models for NbS (F) Lack of funding or investment (F) High upfront costs of NbS (F) Mismatch between short-term investment horizons and long-term economic returns (F) Difficulty monetizing NbS benefits (F)

Enablers	Tools	Barriers
2. Financing mechanisms, economic instruments and incentives	Financing and Business Models Guidebook (CNRC) Time contributions (28, 35) BIOFIN - The Biodiversity Finance Initiative Workbook 2018 (CNRC)	Lack of funding or investment (F) Limited access to and mechanisms for mobilizing investment (F) Lack of financial incentives (subsidies, tax benefits) (F) Dependence on external funding (donors, public) (F)
3. Financial feasibility and economic sustainability of NbS	Financing and Business Models Guidebook (CNRC)	High upfront costs of NbS (F) Mismatch between short-term investment horizons and long-term economic returns (F)
4. Economic valuation of NbS benefits	Capturing the Values and Making the Business Case for Nature-Based Solutions - a step by step guide (CNRC) Formulate value propositions (7)	Difficulty monetizing NbS benefits (F) Prioritizing the allocation of financial resources for conventional projects or other challenges (F)
5. Public-private partnerships (PPP)	Public-private partnerships (PPP) (7, 18, 20, 23, 26, 39, 76, 80, 82, 84)	High upfront costs of NbS (F)
Planning & Spatial (P&S)		
1. Integration of NbS into planning processes	Nature4Cities Platform (CNRC)	Difficulty integrating NbS into legal and planning systems (P&S) Environmental and biophysical constraints on NbS implementation (P&S)
2. Adopting an integrated, context-specific and data-driven approach in planning and implementation n	Database of good practice tools for the design and implementation of NbS, including factsheets for 10 tools (CNRC)	High population density and land-use pressure (P&S)
3. Combining NbS with traditional approaches	System thinking Adaptive Management Living Labs (Chroneer et al., 2019; Lupp et al., 2021) (13)	High population density and land-use pressure (P&S)
4. Land availability, acquisition and management strategies	GIS and Spatial Planning Tools (37)	Limited land availability and plot size constraints (P&S) Land tenure insecurity (P&S) Conflicts with existing land uses and management (housing, infrastructure) (P&S) High population density and land-use pressure (P&S)
5. Supportive planning rules and practices to NbS	Three layers framework (GS2) Planning framework by Albert et al. (2021) (37)	Environmental and biophysical constraints on NbS implementation (P&S)

Enablers	Tools	Barriers
	Connecting Nature Framework (7) Framework for Planning and Evaluation of Nature-Based Solutions for Water in Peri-Urban Areas by de Lima et al. (2022) (16) Database of good practice tools for the design and implementation of NbS, including factsheets for 10 tools (CNRC)	Conflicts with existing land uses and management (housing, infrastructure) (P&S)
Technical & Capacity (T&C)		
1. Technical Knowledge and Expertise in implementing NBS Projects	Report on NBS professional Skill Gaps (CNRC)	Limited technical skills, training and capacity building for NbS (T&C)
2. Guidelines and standards for NbS	Global Standard for Nature-based Solutions (21) CONEXUS Factsheets (CNRC) Urban Nature Gateway (CNRC) Impact Assessment Guidebook (CNRC)	Lack of standardized guidelines and metrics (T&C)
3. Effective monitoring, maintenance and evaluation systems and protocols	Global Standard for Nature-based Solutions (CNRC) Digital twins (13, GS2) Use of technology (AI, monitoring devices and drones) (GS1, GS2 and GN1) Checklist for Nature-based Solutions Monitoring & Indicators (CNRC) Monitoring NbS together for greater uptake and upscaling (CNRC) Database of good practice tools for the design and implementation of NbS, including factsheets for 10 tools (CNRC) A practical guide to using reflexive monitoring for nature-based solutions (CNRC, 7) Reflexive Monitoring Guidebook (CNRC, 7)	Limited monitoring, maintenance and evaluation systems (T&C)
4. Learning-by-doing approach	Multi-actor labs (Tiller et al., 2021)(13) Living Labs (Chroneer et al., 2019; Lupp et al., 2021) (13)	Limited technical skills, training and capacity building for NbS (T&C) Research-practice gap (K&I)

Enablers	Tools	Barriers
	Reflections for establishing Urban Life- Labs for governance of Nature-based Solutions Life-labs (CNRC) CONEXUS Factsheets (CNRC) NL2120 (GN1)	
5. Delivery of multiple benefits	Capturing the Values and Making the Business Case for Nature-Based Solutions - a step by step guide (CNRS) Formulate value propositions (7)	Difficulty monetizing NbS benefits (F) High upfront costs of NbS (F)
6. The integration of technologies and tools in NbS	Database of good practice tools for the design and implementation of NbS, including factsheets for 10 tools (CNRC) Digital twins (13, GS2)	Limited access to appropriate technologies and infrastructure for NbS
7. Education and training programs	Urban Nature Gateway (CNRC)	Limited technical skills, training and capacity building for NbS (T&C)
Social, Engagement & Perception (S, E&P)		
1. Involvement/engagement of the community in the NbS process	Guidelines for Co-Creation and Co-Governance of Nature-based Solutions -Insights from EU-funded Projects Stakeholder mapping (42) Co-production guidebook (CNRC) Citizen science (Restemeyer & Boogaard, 2021) (13) The Multi-Stakeholders Partnerships Guide	Lack of participation or engagement/resistance from community or stakeholders (S,E&P) Conflicts between stakeholders / inequitable distribution of benefits (S,E&P)
2. Stakeholders' engagement and collaboration in NbS planning and development	Guidelines for Co-Creation and Co-Governance of Nature-based Solutions -Insights from EU-funded Projects (CNRC) Stakeholder mapping (42) Co-production guidebook (CNRC) The Multi-Stakeholders Partnerships Guide	Lack of participation or engagement/resistance from community or stakeholders (S,E&P) Conflicts between stakeholders / inequitable distribution of benefits (S,E&P)
3. Partnerships among Stakeholders	Public-private partnerships (PPP) (7, 18, 20, 23, 26, 39, 76, 80, 82, 84) Stakeholder mapping (42) The Multi-Stakeholders Partnerships Guide	Lack of participation or engagement/resistance from community or stakeholders (S,E&P) Conflicts between stakeholders / inequitable distribution of benefits (S,E&P)

Enablers	Tools	Barriers
4. Participatory methods and approaches	Stakeholder mapping (42) A practical guide to using co-production for nature-based solutions (CNRC) Co-production guidebook (CNRC)	Lack of participation or engagement/resistance from community or stakeholders (S,E&P) Conflicts between stakeholders / inequitable distribution of benefits (S,E&P)
5. Enhanced public awareness and capacity building for NBS	Knowledge repositories (37, 42, 82) Effective Storytelling for Nature-based Solutions (CNRC)	Preference for grey infrastructure, conventional solutions, and other alternatives (S,E&P) Low perceived relevance and understanding of NbS among stakeholders (S,E&P)
6. Creation and nurture of NbS networks	City-to-city networks (42) Connecting Nature Enterprise Platform (CNRC)	Lack of participation or engagement/resistance from community or stakeholders (S,E&P)
Knowledge & Information (K&I)		
1. Integration of local knowledge and conditions	Guidelines for Co-Creation and Co-Governance of Nature-based Solutions -Insights from EU-funded Projects (CNRC) Stakeholder mapping (42) Co-production guidebook (CNRC) Citizen science (Restemeyer & Boogaard, 2021) (13) Participatory Rural Appraisal	Lack and underutilization of local and context-specific knowledge (K&I) Limited awareness and understanding of NbS (K&I)
2. Knowledge sharing mechanisms and technologies	Knowledge repositories (37, 42, 82)	Limited knowledge sharing, transfer, and management across sectors and institutions (K&I) Limited awareness and understanding of NbS (K&I)
3. Identifying and communicating benefits, co-benefits of NbS	Effective Storytelling for Nature-based Solutions (CNRC) Formulate value propositions (7)	Limited evidence on NbS effectiveness and long-term benefits (K&I) Limited awareness and understanding of NbS (K&I) Path dependency and resistance to change in planning and decision-making practices (S,E&P) Competing socio-economic priorities over environmental goals (S,E&P)
4. Knowledge production and co-production	A practical guide to using co-production for nature-based solutions (CNRC)	Lack of data on NbS implementation and performance (K&I) Limited awareness and understanding of NbS (K&I)
5. Effective knowledge translation and dissemination	Effective Storytelling for Nature-based Solutions (CNRC)	Lack of data on NbS implementation and performance (K&I)

Enablers	Tools	Barriers
		Limited awareness and understanding of NbS (K&I)
6. Availability of reliable data on NbS	Knowledge repositories (37, 42, 82)	Lack of data on NbS implementation and performance (K&I) Limited awareness and understanding of NbS (K&I)

One of the key findings of this research is the availability of knowledge repositories that provide tools and resources for the implementation and scaling of NbS. Given the time limitations and scope of this thesis, it was not feasible to explore all of these repositories in detail or to integrate them fully into the toolkit. Nonetheless, a list of these repositories is included in Appendix 3, and NbS stakeholders are encouraged to consult them to further complement and expand upon the toolkit proposed in this study.

It is important to note that the toolkit developed within this research should be understood as an initial prototype rather than a definitive or exhaustive resource. Due to the scope and time limitations of this study, it was not possible to incorporate all the available tools and methodologies related to NbS implementation. Therefore, the toolkit is intended to function as a flexible and evolving resource that can continue to be expanded and complemented with additional tools, frameworks, and methodologies in future research and practice.

Nevertheless, the development of this toolkit demonstrates that it is possible to design a practical resource for NbS implementation based on the analysis of existing tools, methodologies, and frameworks. At the same time, the process also revealed the extensive amount of information currently available across different repositories, disciplines, and even languages, highlighting that the construction of a more comprehensive toolkit requires significant time, continuous exploration, and interdisciplinary collaboration. In this sense, the toolkit developed in this research aligns with broader efforts to operationalize and standardize NbS implementation processes, such as those promoted through the IUCN Global Standard for Nature-based Solutions. However, one of the key findings of this research is that NbS are highly context-specific, meaning that a more robust toolkit would ideally continue incorporating tools adapted to different regional realities, governance contexts, and types of NbS interventions.

5. Discussion

One of the main purposes of this research was to identify whether there are differences between the barriers and enablers of NbS in the Global North (GN) and the Global South (GS). In this sense, several insights emerged from the results, confirming that differences do exist in some of these barriers and enablers. At the same time, differences were also observed between the distribution of barriers and enablers within each category, as well as in the way some enablers appear to respond directly to the identified barriers. This relationship can be observed in the toolkit through Table 6, while Appendix 4 presents the complete comparative table between barriers and enablers.

In general terms, one of the main differences identified in the literature between the GN and the GS is the limited number of barriers and enablers coded exclusively within the Global North. Initially, this can be explained by the smaller sample of GN-specific articles compared to GS-focused articles, which represents one of the limitations of this research. However, this also raises questions regarding why the sample produced fewer region-specific findings for the GN, considering that, according to Villamayor-Tomas et al., (2024) and McPhearson et al., (2025), most NbS research is produced in the Global North. This trend is also consistent with the existing N-S divide in relation to broader environmental science topics, such as findings by Blicharska et al., (2017) in relation to climate change policy and practice. One possible explanation might be that studies produced in the GN may

not explicitly frame their analyses under distinctions such as “Global North,” “developed countries” or “high-income countries” but instead refer to specific regions or countries such as Europe or the United States.

A notable difference between the barriers and enablers identified in the literature review is that, for enablers, the contrast between the Global North and the Global South is not as pronounced as it is for barriers. In addition, many of the coded segments identified as enablers originated from the articles classified as General, suggesting a greater degree of overlap across regional contexts. On the other hand, more barriers specific to the GS were identified compared to GS-specific enablers. This may be explained by the fact that barriers are often strongly shaped by territorial and regional conditions, particularly by the specific socio-economic, political, and environmental characteristics of the GS. For example, due to rapid urbanization, informal settlements are very present in the GS and are mentioned in the literature as a challenge for implementing NbS (Abera et al., 2025), also the presence of diverse weather conditions as the monsoon season in countries of the GS add challenges to the Planning & Spatial dimension of NbS (Huynh Thi Ngoc et al., 2024) . In contrast, enablers appear to be more transferable across contexts, as a single enabler may address multiple barriers simultaneously. For example, knowledge sharing mechanisms and technologies like the knowledge repositories found for the toolkit can address barriers like the limited knowledge sharing, transfer, and management across sectors and institutions, the limited awareness and understanding of NbS, the lack of data on NbS implementation and performance and the lack of standardized guidelines and metrics.

The GN-GS distinction is also reflected in related literature as Haque and Sharifi, (2025) identify a divide between the Global North and the Global South in terms of urban ecosystem services, where social barriers tend to be more prominent in the North due to societal variations, while institutional barriers are more dominant in the South, reflecting weaker governance capacities. This is partially consistent with the findings of this research, particularly regarding the prominence of Governance & Institutional barriers in the GS literature.

A separate distinction was also made for the coded category labelled “General,” which included segments extracted from broad or cross-regional studies where it was not possible to identify the specific regional origin of the barrier or enabler. A clear difference emerged between the number of general coded segments identified for barriers and those identified for enablers, with a significantly higher number found for enablers. This finding further suggests that barriers to NbS tend to be more context-dependent and shaped by regional conditions, whereas enablers appear to be more transferable across contexts. This may indicate that while the implementation challenges of NbS are deeply embedded in local socio-political realities, many enabling mechanisms rely on broader principles such as collaboration, participation, knowledge sharing, and institutional support.

This discussion section is structured around the six main categories established in the Results section. Additionally, the results from the interviews and the design of the toolkit were considered in this discussion.

5.1. Governance & Institutional

This category contained the highest number of coded segments in terms of barriers; however, it ranked fourth in relation to enablers. Within the interviews, this was the category least frequently recognized when discussing barriers and enablers, being mentioned mainly by GN1, who also works in a governmental institution.

Consistent with the broader pattern identified between barriers and enablers, a higher number of coded segments from the GS was found for barriers, whereas the distribution of enablers across the three categories (GN, GS, and General) was more balanced.

One of the subcategories that showed a similar pattern across both regions was the issue of policy and regulatory frameworks. In both barriers and enablers, the importance of clear policies supporting

NbS implementation was emphasized in the literature from both the GN and the GS. However, a distinction highlighted in the GN1 interview was that, particularly in Europe, there is stronger environmental legislation and regulation that supports the execution of projects in which NbS can be integrated.

In contrast, one of the barriers identified in both regions was restrictive regulations and conventional planning approaches, suggesting that regulations do not always support NbS implementation and may instead hinder it. This was highlighted mostly in the Global South where for example, according to Zhu et al., (2024), regulations related to organic certifications were described as highly restrictive and often favouring conventional methods, creating challenges for market access and the generation of economic benefits. Similarly, in interview GS1, it was mentioned that public policies in Colombia in the past incentivized deforestation in order to expand productive agricultural land.

Within this category, another enabler identified in both regions was strong and sustained political will and support, which is corresponding to the barrier of lack of long-term political commitment, also identified across both regions. In the interview with GN1, this was reflected in the emphasis placed on the role of institutions in the implementation of NbS. Likewise, interview GN3 highlighted how the current political instability in the US has become a barrier to achieving stable and continuous funding for the implementation of NbS.

Several barriers within this category were more prominent in the GS, and although many of these differences were already presented in the Results section, some are particularly important to revisit here as part of the discussion. One of the most predominant barriers in the GS was weak enforcement of existing policies, which is closely linked to broader issues of transparency and trust within institutions (Abera et al., 2025; McPhearson et al., 2025; Zarei and Shahab, 2025). This may also be connected to the comparatively higher levels of perceived corruption in many GS countries when compared to the GN, as illustrated in the Corruption Perceptions Index (2025) and mentioned by Kowalski (2020). This barrier is further reinforced by the identified need for supportive governance structures, an enabler that was also especially prominent in the GS.

Finally, two barriers identified mainly in the GS were poor coordination across stakeholders and governance levels, as well as the misalignment between national and local policies and decisions. In some GS contexts, these coordination challenges may be linked to institutional legacies of colonial and centralized governance systems, which historically prioritized top-down decision-making structures over participatory and cross-sectoral governance approaches (Abera et al., 2025). At the same time, the enablers collaboration across disciplines and governance levels and inclusive and participatory governance processes were identified not only in GS-specific studies but also across several General articles. This suggests that collaborative processes as facilitators of NbS implementation and scaling are recognized across broader contexts.

In this sense, it can be concluded that Governance & Institutional barriers are among the strongest challenges affecting NbS implementation, as these solutions are deeply influenced by decisions made at macro-governance levels. However, in terms of enablers, many of them emerge more strongly at meso- and micro-levels. This pattern is also reflected in the toolkit, where it was difficult to identify tools that directly address macro-governance and institutional barriers, as these issues are often beyond the control of NbS implementers and depend more heavily on policymakers and institutional structures. Instead, many of the enablers and tools identified across the different categories contribute indirectly to reducing these governance gaps by strengthening collaboration, participation, and coordination among stakeholders, ultimately supporting the development of more supportive policy frameworks and governance processes for NbS.

5.2. Financial

The Financial category represents one of the least prominent categories in terms of the number of identified enablers, while at the same time being one of the most prominent in terms of barriers. This suggests that there are multiple financial constraints limiting NbS implementation, whereas the

recognized enablers tend to be more straightforward and mainly focused on securing funding and recognizing the value of NbS so that these solutions are perceived as economically worthwhile. One of the particularities of this category is that, while barriers showed a greater concentration of coded segments in the GS literature, the enablers were more concentrated within the General articles, and the number of coded segments between the GN and the GS literature was relatively similar. This indicates that the financial enablers identified tend to be more general and transferable across regions, with fewer regional distinctions.

Another important aspect of this category is that although the barrier lack of funding or investment was more prominent in the GS literature, in two of the Global North interviews (GN1 and GN3), funding was also highlighted as a key enabling factor for strengthening capacity building and scaling NbS initiatives. In contrast, the Colombian implementers interviewed (GS1 and GS2) did not explicitly identify funding as one of the main challenges. However, both projects currently receive financial support from European countries, specifically Germany and the Netherlands. This finding is consistent with one of the barriers identified particularly in the GS: dependence on external funding (donors, public).

Within this category, two barriers commonly identified in both regions were the high upfront costs of NbS and the mismatch between short-term investment horizons and long-term economic returns. The latter was also mentioned in interview GN1 as one of the main barriers to NbS implementation. These findings reinforce the idea that NbS often struggle to align with dominant economic and political systems that prioritize immediate returns over long-term environmental and social benefits.

One of the barriers that was more prominent in the GS literature was the lack of viable business models. However, in the enablers section, appropriate business models were identified across all three categories (GN, GS, and General). This exemplifies how the development of suitable business models can function as a broad enabling mechanism capable of addressing several of the financial barriers identified across regions. This is also reinforced in the report *Business Models for Financing Nature-based Solutions in Urban Climate Action* (UNEP-CCC, 2024), which highlights that business models provide a practical framework for organizing projects and investment planning while helping to communicate the value creation potential of NbS to investors, stakeholders, and the public. In particular, value-proposition-based business models can help recognize and communicate the environmental, social, and economic benefits generated by NbS, strengthening stakeholder support and improving the recognition of both monetary and non-monetary values associated with these initiatives.

A commonly highlighted mechanism within this category is public–private partnerships (PPPs), which can also be understood as bridging governance and financial dimensions. PPPs were identified in both regions as an effective approach to financing NbS. Although PPPs were categorized within the Financial section because they represent a strategy for mobilizing funding, they are also strongly connected to collaboration processes, which were consistently recognized throughout the literature review as central to NbS implementation and scaling.

It is important to note that although there was a higher overall concentration of financial barriers in the GS, most of the subcategories were also mentioned at least once in GN-focused articles. This suggests that financial constraints remain a general barrier across regions, despite developed countries having a higher GDP per capita (Kowalski, 2020). In this same sense, estimates by Deutz et al., (2020) indicate that financial efforts up to eight times greater than current levels are needed to scale NbS and achieve global biodiversity targets. This represents a major challenge, particularly considering that NbS initiatives are currently financed predominantly through public resources at the global level, creating a strong dependence on this type of funding. Financing challenges are also affected by the difficulty of monetizing the multiple benefits generated by NbS, as well as by perceived risks associated with limited awareness and experience regarding these approaches, aspects that were also identified by Blicharska et al., (2026).

Recent literature on the economic evaluation of NbS further highlights that, despite the increasing

number of studies on the topic, the field remains fragmented and geographically uneven. Most economic evaluations are concentrated in Europe, North America, and East Asia, while studies in regions such as Africa and the Middle East remain limited due to restricted data availability, funding inequalities, and weaker institutional capacities (Jalilzadehazhari, 2026). Additionally, many evaluations focus primarily on initial investment, construction, operation, and maintenance costs, while broader long-term economic dimensions are often overlooked. The literature also identifies substantial variability in the methodologies and metrics used to evaluate NbS benefits, limiting comparability and creating challenges for policymaking and investment decisions (Jalilzadehazhari, 2026). These findings reinforce the importance of developing more comprehensive and standardized approaches for identifying, communicating, and evaluating the economic benefits and co-benefits generated by NbS.

In this context, the economic valuation of NbS benefits emerged as one of the main financial enablers identified throughout this research. This aspect was also reflected in interview GN2, where it was mentioned that the value generated by NbS is often intangible, and that assigning an economic value to these benefits remains a challenge. In response to this issue, the toolkit proposes tools such as value proposition development as a strategy to help recognize, structure, and communicate the multiple benefits generated by NbS, thereby facilitating their commercialization and broader recognition. This need was also highlighted in interview GS1, where the interviewee emphasized the importance of developing commercialization and marketing strategies for products derived from NbS initiatives. In this sense, tools such as clear business models and value propositions can help define pathways for transforming these products and services into sustainable economic opportunities.

5.3. Planning & Spatial

The Planning & Spatial category was highly influenced by the large number of articles focused on urban NbS, and most of the identified subcategories are therefore closely related to this type of intervention. In both barriers and enablers, this category had the lowest number of coded segments. However, there was a strong concentration of barriers within the GS literature, many of which are directly related to the spatial and territorial conditions of the region. In contrast, the identified enablers were more evenly distributed across both regions, with even a slight predominance of GN-focused articles in relation to strategies for land availability, acquisition, and management. These strategies are particularly relevant in contexts such as the Netherlands, where land scarcity represents a major challenge, as highlighted in interview GN1.

Two of the barriers identified mainly in the GS literature were high population density and land-use pressure, as well as difficulty integrating NbS into legal and planning systems. These barriers are strongly connected to the specific urban dynamics of many GS regions, particularly in Asia, where cities are characterized by high population density and rapid urbanization processes (Huynh Thi Ngoc et al., 2024). These dynamics have also contributed to the expansion of informal settlements, which are often excluded from formal planning systems and create additional pressure on land use and spatial planning processes (Mirsafa, Castaldo and Lemes De Oliveira, 2025).

Additionally, another barrier that was particularly prominent in the GS was environmental and biophysical constraints. However, interview GN1 also highlighted a major biophysical challenge in the Netherlands related to water management and the country's geographical condition of being below sea level. This suggests that although environmental and biophysical constraints were more prominent in the GS literature, these types of challenges are not exclusive to the GS and may emerge differently depending on the territorial and ecological conditions of each context.

One of the most interesting findings emerging from the literature review is that one of the potential advantages identified in the GS is that many infrastructure systems are still under development. As a result, integrating NbS into evolving infrastructure systems may, in some cases, be easier than retrofitting NbS into highly consolidated and already developed systems, as is often the case in the GN (Abera et al., 2025). This finding suggests that, despite the multiple spatial and planning-related

barriers identified in the GS, these rapidly transforming urban and infrastructural contexts may also create opportunities for the integration of NbS from earlier stages of development.

In this sense, Planning & Spatial barriers appear to be particularly context-dependent, as they are strongly shaped by local territorial, ecological, and urban conditions. Unlike other categories where similar barriers emerge across regions, many of the challenges identified in this category are directly linked to the specific spatial realities of each territory.

Finally, similar to the Governance & Institutional category, fewer tools were identified for the toolkit within the Planning & Spatial category, as several of the barriers identified, such as land availability or the integration of NbS with traditional approaches, do not have direct or easily applicable tools. Instead, most of the identified resources are focused on guidelines, planning frameworks, and supportive methodologies that accompany the NbS planning process and support the enabler supportive planning rules and practices for NbS, which was identified across both regions.

5.4. Technical & Capacity

The Technical & Capacity category was one of the least prominent in terms of barriers, but one of the most prominent in relation to enablers. Similarly, while the barriers within this category showed a strong concentration in the GS, the enablers were more evenly distributed across both regions. This may be related to broader structural differences in education and technical capacity between the GN and the GS. Although educational conditions in the GS have improved significantly over the last decades, the region still generally presents lower educational indicators than the GN, and is often associated with poverty, inequality, conflict, and limited access to technology (Kowalski, 2020; Reimers et al., 2026). The issue of education was also highlighted in interview GS2, where it was mentioned that research and education related to specific territorial and biodiversity issues in Colombia remain limited. This is consistent with the fact that the only enabler showing a particularly strong predominance in this category was education and training programs.

Most of the other identified enablers, which were shared across both regions, were associated with the development of clear guidelines, systems, protocols, and standardization processes for NbS implementation. This is reflected in the large number of standardization and measurement tools identified and later incorporated into the toolkit. Interestingly, this also connects with issues discussed within the Financial category, particularly the challenge of monetizing NbS and recognizing their multiple co-benefits, processes that can be facilitated through greater standardization and more systematic evaluation methods.

Another strong pattern emerging from the enablers is the emphasis on knowledge generation and technical learning, reflected in enablers such as technical knowledge and expertise in implementing NbS projects and the learning-by-doing approach. These findings are further reinforced in the Knowledge & Information category, which was the category with the third highest number of coded enabler segments. These results suggest that knowledge generation, together with collaboration, can be recognized in this research as one of the main enabling factors for NbS implementation regardless of region. This is also reflected in the toolkit, particularly in the availability of guidelines and tools supporting effective monitoring, maintenance, and evaluation systems and protocols.

In terms of barriers, one relevant aspect identified in the literature is that a large proportion of the available technical information is produced in English, creating an additional challenge for non-English-speaking countries (Abera et al., 2025). Although this issue is explored more deeply in the Knowledge & Information category, it is important to recognize that it represents a transversal challenge for NbS implementation and emerged in interviews GS2 and GN1.

Another important aspect highlighted in this category is the role of technology in NbS implementation, which appeared both as a barrier and as an enabler. On one hand, limited access to appropriate technologies and infrastructure for NbS was identified mainly in the GS and may be associated with lower levels of technological development when compared to the GN (Reimers et al.,

2026). On the other hand, the integration of technologies and tools in NbS emerged as a broadly recognized enabler across regions. This finding was reinforced by the interviews, where three organizations (GS1, GS2, and GN3) highlighted the importance of technologies for monitoring and measuring project outcomes. Particularly in the GS interviews, the direct use of technology by farmers not only improved monitoring processes but also strengthened stakeholder engagement and participation within NbS implementation.

Another important enabler identified across both regions is the learning-by-doing approach. This enabler was categorized within Technical & Capacity because it relates directly to the practical process of learning how to implement NbS, although it simultaneously contributes to overcoming barriers present across several other categories. Within the toolkit, pilots and living labs were also included as examples of learning-by-doing methodologies. In interview GN2, the project was described as initially starting as a pilot, while interview GN3 emphasized the importance of moving from research to practical application, connecting directly with the research–practice gap discussed later in the Knowledge & Information category. Similarly, interviews GS1 and GS2 highlighted how farmers learn progressively through practice and experimentation, particularly through small pilot initiatives that demonstrate the effectiveness of NbS approaches over time.

5.5. Social, Engagement & Perception

The Social, Engagement & Perception category shows a pattern somewhat similar to that of the Technical & Capacity category. In terms of barriers, it was one of the categories with the lowest number of coded segments, whereas in relation to enablers it was the category with the highest number of coded segments. Similarly, barriers showed a stronger concentration within the GS, and were mainly related to issues of perception and preference regarding NbS. In contrast, the enablers were largely concentrated around themes of collaboration, participation, and stakeholder engagement. This may suggest that one of the main ways to overcome barriers related to perception and preference is through increasing engagement and collaborative processes among stakeholders, as these processes contribute to strengthening trust, knowledge exchange, and collective understanding. As highlighted in the Results section, collaboration and engagement emerged as some of the most prominent themes among the identified enablers and could arguably be considered the main enabler of NbS implementation within this research. Approximately one third of the coded enabler segments were related to knowledge sharing, collaboration, and interaction among stakeholders.

This category was strongly reinforced both through the interviews and through the design of the toolkit. In four out of the five interviews (GS1, GS2, GN2, and GN3), engagement with communities and collaboration among stakeholders were highlighted as central enablers for project implementation. Both GS1 and GS2 emphasized how direct work with farmers and the construction of trust-based relationships have been fundamental for strengthening the projects developed in rural and riparian areas. Similarly, in interview GN2, collaborative work between livestock farmers and technical maintenance workers was described as essential for enabling the successful implementation of the NbS initiative. This aspect was also reinforced in interview GN3, where it was mentioned that working with farmers became relatively straightforward once relationships of trust had been established and once the interests of both the implementing organization and the farmers were recognized and considered. The interview also highlighted how partnerships among stakeholders facilitate implementation by incorporating perspectives from multiple disciplines and by diversifying funding sources among partners rather than depending on a single source of support.

Within the broader themes of engagement and collaboration, interviews GN2 and GN3 highlighted the importance of intermediary actors supporting the implementation of NbS. This was also recognized by Zarei and Shahab, (2025), where intermediaries are described as actors capable of facilitating dialogue, cultivating a shared language, identifying common priorities, and creating bridges of understanding among stakeholders. Finally, the interviews identified passion and personal commitment as additional important enablers that were not strongly reflected in the literature review. Both GN2 and GN3 emphasized how the passion and commitment of project leaders played a fundamental role in the implementation and continuity of the initiatives.

In relation to the toolkit, multiple tools and methodologies supporting participatory and collaborative processes among stakeholders were identified, further reinforcing the importance of collaboration as a central enabling mechanism for NbS implementation and scaling.

Overall, the interviews strongly validate collaboration, engagement, and trust-building processes as some of the main enablers of NbS implementation and suggest that these aspects function as broadly transferable enablers across both the GN and the GS.

5.6. Knowledge & Information

The Knowledge & Information category showed a very similar pattern in both barriers and enablers. In both cases, it represented the third most prominent category, with a similar number of coded segments and a relatively balanced distribution across the three categories (GS, GN, and General). In terms of enablers, however, the category showed a slightly stronger concentration in the GS while still maintaining a considerable representation within the General literature.

One of the barriers identified across both regions was the limited evidence on NbS effectiveness and long-term benefits. Interestingly, this contrasts with the considerable number of knowledge repositories and platforms identified during the development of the toolkit, many of which provide databases, case studies, implementation guidelines, and other technical resources. This suggests that the challenge may not necessarily be the absence of information itself, but rather limitations in knowledge translation and dissemination, an enabler that was identified as particularly relevant in the GS literature. In this sense, the research–practice gap identified throughout the literature appears to be less related to the absence of technical information and more connected to processes of translation, accessibility, contextualization, and stakeholder engagement.

A topic that emerged strongly in two of the interviews, although not highly represented in the literature review, was the role of terminology as a barrier to NbS implementation and understanding. In interview GS2, it was highlighted how NbS terminology, which is largely generated within the GN (McPhearson et al., 2025), can create additional gaps in the GS. The interviewee explained that when communities are approached using the term “Nature-based Solutions,” they often do not immediately recognize the concept. However, once the idea is explained in more practical terms, communities frequently realize that they have already been implementing similar practices long before these approaches were labelled as NbS. Furthermore, some initiatives that effectively function as NbS may become excluded from the concept due to terminological or classification issues rather than because of the actual practices being implemented. In some cases, these concepts may also face resistance simply because they are perceived as originating from the GN, despite the fact that many of these solutions have historically existed within local practices and knowledge systems.

This issue of terminology was also mentioned in interview GN1, where it was noted that many NbS-related concepts arrive in the Netherlands in English and are not always translated into local languages, creating ambiguity regarding how the concepts are interpreted and what should be included or excluded under these definitions. These findings suggest that language and terminology can themselves become barriers to the adoption and mainstreaming of NbS across contexts.

Closely related to this, knowledge translation emerged as a particularly important theme in interviews GS1 and GS2. In both cases, technical and scientific concepts were “translated” into more practical and accessible language so that farmers and local communities could better understand, internalize, and engage with the processes being implemented. This translation process was described as fundamental for building trust and establishing long-term relationships with stakeholders.

Another important aspect highlighted in interview GN1 was how higher levels of awareness regarding climate change can facilitate the adoption of NbS. The interview emphasized that recognizing the existence of a climate crisis can transform environmental threats into motivation for implementing NbS initiatives. This finding is consistent with Velayo et al., (2024), where both GN and GS cases

showed a correlation between awareness of climate change and a greater willingness to adopt actions such as NbS. Although this specific enabler was not highly prominent within the literature review, it emerged as a relevant insight through the interviews.

Despite the limitations of time in the design of the toolkit, which did not allow for an exhaustive exploration of all existing knowledge-sharing platforms, a large number of repositories and resources were still identified. This further reinforces enablers such as knowledge-sharing mechanisms and technologies and the availability of reliable data on NbS, suggesting that a considerable amount of information and technical support already exists. This is also reflected in recent initiatives such as the IUCN NbS Knowledge Hub (IUCN NbS, 2026) , which aims to become a central reference point for NbS practitioners by progressively compiling guidance documents, case studies, training materials, and practical tools for NbS implementation. Interestingly, although the platform is still under development and currently does not yet provide accessible resources, its creation itself highlights the recognized need for more centralized, practical, and accessible NbS implementation support, further reinforcing the relevance of the toolkit proposed in this research.

Overall, the findings of this research suggest that while many of the barriers to NbS implementation are highly context-dependent and shaped by territorial, institutional, socio-economic, and ecological realities, many of the identified enablers appear to be more transferable across regions. In particular, collaboration, stakeholder engagement, knowledge sharing, continuous support, and capacity-building processes emerged as transversal enablers across both the Global North and the Global South. At the same time, the research highlights that the implementation and scaling of NbS cannot rely solely on technical solutions or isolated tools, as many of the identified barriers are structural and linked to governance systems, financial priorities, planning processes, and historical inequalities. In this sense, the toolkit developed through this research should not be understood as a definitive solution to all barriers surrounding NbS implementation, but rather as a supporting resource that can help strengthen enabling conditions, facilitate collaboration and knowledge exchange, and support practitioners in navigating the complexity of NbS processes. Ultimately, the findings reinforce the idea that successful NbS implementation depends not only on ecological or technical considerations, but also on the capacity to build trust, translate knowledge into practice, foster long-term relationships among stakeholders, and adapt solutions to the specific realities of each territory.

6. Conclusion

This research explored how barriers and enablers for implementing Nature-based Solutions (NbS) differ between the Global North (GN) and the Global South (GS), while also examining whether a practical toolkit for NbS implementation can be designed based on existing tools, methodologies, and frameworks. Through a mixed-methods approach combining a literature review, semi-structured interviews, and the development of a toolkit prototype, the study identified important regional differences as well as several transversal dynamics shaping NbS implementation across contexts.

Regarding the first research question, the findings indicate that there are indeed important differences between the barriers and enablers affecting NbS implementation in the GN and the GS. Barriers identified in the GS were more frequently associated with governance weaknesses, policy misalignment, financial dependency, rapid urbanization, limited technical capacity, and lower access to technology, education, and institutional support. In contrast, the GN literature more commonly highlighted barriers related to institutional inertia, highly regulated systems, lack of space, and the complexity of integrating NbS into already consolidated infrastructural and governance frameworks. These findings suggest that barriers to NbS are highly context-dependent and shaped by territorial, institutional, socioeconomic, and ecological realities.

At the same time, the results showed that enablers were generally more transferable across regions than barriers. While some regional distinctions were identified, particularly in governance and technical capacity, many of the most prominent enablers appeared consistently across both the GN

and the GS. Among these, collaboration, stakeholder engagement, trust-building, knowledge sharing, participatory governance, and continuous support emerged as some of the most important enabling conditions for NbS implementation. This suggests that although many barriers originate at structural and institutional levels, successful implementation processes are often strengthened through relational and collaborative dynamics operating at local and organizational scales.

Another important finding of this research is that NbS implementation cannot be understood solely as a technical or ecological process. Instead, the findings repeatedly highlight the importance of social and relational dimensions such as trust, communication, co-production of knowledge, and long-term stakeholder engagement. In this sense, the study reinforces the idea that successful NbS implementation depends not only on scientific or technical expertise, but also on the ability to translate knowledge into practice, adapt solutions to local realities, and foster collaborative relationships among stakeholders.

The research also identified the importance of a balance between standardization and contextualization in NbS implementation. On one hand, frameworks such as the IUCN Global Standard and the large number of repositories, methodologies, and operational tools identified throughout the study demonstrate ongoing efforts to operationalize and standardize NbS processes. On the other hand, one of the strongest findings of this research is that NbS are highly context-specific, meaning that implementation strategies and tools must remain flexible and adaptable to different ecological, social, cultural, and governance realities.

In relation to the second research question, the study demonstrates that it is possible to design a practical toolkit for implementing NbS based on the analysis of existing tools, methodologies, and frameworks. The toolkit developed through this research functions as an initial prototype that compiles practical resources associated with the barriers and enablers identified throughout the study. The process revealed that a significant number of tools and repositories related to NbS implementation already exist, including implementation guides, governance frameworks, participatory methodologies, monitoring systems, and business and innovation tools. However, the research also highlighted that this information is divided across repositories, requiring considerable time and effort to identify, organize, and contextualize.

Therefore, the toolkit developed in this thesis should not be understood as an exhaustive or definitive resource, but rather as an evolving and expandable support mechanism that can continue incorporating additional tools, regional perspectives, and types of NbS interventions over time. In this sense, the toolkit contributes to ongoing efforts to operationalize NbS implementation while also acknowledging that effective implementation requires context-sensitive and adaptable approaches rather than one-size-fits-all solutions.

Finally, this research contributes to the growing body of literature seeking to better understand NbS implementation across different regional contexts. It also highlights the need for more comparative studies explicitly examining the differences and similarities between the Global North and the Global South, as well as greater integration of local and Indigenous knowledge systems, stronger processes of knowledge translation and dissemination, and continued efforts to bridge the gap between research and implementation.

7. Limitations

This research presents several limitations that should be acknowledged. First, the literature review sample included a larger number of articles focused on the Global South compared to the Global North. Although this imbalance itself generated important insights regarding the current state of NbS research, it also limits the extent to which direct comparisons between the two regions can be generalized. In addition, several of the reviewed articles classified as “General” did not explicitly specify a regional context, making it difficult to determine whether some barriers and enablers were region-specific or more universally applicable.

Another limitation of this research is the difference in thematic focus between the literature review and the interview component. A significant proportion of the NbS articles analysed were focused on urban Nature-based Solutions, particularly within the context of urban planning and green infrastructure. In contrast, most of the interviews were centred on ecosystem conservation, biodiversity restoration, agricultural landscapes, and rural or riverine contexts. This difference may have influenced the types of barriers and enablers identified across the two methods and limited the comparability between the literature review findings and the interview results in some of the categories.

Regarding the toolkit, the scope and time limitations of this research did not allow for an exhaustive exploration of all available NbS tools, repositories, and methodologies. A large number of resources were identified during the research process, many of them distributed across different platforms and languages, suggesting that the toolkit developed in this study represents only a partial compilation of the tools currently available. Furthermore, some of the identified barriers, particularly those related to governance, institutional structures, and spatial constraints, do not easily translate into direct operational tools, highlighting the limitations of tool-based approaches for addressing deeply structural challenges.

Finally, given the complexity and context-specific nature of NbS, the findings of this research should not be interpreted as universally applicable across all contexts. Instead, they should be understood as an exploratory and comparative contribution intended to support future research and practice on NbS implementation across different regional realities.

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Note: ChatGPT (OpenAI) was used to support the editing of language, grammar, and overall coherence of the text. All ideas, analysis, and interpretations presented in this draft are the author's own. The use of this tool was limited to improving clarity and readability and did not involve the generation of original content or findings.

10. Appendix

10.1. Appendix 1. Interview guide

Interview Guide 1.0

1. Could you briefly describe the project you are involved in and your role in it?
    a. What is the main purpose of the project?
    b. Do you consider this project to be an NbS?
    c. Where is the project located?
2. Who are the actors/collaborators (stakeholders) involved in the project?
3. Since you have been working on this project, what are the main challenges the project has faced or is currently facing?
    a. From the provided list of barriers, which ones have affected or are affecting your project?
    b. Would you like to mention other barriers that may affect NbS in general and are not listed here?
    c. Do you think there is a main barrier that needs to be addressed?
4. On the other hand, can you identify some enablers, elements, or drivers that have supported the development of your project?
    a. From the list of enablers, are there any that resonate with your project?
    b. Would you like to mention other enablers that may affect NbS in general and are not listed?
    c. Do you feel there is a key enabler that has been most important for your project and/or for NbS projects in general?
5. Do you think some of these barriers or enablers are specific to your regional context (e.g., Global North or Global South)? If so, how?
6. In your opinion, what is most urgently needed to scale, implement, or further develop NbS in your region?
    a. Do you have any other experience working with NbS projects that you will like to share?
7. Have you used any tools, frameworks, or methodologies that have supported the development or implementation of this NbS project? If yes, which ones?
    a. What types of tools, support, resources, or guidance would make it easier to implement Nature-based Solutions in your context?
    b. Do you think a general toolkit with different tools and frameworks could be useful for NbS practitioners across different regions? Why or why not?
8. We are developing a toolkit to support NbS practitioners—would you be interested in participating in or contributing to its design?
    a. Do you know others who will also like to contribute?
9. Is there anything else you would like to share about this topic that we have not covered today?

Interview Guide 2.0

1. Could you briefly describe the project you are involved in and your role in it?
    a. What is the main purpose of the project?
    b. Where is the project located?
2. Who are the actors/collaborators (stakeholders) involved in the project?
3. Since you have been working on this project, what are the main challenges the project has faced or is currently facing?
4. On the other hand, can you identify some enablers, elements, or drivers that have supported the development of your project?

5. Do you think some of these barriers or enablers are specific to your regional context (e.g., Global North or Global South)? If so, how?
6. In your opinion, what is most urgently needed to scale, implement, or further develop NbS in your region?
7. Have you used any tools, frameworks, or methodologies that have supported the development or implementation of this NbS project? If yes, which ones?
    a. What types of tools, support, resources, or guidance would make it easier to implement Nature-based Solutions in your context?
    b. Do you think a general toolkit with different tools and frameworks could be useful for NbS practitioners across different regions? Why or why not?
8. We are developing a toolkit to support NbS practitioners—would you be interested in participating in or contributing to its design?
9. Is there anything else you would like to share about this topic that we have not covered today?

10.2. Appendix 2: Articles Literature Review

Code	Authors	Article Title	Where
1	Fu, HP	A Comprehensive Review of Nature-Based Solutions: Current status and future research	General
2	Bousquet, M; Kuller, M; Lacroix, S; Vanrolleghem, PA	A critical review of MCDA practices in planning of urban green spaces and NBS	General
3	Leknoi, U; Duangden, B; Tanapalungkorn, W; Likitlersuang, S	A multi-level perspective on implementing vetiver grass as a nature-based and social innovation for landslide risk reduction	GS
4	Wolff, E; French, M; Naulumatua, M; Mcevoy, D	Advancing just climate adaptation in the global South: socio-technical assistance for nature-based solutions in the Pacific	GS
5	Rauf, HA; Wolff, E; Natakun, B; Marome, W; Hamel, P	Aligning nature-based solutions and housing policy: A study on stakeholders' perceptions of nature within informal settlement upgrading projects	GS
6	Tyllianakis, E; Martin-Ortega, J; Banwart, SA	An approach to assess the world's potential for disaster risk reduction through nature-based solutions	General
7	Collier, MJ; Frantzeskaki, N; Connop, S; Dick, G; Dumitru, A; Dziubala, A; Fletcher, I; Georgiou, P; Hölscher, K; Kooijman, E; Lodder, M; Madajczyk, N; McQuaid, S; Nash, C; Osipiuk, A; Quartier, M; Reil, A; Rhodes, ML; Rizzi, D; Vandergert, P; van de Sijpe, K; Vos, P; Xidousi, D	An integrated process for planning, delivery, and stewardship of urban nature-based solutions: The Connecting Nature Framework	GN
9	Velayo, ARA; Suson, PD; Aguilos, MM; Bacosa, HP	Building Urban Climate Resilience: Assessing Awareness, Perception, and Willingness regarding Nature-Based Solutions and Climate Change among Stakeholders in Iligan City, Philippines	GS
10	Boafo, YA; Enu, KB; Agyei, KD; Kolog, JD	Citizen-led initiatives and hydro-meteorological risks reduction: Who is implementing nature-based solutions?	GS

Code	Authors	Article Title	Where
11	Nizamani, AG; Torgersen, G	Climate as the primary moderator: towards context-driven design and implementation of vertical greenery systems for stormwater management	GN and GS
12	Woroniecki, S; Spiegelenberg, FA; Chausson, A; Turner, B; Key, I; Irfanullah, HM; Seddon, N	Contributions of nature-based solutions to reducing people's vulnerabilities to climate change across the rural Global South	GS
13	Mirsafa, M; Castaldo, AG; de Oliveira, FL	Enabling nature-based solutions for climate adaptation in cities of the Global South: planning dimensions and cross-cutting pathways for implementation	GS
14	Boateng, EA; Asibey, MO; Cobbinah, PB; Adutwum, IO; Blija, DK	Enabling nature-based solutions: Innovating urban climate resilience	GS
16	Hale, SE; von der Tann, L; Rebelo, AJ; Esler, KJ; de Lima, APM; Rodrigues, AF; Latawiec, AE; Ramírez-Agudelo, NA; Bosch, ER; Suleiman, L; Singh, N; Oen, AMP	Evaluating Nature-Based Solutions for Water Management in Peri-Urban Areas	GN GS
19	McPhearson, T; Frantzeskaki, N; Ossola, A; Diep, L; Anderson, PML; Blatch, T; Collier, MJ; Cook, EM; Fatti, CC; Grabowski, ZJ; Grimm, NB; Haase, D; Herreros-Cantis, P; Kavonic, J; Lin, BB; Meneses, DHL; Matsler, AM; Moglia, M; Morató, J; O'Farrell, P; Roy, P; Singh, C; Wang, J; Zhou, WQ	Global synthesis and regional insights for mainstreaming urban nature-based solutions	General
20	Ngoc, CHT; Back, Y; Funke, F; Hauser, M; Kleidorfer, M	Implementation of Nature-Based Solutions in Urban Water Management in Viet Nam, a Comparison among European and Asian Countries	GS
21	Lee, WJ; Song, JRY	Innovative strategy for enhancing nature-based solutions during climate technology transfer process	GS
22	Pan, CY; Shrestha, A; Li, C; Ying, MJ; Duan, J; He, M; Zhu, WR; Chai, QH; Innes, JL; Kozak, R; Zhou, GM; Wang, GY	Integrating bamboo forests into the carbon markets: Insights from China	GS
23	Rivera, LM; Faggian, R	Integrating Blue-Green Infrastructure strategies to enhance climate resilience in Colombia	GS

Code	Authors	Article Title	Where
24	Abera, LE; Jumani, S; van Rees, CB; Krishnaswamy, J; Seigerman, CK; Nelson, DR; Hallemeier, J; Mulatu, DW; Arora, R; Sesma, MPV; Asnake, K; Aliu, J; Quayee, A; Myeong, S; Echevarria, M; Stern, M; van Breda, A; Hettiarachchi, M; Becerra, AC; Quick, A; McKay, SK	Integrating Nature-based Solutions for urban water security in global south	GS
26	Bauer, W; Titz, A	Manoeuvring barriers: Assessing adaptive strategies for and persistent barriers to urban Nature-based Solutions in Lilongwe, Malawi	GS
28	Castaldo, AG; Nocentini, MG; de Oliveira, FL; Mahmoud, IH	Nature-based solutions and urban planning in the Global South: Challenge orientations, typologies, and viability for cities	GS
30	Simelton, E; Carew-Reid, J; Coulier, M; Damen, B; Howell, J; Pottinger-Glass, C; Tran, HV; Van der Meiren, M	NBS Framework for Agricultural Landscapes	General
34	Thuy, BT; Dao, AD; Han, M; Nguyen, DC; Nguyen, VA; Park, H; Luan, PDMH; Duyen, NTT; Nguyen, HQ	Rainwater for drinking in Vietnam: barriers and strategies	GS
36	Zyoud, S; Zyoud, AH	Revealing global trends on nature-based solutions: Mapping and visualizing research landscapes	General
37	Regmi, S; Gheewala, SH	Scaling up nature-based solutions for food security and climate resilience in wetlands: typologies, barriers and opportunities	General
38	Liu, HH; Martens, P	Stakeholder Participation for Nature-Based Solutions: Inspiration for Rural Area's Sustainability in China	GS
39	Enu, KB; Zingraff-Hamed, A; Lupp, G; Raum, S; Moors, E; Pauleit, S	Stakeholder priorities and navigating barriers in urban river restoration: Comparative insights from Germany and Ghana	GN GS

Code	Authors	Article Title	Where
40	Agaton, CB; Guila, PMC	Success Factors and Challenges: Implications of Real Options Valuation of Constructed Wetlands as Nature-Based Solutions for Wastewater Treatment	GS
41	Debie, E	Synergistic integration of urban agriculture and green infrastructure to enhance urban sustainability in Bahir Dar, Ethiopia	GS
42	Marchetti, L; Kauark-Fontes, B; Guerrero, CO; Hernandez-Garcia, J; Wild, T; Salbitano, F	The emergence of nature-based solutions in Latin American urban policies: A critical review. The case of Bogota, Buenos Aires, Santiago de Chile and Sao Paulo	GS
43	Artmann, M; Sartison, K	The Role of Urban Agriculture as a Nature-Based Solution: A Review for Developing a Systemic Assessment Framework	GN
44	Villarroel-Lamb, D; Ganpat, S; Ville, AS; Williams, A; Kanhai, L; Aziz, F; Rouwette, EAJA; Davis, D; Mohan, P	Transcending disciplinary boundaries: Nature based solutions (NbSs) for Caribbean coastal resilience	GS
45	Essuman-Quainoo, B; Jim, CY	Understanding the drivers of green roofs and green walls adoption in Global South cities: Analysis of Accra, Ghana	GS
47	Telwala, Y	Unlocking the potential of agroforestry as a nature-based solution for localizing sustainable development goals: A case study from a drought-prone region in rural India	GS
59	Bustamante, MMC; Silva, JS; Scariot, A; Sampaio, AB; Mascia, DL; Garcia, E; Sano, E; Fernandes, GW; Durigan, G; Roitman, I; Figueiredo, I; Rodrigues, RR; Pillar, VD; de Oliveira, AO; Malhado, AC; Alencar, A; Vendramini, A; Padovezi, A; Carrascosa, H; Freitas, J; Siqueira, JA; Shimbo, J; Generoso, LG; Tabarelli, M; Biderman, R; Salomao, RD; Valle, R; Junior, B; Nobre, C	Ecological restoration as a strategy for mitigating and adapting to climate change: lessons and challenges from Brazil	GS
63	du Plessis, NS; Rebelo, AJ; Richardson, DM; Esler, KJ	Guiding restoration of riparian ecosystems degraded by plant invasions: Insights from a complex social-ecological system in the Global South	GS
64	Zhu, Z; Chan, FKS; Li, G; Xu, MX; Feng, ML; Zhu, YG	Implementing urban agriculture as nature-based solutions in China: Challenges and global lessons	GS
77	Ojeda-Revah, L; Muñoz-Meléndez, G	Ecoparque: An Example of Nature-Based Solutions Implementation at Tijuana a Global South City	GS

Code	Authors	Article Title	Where
78	Niki Frantzeskaki , Timon McPhearson , Marcus J Collier , Dave Kendal , Harriet Bulkeley , Adina Dumitru , Claire Walsh , Kate Noble , Ernita van Wyk , Camilo Ordóñez , Cathy Oke , László Pintér	Nature-Based Solutions for Urban Climate Change Adaptation: Linking Science, Policy, and Practice Communities for Evidence-Based Decision-Making	General
79	Alexander P.N. van der Jagt, Luca R. Szaraz,Tim Delshammar, Rozalija Cvejić, Artur Santos, Julie Goodness, Arjen Buijs	Cultivating nature-based solutions: The governance of communal urban gardens in the European Union	GN
80	Björn Wickenberg, Kes McCormick, Johanna Alkan Olsson	Advancing the implementation of nature-based solutions in cities: A review of frameworks	General
81	Shahryar Sarabi, Qi Han a, A. Georges L. Romme, Bauke de Vries, Rianne Valkenburg, Elke den Ouden	Uptake and implementation of Nature-Based Solutions: An analysis of barriers using Interpretive Structural Modeling	GN
82	Shahryar Ershad Sarabi, Qi Han, A. Georges L. Romme, Bauke de Vries, Laura Wendling	Key Enablers of and Barriers to the Uptake and Implementation of Nature-Based Solutions in Urban Settings: A Review	General
84	Melika Zarei and Sina Shahab	Nature-Based Solutions in Urban Green Infrastructure: A Systematic Review of Success Factors and Implementation Challenges	General

10.3. Appendix 3: Knowledge repositories

- Naturaleza Transformativa: https://naturaleza-transformativa.com/
- Oppla: http://oppla.eu/
- ThinkNature: https://think-nature.eu/
- Natural Infrastructure for Business platform: https://naturalinfrastructureforbusiness.org/
- Repository of NbS guidance material: https://oppla.eu/conexus/resource/repository-nbs-guidance-material
- NetworkNature Resources: https://networknature.eu/networknature/networknature-resources
- Urban Nature Gateway: https://interlace-hub.com/urban-nature-gateway
- Conexus resources. https://www.conexusnbs.com/resources

10.4. Appendix 4: Enablers and barriers comparison

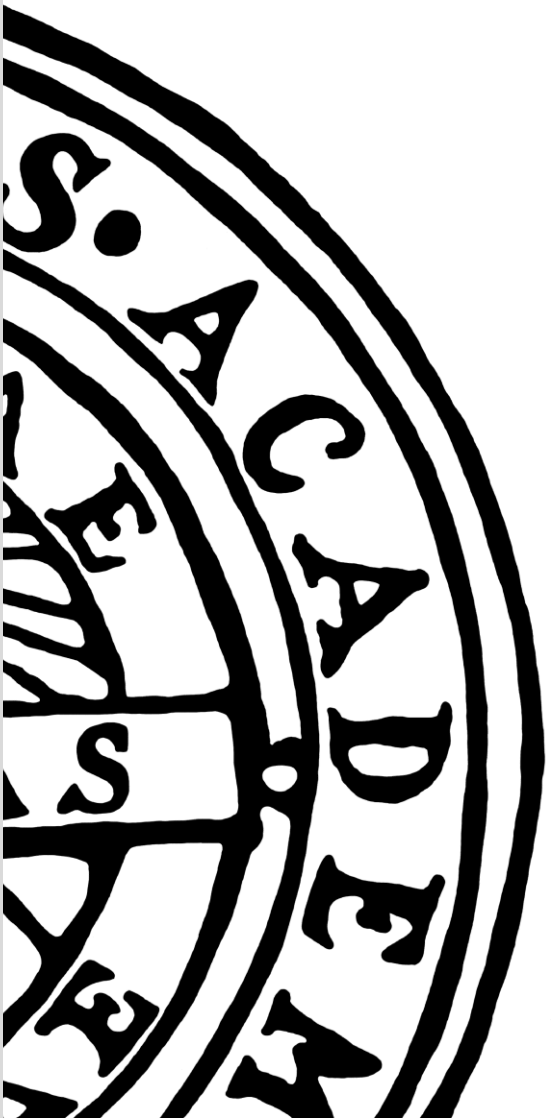
Barriers	Articles mentioning this*	Prevalence	Enablers	Articles mentioning this*	Prevalence
Governance & Institutional			**Governance & Institutional**		
1. Lack of clear policies and regulatory frameworks	11, 21, 23, 26, 28, 36, 37, 41, 64, 82, 84	**GS, GN and General**	1. Supportive policy frameworks and regulations	7, 11, 12, 14, 16, 20, 21, 23, 24, 26, 28, 30, 36, 37, 39, 40, 42, 43, 59, 64, 79, 80, 82, 84	**GS, GN and General**
2. Weak enforcement of existing policies	14, 19, 28, 37, 39, 42	**GS**	3. Supportive governance structures to NbS processes	9, 11, 12, 13, 20, 28, 42	**GS**
3. Fragmented and siloed institutional structures	19, 21, 23, 26, 28, 30, 42, 64, 81, 82, 84	**GS, GN and General**			
4. Poor coordination across stakeholders and governance levels	2, 5, 13, 14, 19, 20, 21, 24, 26, 28, 37, 38, 39, 42, 44, 80, 84	**GS**	2. Collaboration across disciplines and governance levels	1,7, 13, 21, 24, 28, 37, 42, 80	**GS and General**
			5. Inclusive and participatory governance processes	5, 10, 19, 30, 42	**GS and general**
5. Misalignment between national and local policies and decisions	5, 13, 19, 28, 82	**GS**	3. Supportive governance structures to NbS processes	9, 11, 12, 13, 20, 28, 42	**GS**
6. Restrictive regulations and conventional planning approaches	11, 13, 14, 21, 26, 30, 36, 39, 63, 64, 84	**GS, GN and General**			
7. Lack of long-term political commitment	5, 6, 19, 23, 26, 28, 30, 36, 38, 39, 43, 44, 64, 81, 84	**GS, GN and General**	4. Strong and sustained political will and support	3, 4, 11, 13, 20, 21, 30, 40 42, 64, 77, 79, 80	**GS, GN and General**
Financial			**Financial**		
1. Lack of funding or investment	5, 14, 16, 19, 20, 21, 26, 28, 36, 37, 38, 41, 42, 47, 63, 64, 77, 76	**GS> GN and General**	2. Financing mechanisms, economic instruments and incentives	7, 10, 14, 19, 20, 24, 28, 30, 36, 37, 38, 42, 43, 63, 82, 84	**GS, GN and General**
2. High upfront costs of NbS	11, 14, 30, 34, 36, 37, 39, 43,	**GS, GN and General**	3. Financial feasibility and economic sustainability of NbS	7, 13, 19, 20, 30 37, 76, 80, 84	**GS, GN and General**

Barriers	Articles mentioning this*	Prevalence	Enablers	Articles mentioning this*	Prevalence
	44, 47, 59, 84				
3. Limited access to and mechanisms for mobilizing investment	21, 26, 37, 42, 44, 81, 84	**GS, GN and General**	2. Financing mechanisms, economic instruments and incentives	7, 10, 14, 19, 20, 24, 28, 30, 36, 37, 38, 42, 43, 63, 82, 84	**GS, GN and General**
4. Lack of financial incentives (subsidies, tax benefits)	11, 19, 21, 44, 81	**GS, GN and General**			
5. Difficulty monetizing NbS benefits	11, 19, 21, 22, 42, 76, 81	**GS, GN and General**	1. Clear and appropriate business models	3, 7, 11, 21, 37, 64, 76, 79, 80	**GS, GN and General**
6. Dependence on external funding (donors, public)	5, 26, 39, 77, 82	**GS**	5. Public-private partnerships (PPP)	7, 18, 20, 23, 26, 39, 76, 80, 82, 84	**GS, GN and General**
7. Mismatch between short-term investment horizons and long-term economic returns	36, 37, 47, 82, 84	**General**	4. Economic valuation of NbS benefits	11, 13, 76, 80	**GS, GN and General**
8. Prioritizing the allocation of financial resources for conventional projects or other challenges (2)	5, 84	**GS and General**			
9. Lack of viable business models for NbS	21, 37, 42, 63, 64, 81	**GS**	1. Clear and appropriate business models	3, 7, 11, 21, 37, 64, 76, 79, 80	**GS, GN and General**
Planning & Spatial			**Planning & Spatial**		
1. Limited land availability and plot size constraints	5, 13, 14, 24, 26, 41, 43, 47, 64, 81, 84	**GS, GN, General**	4. Land availability, acquisition and management strategies	6, 13, 43, 63, 64	**GN>GS and General**
2. Land tenure insecurity	24, 26, 39	**GS**	3. Combining NBS with traditional approaches	14, 20, 24, 82	**GS**
			4. Land availability, acquisition and management strategies	6, 13, 43, 63, 64	**GN>GS and General**
3. Conflicts with existing land uses and management (housing, infrastructure)	5, 13, 19, 26, 39, 64, 81, 84	**GS, GN and General**	2. Adopting an integrated, context-specific and data-driven approach in planning and implementation n	3, 7, 10, 11, 13, 19, 20, 24, 42, 47	**GS**
			4. Land availability, acquisition and management strategies	6, 13, 43, 63, 64	**GN>GS and General**
4. Difficulty integrating NbS into legal and planning systems	5, 13, 14, 21, 28, 64, 84	**GS**	5. Supportive planning rules and practices to NbS	7, 19, 20, 24, 26, 28, 44, 79, 84	**GS, GN and General**
			3. Combining NBS with traditional approaches	14, 20, 24, 82	**GS**
			1. Integration of NbS into planning processes	7, 19, 23, 26, 64, 77, 82	**GS, GN and General**
5. Environmental and biophysical constraints on NbS implementation	13, 20, 21, 24, 26, 41,	**GS**	4. Land availability, acquisition and management strategies	6, 13, 43, 63, 64	**GN>GS and General**

Barriers	Articles mentioning this*	Prevalence	Enablers	Articles mentioning this*	Prevalence
	43, 64, 77		5. Supportive planning rules and practices to NbS	7, 19, 20, 24, 26, 28, 44, 79, 84	**GS, GN and General**
6. High population density and land-use pressure	13, 20, 39, 42, 84	**GS**	4. Land availability, acquisition and management strategies	6, 13, 43, 63, 64	**GN>GS and General**
			5. Supportive planning rules and practices to NbS	7, 19, 20, 24, 26, 28, 44, 79, 84	**GS, GN and General**
Technical & Capacity			**Technical & Capacity**		
1. Limited technical skills, training and capacity building for NbS	4, 5, 11, 14, 19, 21, 24, 34, 36, 37, 38, 39, 44, 63, 64, 77, 81, 84	**GS**	1. Technical Knowledge and Expertise in implementing NBS Projects	7, 16, 19, 24, 42, 84	**GS, GN and General**
			4. Learning-by-doing approach	1, 3, 7, 11, 13, 14, 19, 30, 80, 82	**GS, GN and General**
			7. Education and training programs	3, 4, 11, 14, 16, 20, 24, 42, 44, 63, 64, 82	**GS**
2. Lack of standardized guidelines and metrics	1, 19, 20, 37, 41, 64, 84	**GS and General**	2. Guidelines and standards- for NbS	1, 7, 11, 13, 19, 20, 24, 37	**GS, GN and General**
3. Limited monitoring, maintenance and evaluation systems	5, 11, 13, 19, 24, 36, 37, 38, 40, 41, 77, 81, 84	**GS, GN and General**	3. Effective monitoring, maintenance and evaluation systems and protocols	1, 2, 4, 13, 19, 20, 24, 37	**GS, GN and General**
			5. Delivery of multiple benefits	13, 24, 26, 39, 41, 43, 45, 84	**GS**
4. Limited access to appropriate technologies and infrastructure for NbS	23, 28, 37, 41, 43, 64	**GS**	6. The integration of technologies and tools in NbS	19, 21, 37, 64	**General**
Social, Engagement & Perception			**Social, Engagement & Perception**		
1. Preference for grey infrastructure, conventional solutions, and other alternatives	10, 23, 24, 44, 77, 81	**GS**	5. Enhanced public awareness and capacity building for NBS	6, 7, 9, 10, 11, 13, 42 ,63	**GS**
2. Low perceived relevance and understanding of NbS among stakeholders	4, 5, 11, 13, 14, 19, 21, 23, 24, 26, 36, 38, 42, 43, 81, 84	**GS, GN and General**			
3. Lack of participation or engagement/resistance from community or stakeholders	2, 5, 19, 24, 26, 36, 38, 41, 42, 43, 44, 63, 64, 81, 84	**GS, GN and General**	1. Involvement/engagement of the community in the NbS process	3, 4, 5, 6, 7, 10, 13, 16, 19, 21, 24, 37, 38, 40, 42, 43, 44, 47, 63, 64, 81, 82, 84	**GS, GN and General**
			2. Stakeholders' engagement and collaboration in NbS planning and development	1, 6, 7, 10, 13, 16, 19, 21, 24, 37, 39, 42, 79, 80, 82, 84	**GS, GN and General**

Barriers	Articles mentioning this*	Prevalence	Enablers	Articles mentioning this*	Prevalence
			3. Partnerships among Stakeholders	4, 7, 24, 36, 79, 80, 82	**GS, GN and General**
			4. Participatory methods and approaches	7, 10, 13, 19, 21, 40, 42, 44, 81	**GS, GN and General**
			6. Creation and nurture of NbS networks	6, 19, 30, 63, 79	**GS, GN and General**
4. Conflicts between stakeholders / inequitable distribution of benefits	5, 19, 21, 24, 26, 36, 38, 41, 42, 43, 44, 63, 64, 81, 84	**GS>GN and General**	1. Involvement/engagement of the community in the NbS process	3, 4, 5, 6, 7, 10, 13, 16, 19, 21, 24, 37, 38, 40, 42, 43, 44, 47, 63, 64, 81, 82, 84	**GS, GN and General**
			2. Stakeholders' engagement and collaboration in NbS planning and development	1, 6, 7, 10, 13, 16, 19, 21, 24, 37, 39, 42, 79, 80, 82, 84	**GS, GN and General**
			3. Partnerships among Stakeholders	4, 7, 24, 36, 79, 80, 82	**GS, GN and General**
			4. Participatory methods and approaches	7, 10, 13, 19, 21, 40, 42, 44, 81	**GS, GN and General**
			6. Creation and nurture of NbS networks	6, 19, 30, 63, 79	**GS, GN and General**
5. Path dependency and resistance to change in planning and decision-making practices	7, 11, 82	**GN and General**	5. Enhanced public awareness and capacity building for NBS	6, 7, 9, 10, 11, 13, 42 ,63	**GS**
6. Competing socio-economic priorities over environmental goals	24, 42	**GS**			
Knowledge & Information			**Knowledge & Information**		
1. Lack of data on NbS implementation and performance	2, 11, 13, 14, 19, 20, 21, 23, 26, 28, 34, 36, 42, 64, 82	**GS and General**	6. Availability of reliable data on NbS	7, 13, 20, 21, 30, 42, 44, 84	**GS and General**
2. Limited evidence on NbS effectiveness and long-term benefits	1, 7, 11, 19, 23, 24, 26, 80, 81	**GS, GN and General**	3. Identifying and communicating benefits, co-benefits of NbS	3, 7, 10, 11, 16, 19, 20, 26, 28, 42, 45, 64, 79, 80, 84	**GS, GN and General**
			4. Knowledge production and co-production	1, 19, 20, 21, 36, 42, 44, 80	**GS and General**
3. Lack and underutilization of local and context-specific knowledge	11, 19, 37, 42, 84	**GS and General**	1. Integration of local knowledge and conditions	7, 10, 19, 24, 26, 28, 30 37, 42, 81	**GS and General**
4. Limited knowledge sharing, transfer, and management across sectors and institutions	19, 24, 38, 84	**GS, GN and General**	2. Knowledge sharing mechanisms and technologies	3, 4, 11, 13, 19, 20, 24, 28, 37, 40,	**GS and General**

Barriers	Articles mentioning this*	Prevalence	Enablers	Articles mentioning this*	Prevalence
				42, 44, 79, 80, 82, 84	
			4. Knowledge production and co-production	1, 19, 20, 21, 36, 42, 44, 80	**GS and General**
5. Research-practice gap	1, 26, 44, 81, 82	**GS, GN and General**	5. Effective knowledge translation and dissemination	4, 16, 37, 40, 42, 44, 80	**GS**
6. Limited awareness and understanding of NbS	9, 11, 14, 16, 19, 21, 23, 26, 28, 34, 36, 37, 38, 41, 42, 47, 82	**GS and General**	3. Identifying and communicating benefits, co-benefits of NbS	3, 7, 10, 11, 16, 19, 20, 26, 28, 42, 45, 64, 79, 80, 84	**GS, GN and General**


ACADEMY